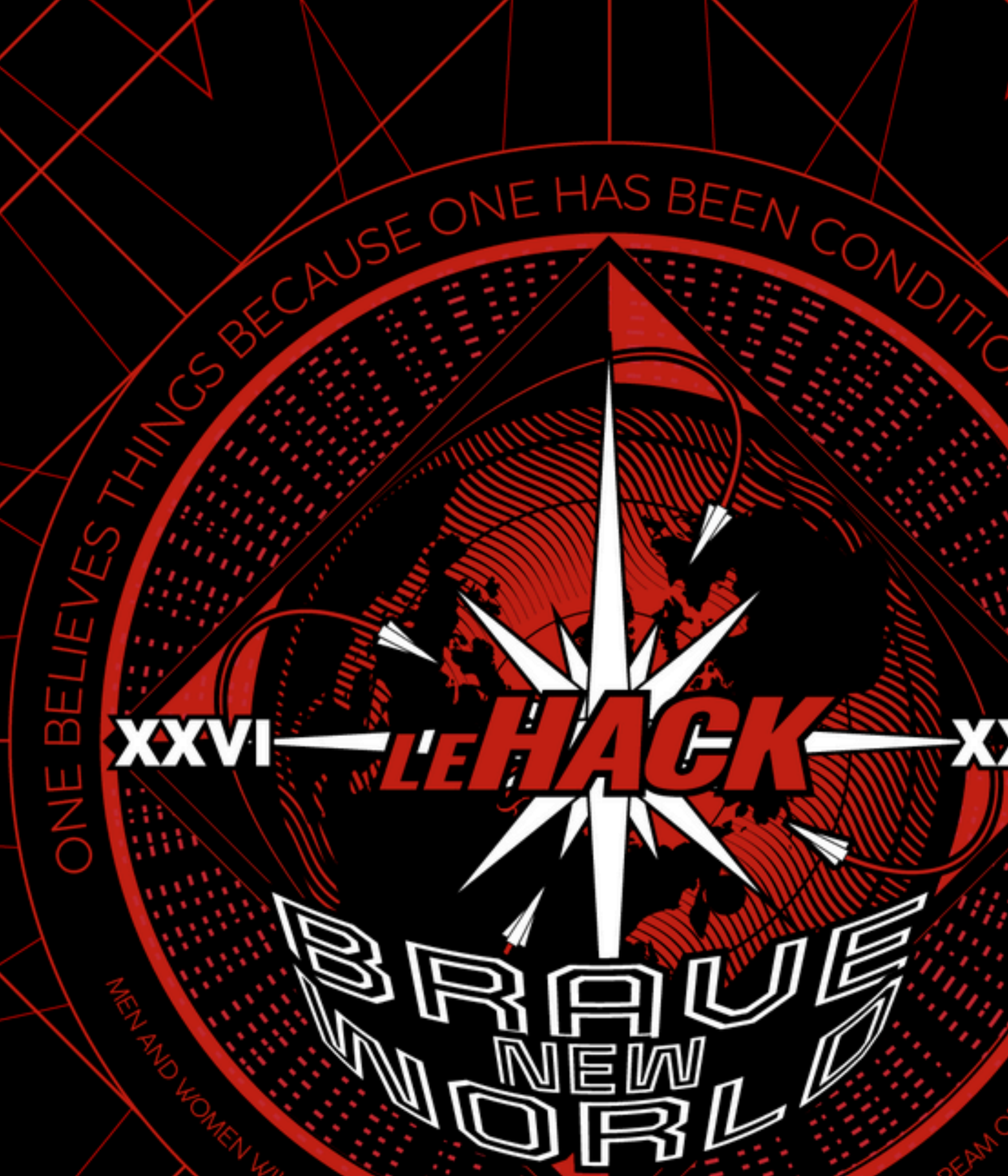
DEXCALIBUR 2.0 | CODE NAME :

REVERSENSE



Georges-Bastien Michel

LeHack - June 2026

</ɹǝʌǝɹSense>

Who am I ?



@FrenchYeti

Georges-Bastien Michel

Reverser / Security Researcher / Developer

https://github.com/FrenchYeti

Founder of	</ɹǝvǝʁSense>
Author of	Dexcalibur, Interruptor, …
Contributor to	*Radare2, frida, Ghidra, APKid, r2hermes, r2pipe…*

AGENDA

Intro

Principle, Features, Demo

ElixIR : Universal representation

Workshop Info

FILE FORMAT DETECTION

Mach-O not detected

```
[→  SideStore.app binwalk SideStore

                                                                                                                /Users/ge
--------------------------------------------------------------------------------------------------------------------------
DECIMAL                                HEXADECIMAL                            DESCRIPTION
--------------------------------------------------------------------------------------------------------------------------
4770112                                0x48C940                               CRC32 polynomial table, little endian
4783968                                0x48FF60                               CRC32 polynomial table, little endian
5441728                                0x5308C0                               SHA256 hash constants, little endian
--------------------------------------------------------------------------------------------------------------------------

Analyzed 1 file for 85 file signatures (187 magic patterns) in 50.0 milliseconds
```

PNG Confusion / Hallucination | PLT:



PNG



Apple's PNG (CGBI)

Unknown NIB, XIB, plist …

FILE FORMAT DETECTION

Mach-O not detected

```
[→  SideStore.app binwalk SideStore

                                                                                /Users/ge
----------------------------------------------------------------------------------------
DECIMAL                                                                       SCRIPTION
----------------------------------------------------------------------------------------
4770112                                                                       C32 polynomial table, little endian
4783968                                                                       C32 polynomial table, little endian
5441728                                                                       A256 hash constants, little endian
----------------------------------------------------------------------------------------

Analyzed 1 file f                                                             iseconds
```

binwalk / src / magic.rs

devttys0 Added MD5 hash constant signatures ✓

Code | Blame | 1235 lines (1233 loc) · 48.4 KB

```rust
use crate::extractors;
use crate::signatures;

/// Returns a list of all supported signatures, including their "magi
pub fn patterns() -> Vec<signatures::common::Signature> {
    let binary_signatures: Vec<signatures::common::Signature> = vec![
        // gzip
        signatures::common::Signature {
            name: "gzip".to_string(),
            short: false,
            magic_offset: 0,
            always_display: false,
            magic: signatures::gzip::gzip_magic(),
            parser: signatures::gzip::gzip_parser,
            description: signatures::gzip::DESCRIPTION.to_string(),
            extractor: Some(extractors::gzip::gzip_extractor()),
        },
```

PNG Confusion /



pple's PNG (CGBI)

Unknown NIB, XIB, plist …

Single byte array pattern
no multiple choice

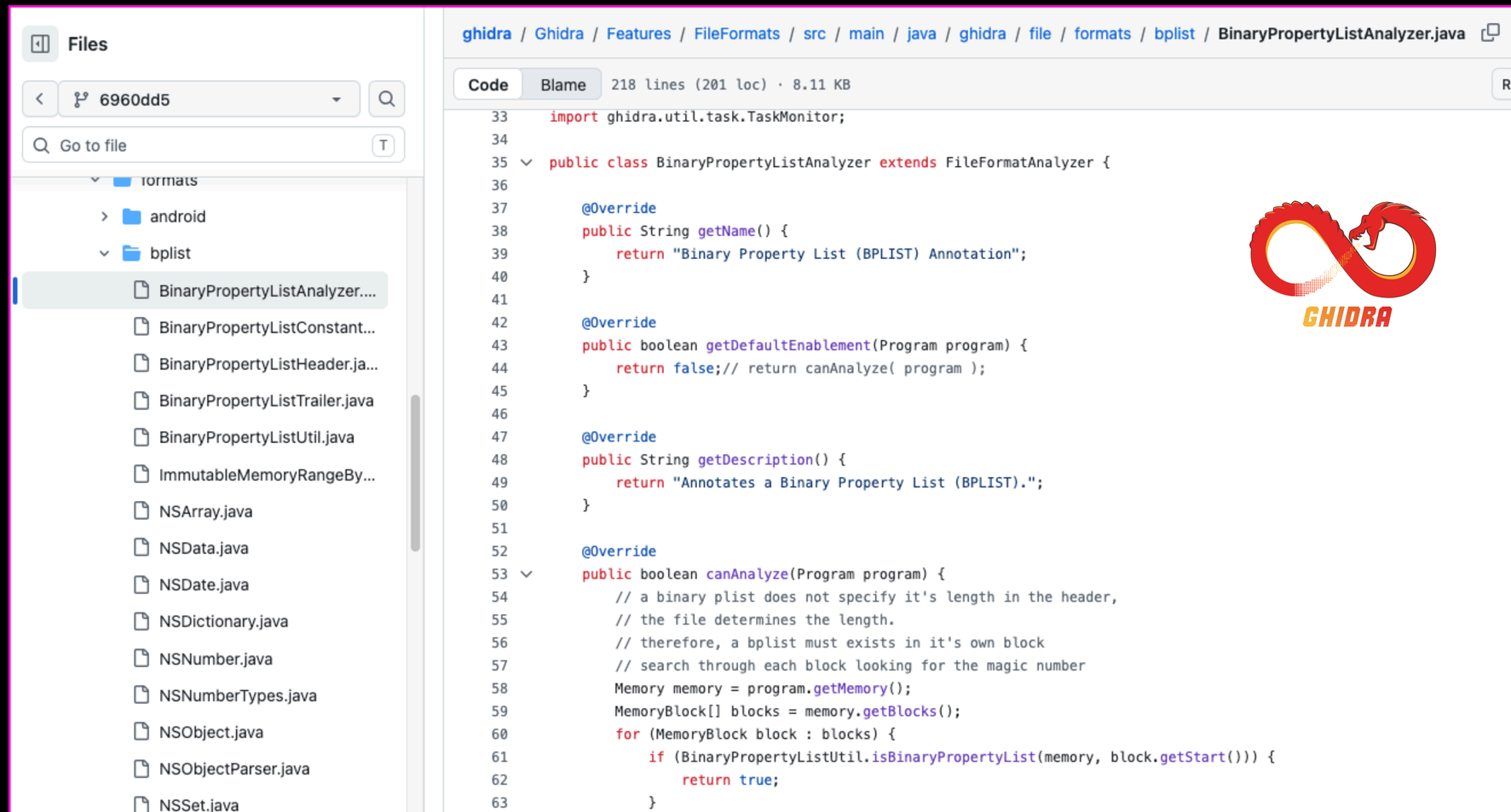
DATA PARSING

Chinese XML files not always parsed by Sax

Efficient tools have own

Internal parsers : Ghidra,

Radare2, IDA, …



DECOMPILER

Breaking stack analysis




```
001b6068 e0 1b 80 3d     str        q0,[sp, #local_180[0]]
001b606c ef bb 06 a9     stp        x15,x14,[sp, #local_180[8]]
001b6070 eb 5f 00 b9     str        w11,[sp, #local_184]
001b6074 ef 53 00 f9     str        x15,[sp, #local_140[0]]
001b6078 f1 43 00 f9     str        x17,[sp, #local_160[0]]
001b607c f1 33 00 f9     str        x17,[sp, #local_180[0]]
001b6080 05 00 00 14     b          LAB_001b6094

                     virt_next_step
001b6084 f0 03 1e aa     mov        x16,x30
001b6088 0e 02 40 f9     ldr        x14,[x16]
001b608c 0f 06 40 f9     ldr        x15,[x16, #0x8]
001b6090 06 00 00 14     b          LAB_001b60a8

                     LAB_001b6094
001b6094 fc ff ff 97     bl         virt_next_step
001b6098 a9              ??         A9h
001b6099 cc              ??         CCh
001b609a eb              ??         EBh
001b609b f6              ??         F6h
001b609c 00 00 00 00     udf        0x0
001b60a0 17              ??         17h
001b60a1 a7              ??         A7h
001b60a2 4d              ??         4Dh    M
001b60a3 69              ??         69h    i
001b60a4 00 00 00 00     udf        0x0

                     LAB_001b60a8
001b60a8 ee 2b 00 f9     str        x14,[sp,
001b60ac ef 27 00 f9     str        x15,[sp,
001b60b0 ce 01 00 0b     add        w14,w14,param_
001b60b4 ef 01 01 0b     add        w15,w15,param_2
001b60b8 d1 09 8e 13     ror        w17,w14,#0x2
001b60bc ce 79 8e 13     ror        w14,w14,#0x1e
001b60c0 e0 21 8f 13     ror        param_1,w15,#0x8
001b60c4 ef 61 8f 13     ror        w15,w15,#0x18
001b60c8 31 e6 01 12     and        w17,w17,#0x99999999
001b60cc ce e5 03 12     and        w14,w14,#0x66666666
001b60d0 00 9c 03 12     and        param_1,param_1,#0xe01fe01f
```

Data inserted inside code

Hallucinations




```
:goto_1
invoke-static/range {v0 .. v0}, Lcom/eshard/crackme/activities/CrackMeChallenge;->HVRQbvhaFlYdeGD(Ljava/lang/String;)Ljava/lang/S

const/4 v0, 0x1

move-result-object v0

goto/32 :goto_3

:goto_2
const-string v0, "3780b8133459f5a028d742efbcfc7d2d"

goto/32 :goto_1

:goto_3
invoke-static {v0}, Lcom/eshard/crackme/activities/CrackMeChallenge;->cVEqFjJdYlzfiuy(Ljava/lang/String;)V

goto/32 :goto_4
```

Always throws AndroidVerifier exception at runtime

JADX

```
static {
    HVRQbvhaFlYdeGD("3780b8133459f5a028d742efbcfc7d2d");
    cVEqFjJdYlzfiuy(1);
}
```

WRONG

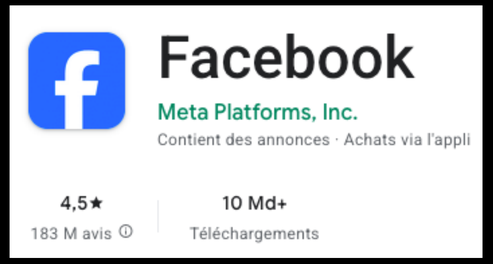
https://www.sstic.org/media/SSTIC2020/SSTIC-actes/dexcalibur_hook_it_yourself/SSTIC2020-Slides-dexcalibur_hook_

HOOKING

- Syscall tracing & instruction level hooking have caveats
- No follow thread, No builtin code coverage export
- Bad hook scheduling -> overhead
- Portability limited



PROBLEM ORIGIN



« You've got two days to build a script that generates valid FB-SESSION-IDs. »

PROBLEM ORIGIN



270 000
Files

187k *Classes* **612k** *Fields* **656k** *Methods*

Java

147 *Libs* **531k** *Functions* **?** *Classes*

Native

?
Hermes Function

React

5 700 000
Cross References (callsite)

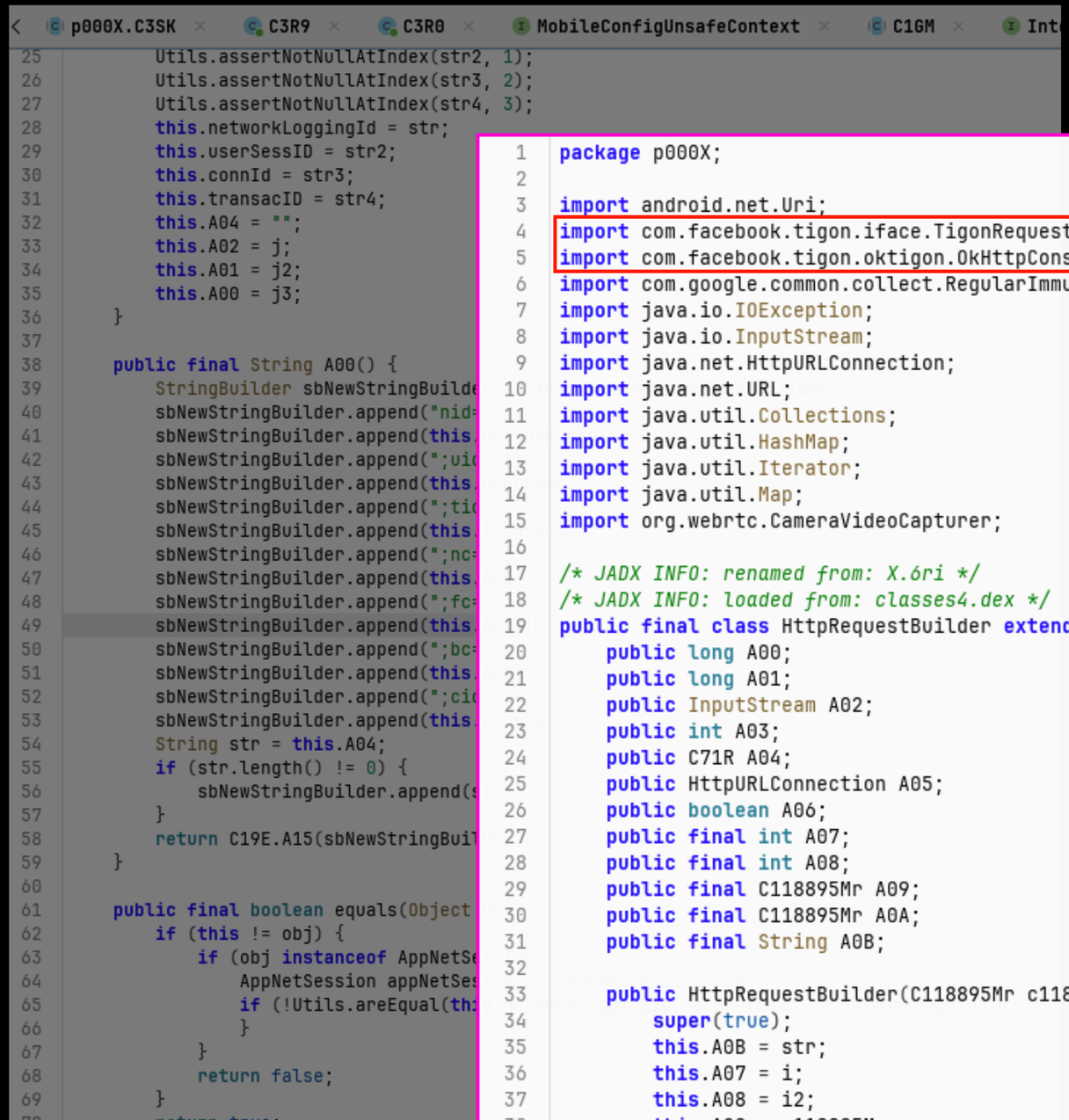
PROBLEM ORIGIN



p000X.C3SK | C3R9 | C3R0 | MobileConfigUnsafeContext | C1GM

```
25          Utils.assertNotNullAtIndex(str2, 1);
26          Utils.assertNotNullAtIndex(str3, 2);
27          Utils.assertNotNullAtIndex(str4, 3);
28          this.networkLoggingId = str;
29          this.userSessID = str2;
30          this.connId = str3;
31          this.transacID = str4;
32          this.A04 = "";
33          this.A02 = j;
34          this.A01 = j2;
35          this.A00 = j3;
36      }
37
38      public final String A00() {
39          StringBuilder sbNewStringBuilder = StringHelper.newStringBuilder();
40          sbNewStringBuilder.append("nid=");
41          sbNewStringBuilder.append(this.networkLoggingId);
42          sbNewStringBuilder.append(";uid=");
43          sbNewStringBuilder.append(this.userSessID);
44          sbNewStringBuilder.append(";tid=");
45          sbNewStringBuilder.append(this.transacID);
46          sbNewStringBuilder.append(";nc=");
47          sbNewStringBuilder.append(this.A02);
48          sbNewStringBuilder.append(";fc=");
49          sbNewStringBuilder.append(this.A01);
50          sbNewStringBuilder.append(";bc=");
51          sbNewStringBuilder.append(this.A00);
52          sbNewStringBuilder.append(";cid=");
53          sbNewStringBuilder.append(this.connId);
54          String str = this.A04;
55          if (str.length() != 0) {
56              sbNewStringBuilder.append(str);
57          }
58          return C19E.A15(sbNewStringBuilder);
59      }
60
61      public final boolean equals(Object obj) {
62          if (this != obj) {
63              if (obj instanceof AppNetSession) {
64                  AppNetSession appNetSession = (AppNetSession) obj;
65                  if (!Utils.areEqual(this.networkLoggingId, appNetSession.networkLoggingId) ||
66                  }
67              }
68              return false;
69          }
```

PROBLEM ORIGIN





```java
package p000X;

import android.net.Uri;
import com.facebook.tigon.iface.TigonRequest;
import com.facebook.tigon.oktigon.OkHttpConstants;
import com.google.common.collect.RegularImmutableMap;
import java.io.IOException;
import java.io.InputStream;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.Collections;
import java.util.HashMap;
import java.util.Iterator;
import java.util.Map;
import org.webrtc.CameraVideoCapturer;

/* JADX INFO: renamed from: X.6ri */
/* JADX INFO: loaded from: classes4.dex */
public final class HttpRequestBuilder extends C5UN implements C6W3 {
    public long A00;
    public long A01;
    public InputStream A02;
    public int A03;
    public C71R A04;
    public HttpURLConnection A05;
    public boolean A06;
    public final int A07;
    public final int A08;
    public final C118895Mr A09;
    public final C118895Mr A0A;
    public final String A0B;

    public HttpRequestBuilder(C118895Mr c118895Mr, String str, int i, int i2) {
        super(true);
        this.A0B = str;
        this.A07 = i;
        this.A08 = i2;
```

PROBLEM ORIGIN

```
p000X.C3SK   C3R9   C3R0   MobileConfigUnsafeContext   C1GM   Int
25          Utils.assertNotNullAtIndex(str2, 1);
26          Utils.assertNotNullAtIndex(str3, 2);
27          Utils.assertNotNullAtIndex(str4, 3);
28          this.networkLoggingId = str;
29          this.userSessID = str2;
30          this.connId = str3;
31          this.transacID = str4;
32          this.A04 = "";
33          this.A02 = j;
34          this.A01 = j2;
35          this.A00 = j3;
36      }
37
38      public final String A00() {
39          StringBuilder sbNewStringBuilde
40          sbNewStringBuilder.append("nid=
41          sbNewStringBuilder.append(this.
42          sbNewStringBuilder.append(";uic
43          sbNewStringBuilder.append(this.
44          sbNewStringBuilder.append(";tic
45          sbNewStringBuilder.append(this.
46          sbNewStringBuilder.append(";nc=
47          sbNewStringBuilder.append(this.
48          sbNewStringBuilder.append(";fc=
49          sbNewStringBuilder.append(this.
50          sbNewStringBuilder.append(";bc=
51          sbNewStringBuilder.append(this.
52          sbNewStringBuilder.append(";cic
53          sbNewStringBuilder.append(this.
54          String str = this.A04;
55          if (str.length() != 0) {
56              sbNewStringBuilder.append(s
57          }
```

```
1   package p000X;
2
3   import android.net.Uri;
4   import com.facebook.tigon.iface.TigonRequest;
5   import com.facebook.tigon.oktigon.OkHttpConstants;
6   import com.google.common.collect.RegularImmutableMap;
7   import java.io.IOException;
8   import java.io.InputStream;
9   import java.net.HttpURLConnection;
10  import java.net.URL;
11  import java.util.Collections;
12  import java.util.HashMap;
13  import java.util.Iterator;
14  import java.util.Map;
15  import org.webrtc.CameraVideoCapturer;
16
17  /* JADX INFO: renamed from: X.6ri */
18  /* JADX INFO: loaded from: classes4.dex */
19  public final class HttpRequestBuilder extends C5UN implements C6W3 {
20      public long A00;
21      public long A01;
22      public InputStream A02;
23      public int A03;
24      public C71R A04;
25      public HttpURLConnection A05;
26      public boolean A06;
```

```
→  700d148d-afce-4b1a-b8e6-4f9f6dba86a1 ls ~/0d/fb/lib/arm64-v8a/
libachilles-jni.so                  libbreakpad_cpp_helper.so      libfbunwindstack.so      libsuperpack-jni.so
libandroidx.graphics.path.so        libc++_shared.so               libgraphics-core.so      libsuperpack_common.so
libappcomponentfactory-jni.so       libdextricks-early.so          libsqlitejni.so          libunwindstack_binary.so
libbreakpad.so                      libdistract-config.so          libsuperattic.so         libxplat_superpack_portability_c_portabilityAndroid.so
```

16 Libs

```
62          if (this != obj) {
63              if (obj instanceof AppNetSe
64                  AppNetSession appNetSes
65                  if (!Utils.areEqual(thi
66                  }
67              }
68              return false;
69          }
```

```
30      public final C118895Mr A0A;
31      public final String A0B;
32
33      public HttpRequestBuilder(C118895Mr c118895Mr, String str, int i, int i2) {
34          super(true);
35          this.A0B = str;
36          this.A07 = i;
37          this.A08 = i2;
```

PROBLEM ORIGIN

```
p000X.C3SK  C3R9  C3R0  MobileConfigUnsafeContext  C1GM  Int
25        Utils.assertNotNullAtIndex(str2, 1);
26        Utils.assertNotNullAtIndex(str3, 2);
27        Utils.assertNotNullAtIndex(str4, 3);
28        this.networkLoggingId = str;
29        this.userSessID = str2;
30        this.connId = str3;
31        this.transacID = str4;
32        this.A04 = "";
33        this.A02 = j;
34        this.A01 =
35        this.A00 =
36    }
37
38    public final S
39        StringBuil
40        sbNewStrin
41        sbNewStrin
42        sbNewStrin
43        sbNewStrin
44        sbNewStrin
45        sbNewStrin
46        sbNewStrin
47        sbNewStrin
48        sbNewStrin
49        sbNewStrin
50        sbNewStrin
51        sbNewStrin
52        sbNewStrin
53        sbNewStrin
54        String str
55        if (str.le
56            sbNewS
57        }
58        return C19
59    }
60
61    public final b
62        if (this !
63            if (ob
64                Ap
65                if
66                }
67            }
68            return false;
69        }
```

```
libachilles-jni.so
libandroidx.graphics.path.so
libappcomponentfactory-jni.so
libbreakpad.so
```

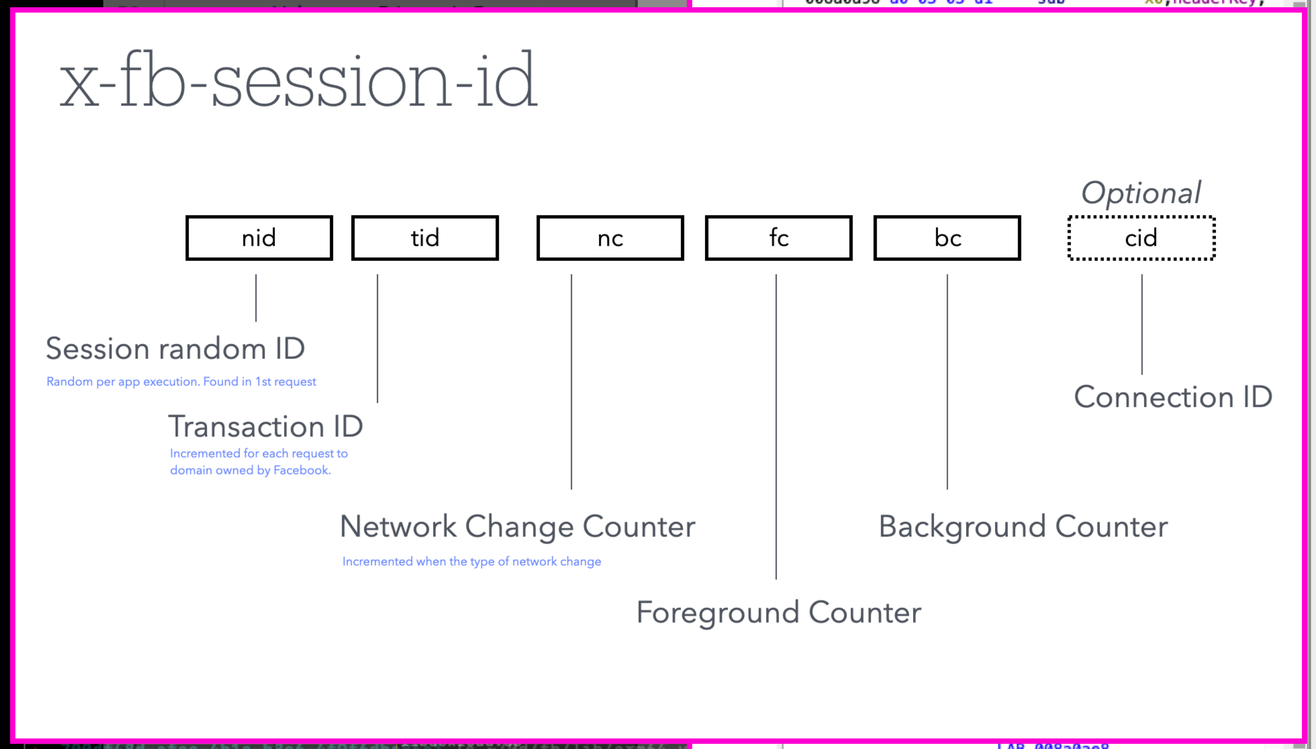
```
1  package p000X;
2
3  import android.net.Uri;
4  import com.facebook.tigon.iface.TigonRequest;
5  import com.facebook.tigon.oktigon.OkHttpConstants;
```

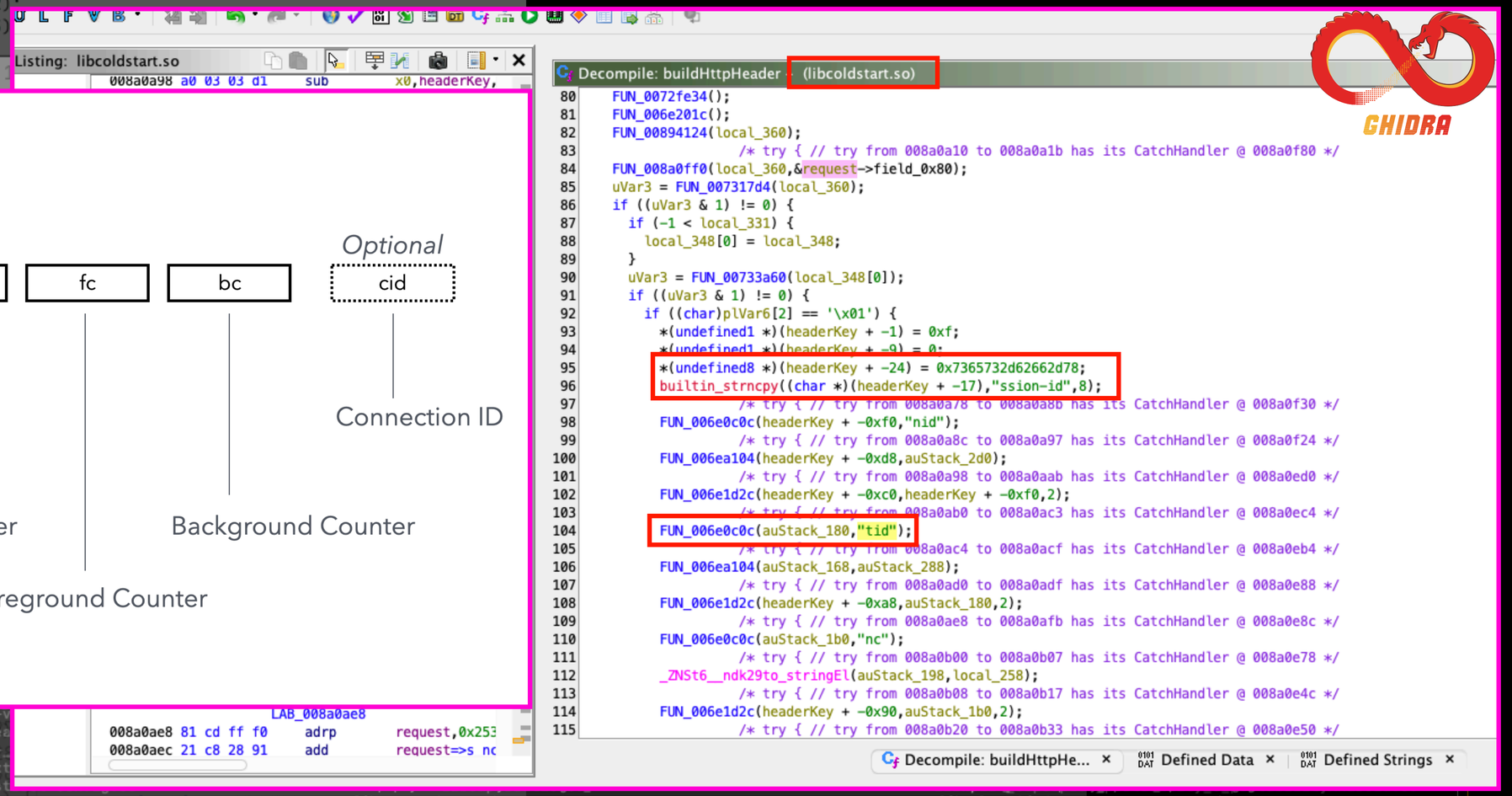
```
➜  700df48d-afce-4b1a-b8e6-4f9f6dba86a1 ls ~/0d/fb_data/com.facebook.katana/lib-compressed
dso_deps
dso_lock
dso_state
libacodechooks_jni.so
libandroid_c2pa.so
libanimatedcxxreactpackage.so
libappdrivenaudiomerged.so
libardelivery-merged.so
libarengine-merged.so
libarpersistenceserviceconfig.so
libarserviceinterfacemerged.so
libarservicesoptionalcaffe2merged.so
libarservicesoptionalmerged.so
libarstandalonetracking-native-android.so
libautogen_frameprocessor.so
libauxiliary_database_configMCFBridgejni.so
libavatarembodiment.so
libavatarsdkbridgeholder.so
libbacktrace-jni.so
libbreakpad_extra.so
libbsdiff.so
libc2pa_jni.so
libcache_key_generator_jni.so
libcalmerged.so
libcamera-xplat-spars-jni.so
libcgmerged.so
libcoldstart.so
libcomet-native-jni.so
libcommonpatchjni.so
libcompphoto-sdk-pipeline-graphs-common-autoenhanceinfo-native-android.so
libctmerged.so
libctmerged_1.so
libdexload.so
libdownloadservice-jni.so
libdynamic_executorch.so
libearlystartupperf-jni.so
libeffectservicehostmerged.so
libessim.so
libetsegmentationmodelshim.so
libfacebookapplicationreliabilityexperiments.so
libfalco-e2e-logger-jni.so
libfb_jpegturbo.so
libfb_libyuv_jni.so
libfb_mboost-lite.so
libfbandroid_native_msys_pandoblobjni_util_util.so
libfbmaps.so
libfilestathelper.so
libfosdogfoodinterceptor.so
libgc_interceptor.so
libgcpriority.so
libglobalrefmonitor.so
libgltfrenderer-native-android.so
libgraphservice-jni-loggedoutservice.so
libhealthcounters.so
libhtmltrimmer_jni.so
libimage_quality_utils.so
libjava2js-slow.so
libkaleidoscope.so
liboopermerged.so
libmagimerged.so
libmantlemerged.so
libmapsreader.so
libmediamessagingsdk.so
libmediapipeline-iglufilter-detection.so
libmediapipeline-iglufilter-facebook.so
libmediapipeline-iglufilter-impl-basic.so
libmediapipeline-iglufilter-messenger.so
libmediapipeline-iglufilter-textureloader.so
libmediapipelinemerged.so
libmessengerfileencryptionorcajni.so
libmonitorcontention.so
libnative-bitmaps.so
libnative_threadprioritymonitor.so
liboceanqrcodedetectorjni.so
liboktigon.so
libonecamera-autoenhance-native-android.so
libonecamera-iglufilter-graph.so
libopusJNIExo2.so
libpae.so
libpandoblobserializerjni.so
libpapayatorchanalyticsmerged.so
libpapayatorchanalyticsmerged_1.so
libpregensdpmerged.so
libprofilo_gcstats.so
libprofilo_memory.so
libprofilo_memory_mapping_actions.so
libprofilo_native_memory.so
libprofilo_network.so
libprofilo_thread_lifecycle.so
libproxygen-eventbase_jni.so
libpublickeycryptomerged.so
libqplfilelogger.so
librenderthread.so
libroomsrsysjniLatest.so
librssignalingtransportplatform_dgw_jniLatest.so
librsysmerged.so
librsysmergeddevxagent.so
librsysmergedthirdparty.so
librtcdgwserviceholderjni.so
libscrollmerged.so
libsessionmerged.so
libsigcoremerged.so
libslamfactoryprovider.so
libsmartcapture_id.so
libsodiumhpke.so
libsonic-native.so
libspark-qpluserflow-native.so
libsphericalmerged.so
libstatic-webp.so
libsuperattic_extra.so
libsurfacenativemem.so
libthird-party_c2pa_c2paAndroid.so
libthird-party_cborg_cborgAndroid.so
libthreadutils-jni.so
libvesta_client_jni.so
libvmasaver.so
libvmmerged.so
libvoprf-ed25519.so
libxplat_c2pautils_c2pautilsAndroid.so
libxplat_compphoto_sdk_common_perfStatsComputerAndroid.so
libxplat_compphoto_sdk_common_textureUtilsAndroid.so
libxplat_media_media_messaging_sdk_mms_cipher_MMSCipherAndroid.so
libxplat_rssignalingtransportplatform_dgw_rp_sdkAndroid.so
libxplat_structuredlogger_events_gen_FflStreamEvent_genAndroid.so
libxplat_third-party_opencv_opencv-core.so
libxplat_third-party_opencv_opencv-highgui.so
libxplat_third-party_opencv_opencv-imgcodecs.so
libxplat_third-party_opencv_opencv-imgproc.so
libxplat_third-party_opencv_opencv-videoio.so
libzstddecoder.so
libzstdencoder.so
```

+ 131 Libs !!

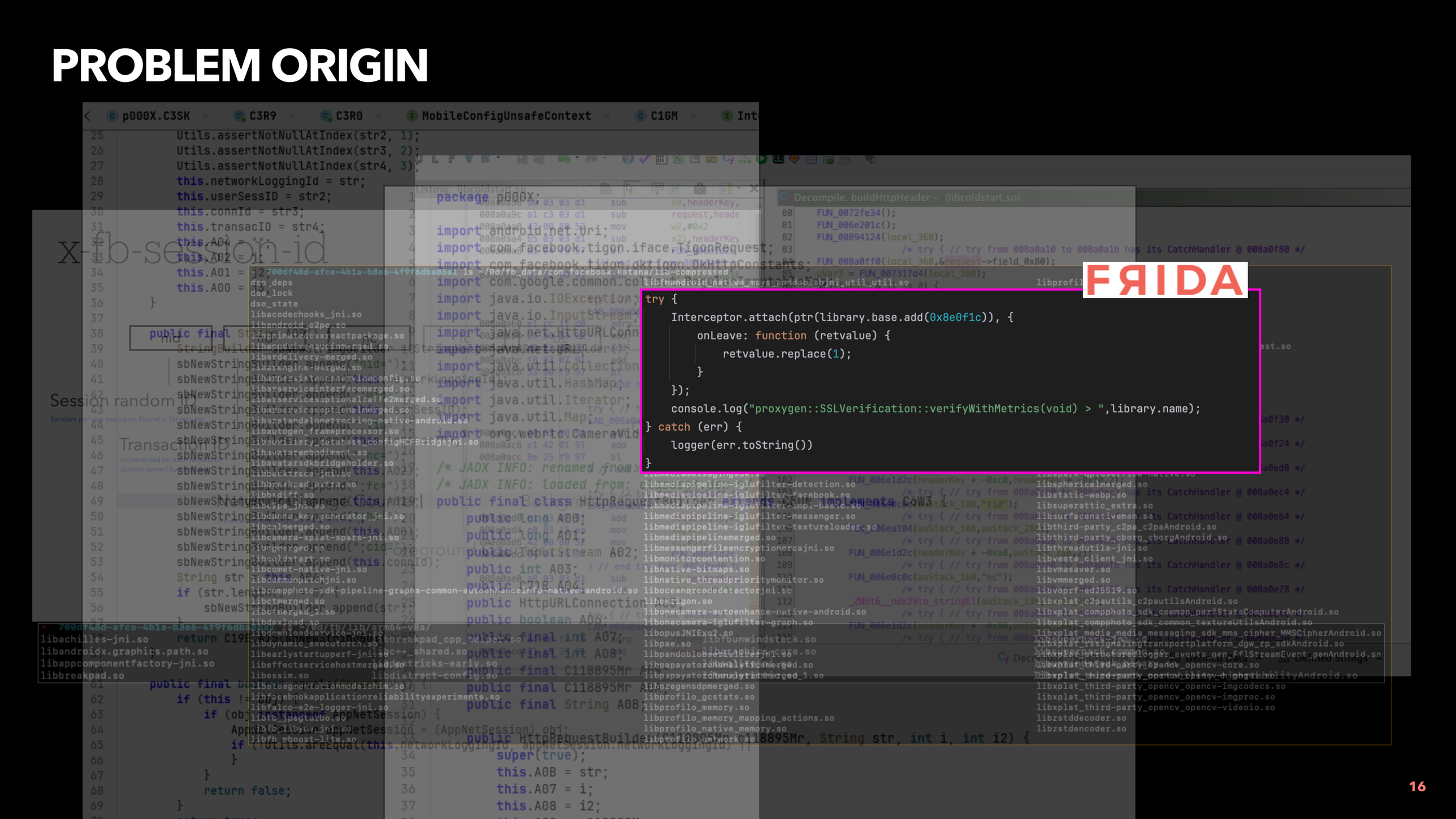
```
34        super(true);
35        this.A0B = str;
36        this.A07 = i;
37        this.A08 = i2;
```

PROBLEM ORIGIN





PROBLEM ORIGIN



PROBLEM ORIGIN

mitmproxy

```
15:22:29 HTTPS POST          graph.facebook.com /graphql
15:22:29 HTTPS POST          graph.facebook.com /graphql
15:22:29 HTTPS POST          graph.facebook.com /graphql
15:22:29 HTTPS POST          graph.facebook.com /rmd?access_token=350685531728%7C62f8ce9f74b12f84c123cc23437a4a32&rule_context=facebook_prod_v2&net_iface=Wifi&reason=NETWORK_SWITCH
15:22:29 HTTPS POST        gateway.facebook.com /streamcontroller
15:22:29 HTTPS POST          graph.facebook.com /graphql
15:22:29 HTTPS POST          graph.facebook.com /graphql
```

Flow Details

2026-04-08 17:49:19 POST https://graph.facebook.com/graphql HTTP/2.0

← 200 text/html [no content] 86ms

Request	Response	Detail

x-fb-request-analytics-tags:	{"network_tags":{"product":"350685531728","request_category":"graphql","purpose":"none","retry_attempt":"0"},"application_tags":"graphservice"}
x-fb-rmd:	state=URL_ELIGIBLE
x-fb-network-properties:	Wifi;
content-encoding:	gzip
user-agent:	[FBAN/FB4A;FBAV/554.0.0.57.70;FBBV/918990857;FBDM/{density=2.625,width=1080,height=2424};FBLC/en_US;FBRV/0;FBCR/T-Mobile;FBMF/Google;FBBD/google;FBPN/com.1 phone64_arm64;FBSV/16;FBOP/1;FBCA/arm64-v8a:;]
x-fb-friendly-name:	NotificationsPageApiBadgeCountQuery
x-zero-f-device-id:	ff41efd3-9967-4d18-9757-b4ea8841e859
x-fb-device-group:	4399
x-fb-integrity-machine-id:	NW_NaduImCtPbxB7pVDA4_87
x-zero-state:	unknown
x-graphql-client-library:	graphservice
x-fb-appnetsession-sid:	89a227c534128a8e3a4c0ef04f907c7e
x-fb-sim-hni:	310260
content-type:	application/x-www-form-urlencoded
x-fb-tasos-experimental:	1
x-fb-net-hni:	310260
x-fb-session-id:	nid=U84UMcy6+XDz;tid=3;nc=0;fc=0;bc=0
x-meta-zca:	empty_token
app-scope-id-header:	4fa6f117-fafd-4378-af9e-bcd72d8b5930
x-fb-connection-type:	WIFI
x-fb-appnetsession-nid:	70a8d53ad075b871580a17c5708e4247,Unknown
authorization:	OAuth EAAAAUaZA8jlABRNrHjZC6Gf9mx9ZAlQYwUZBXou4U09pbeGwJ0IKmve2QZBSJvnlX53CQbNGCQj3uK6K3XqDH6OZAZANmtl7UKZCYHxr3AnpykSxorYp7MZA9IXtlH1mfn1MPWu7KKec54BtXSGr Aqc4i8n3vx18OlL5vluUZB1yWx3PCUNpF7yYRh9eHZAmgS7XTgvYwZDZD
priority:	u=3, i
x-fb-qpl-active-flows-json:	{"schema_version":"v3","inprogress_qpls":[{"marker_id":25952257,"annotations":{"current_endpoint":"IXTEnrollmentActivity"}}],"snapshot_attributes":{}}
x-meta-enable-tasos-ss-bwe:	1
x-tigon-is-retry:	False
accept-encoding:	zstd, gzip, deflate
x-fb-http-engine:	Tigon/Liger
x-fb-client-ip:	True
x-fb-server-cluster:	True
x-fb-conn-uuid-client:	sAE/0gc2clo8eC9TDackEA==

mitmproxy

PROBLEM ORIGIN

Trace HTTP header building

```
try {
    const px = ptr(library.base.add(0x5e1d2c));
    Interceptor.attach(px, {
        onEnter: function (a) {
            const ss = [];
            for(let i=0; i<a[2]; i++){
                ss.push(readStdStr(a[1].add(i*0x18)));
            }
            console.log(`[${Process.getCurrentThreadId()}] Tigon.processHeaderKey (${a[2]}) : ${ss.join(',')} `);
        }
    });
    console.log(`[*][+] Hooked function processHeaderKey`);
} catch (err) {
    console.log(`[*][-] Failed to hook function processHeaderKey`);
}
```

FRIDA

Print internal FB logs

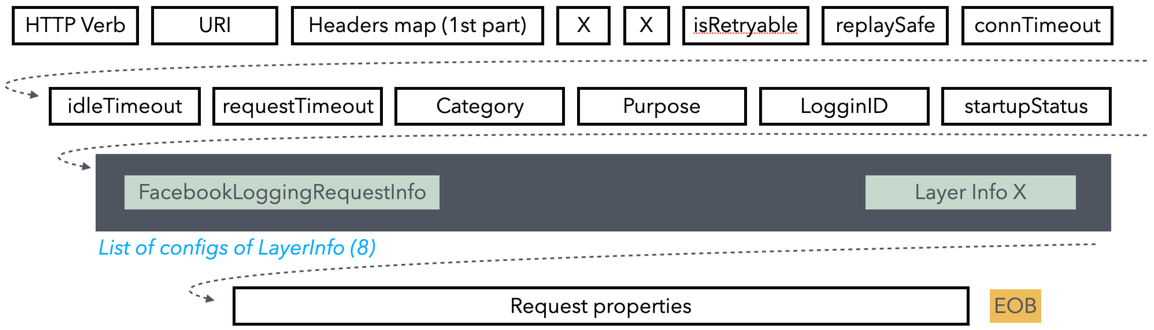
```
const fbsys = library.findExportByName("fbsystrace_begin_section");
if(fbsys){
    console.log("fbsystrace_begin_section IN ",library.name  );
    Interceptor.attach(fbsys, {
        onEnter: function (a) {
            console.log(`[${Process.getCurrentThreadId()}] fbsystrace: `, a[1].readCString());
        }
    });
}
```

Frida Output

```
fbsystrace:  JExecutor - execute
fbsystrace:  JExecutor - execute
fbsystrace:  GraphQLLayeredInformation - getLayerInformation
x-fb-sessions-id >                0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F  0123456789ABCDEF
7d01bddb50  30 31 31 4d 54 70 6b 6a 2b 51 75 63 00 00 00 00  011MTpkj+Quc....
7d01bddb60  48 74 eb 44 7b 00 00 0c 00 f2 55 4a 7b 00 00 00  Ht.D{.....UJ{...
7d01bddb70  b4 ef c5 fd 7b 00 00 00 e8 f4 55 4a 7b 00 00 00  ....{.....UJ{...
fbsystrace:  HTTPTransactionAdaptor::sendRequestWithBodyAndEOM
fbsystrace:  HTTPTransactionAdaptor::sendRequest
fbsystrace:  HTTPTransactionAdaptor::messageFromRequest
fbsystrace:  HTTPTransactionAdaptor::setTransaction
fbsystrace:  HTTPTransactionAdaptor::sendBody
fbsystrace:  HTTPTransactionAdaptor::sendEOM
fbsystrace:  JExecutor - execute
fbsystrace:  JExecutor - execute
fbsystrace:  JExecutor - execute
```

PROBLEM ORIGIN

FB build HTTP request continuously through an output stream (kind of egress)

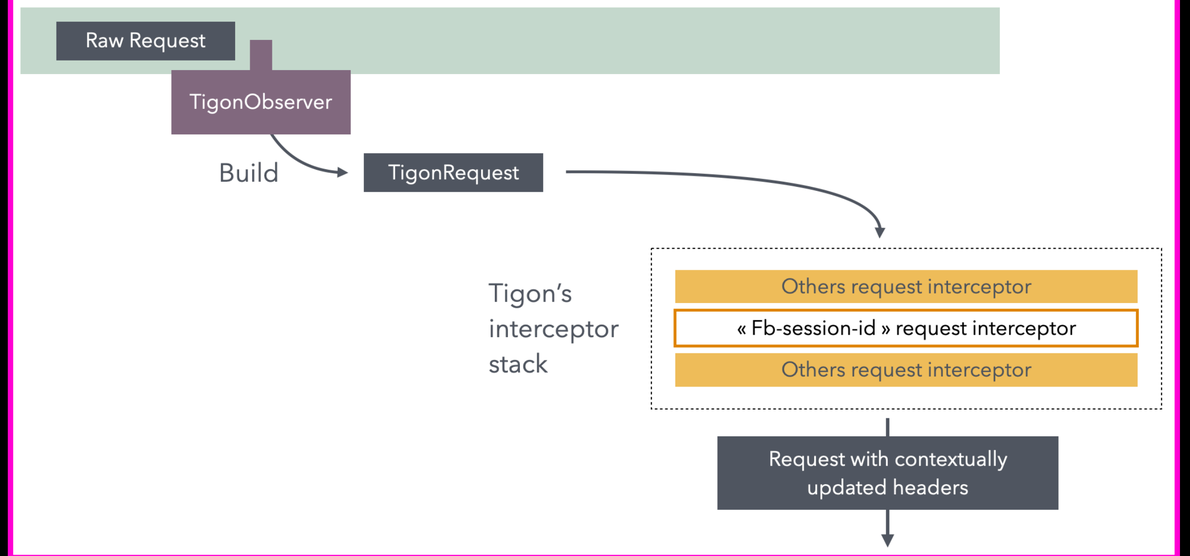


TigonObserver, as some others listeners, observe this stream in Java *X.3SJ* class.
This stream use **Length-Value** encoding for string

```
Tigon.addHeader >  ▯▯A| {"schema_version":"v3","inprogress_qpls":[],"snapshot_attributes":{}}
Tigon.addHeader >  X-FB-HTTP-Engine Tigon/Liger
x-fb-sessions-id >  0x7aa25b71e0 0x78
Tigon.addHeader >  X-FB-AppId 350685531728
Tigon.addHeader >  Host gateway.facebook.com
Tigon.addHeader >  User-Agent [FBAN/FB4A;FBAV/554.0.0.57.70;FBBV/918990857;FBDM/{density=2.625,
Tigon.addHeader >  x-fb-net-hni 310260
Tigon.addHeader >  x-fb-sim-hni 310260
Tigon.addHeader >  app-scope-id-header b1da0934-6076-4539-a74e-84d93643a963
Tigon.addHeader >  X-ZERO-F-DEVICE-ID ff41efd3-9967-4d18-9757-b4ea8841e859
Tigon.addHeader >  X-Meta-ZCA empty_token
Tigon.addHeader >  x-fb-session-id nid=/VoQUkq7lf3d;tid=5;nc=0;fc=0;bc=0
Tigon.addHeader >  X-FB-Connection-Type WIFI
Tigon.addHeader >  ▯▯A| NW_NaduImCtPbxB7pVDA4_87
Tigon.addHeader >  x-fb-device-group 4399
```

FRIDA

Tigon Interceptors



PROBLEM ORIGIN

```
dl_open = /data/data/com.facebook.katana/lib-compressed/libcoldstart.so
loading libcoldstart.so - base @ 0x6dfd128000
onEnter [ 1648 ]
Deploying hooks of  libcoldstart.so [s=0x6dfd128000, len=21803008]
[*][+] Hooked function add fb session id
[*][+] Hooked function Tigon.addHeader
[*][+] Hooked function processHeaderKey
[*][+] Hooked function Tigon.buildFbId
[1649] Tigon.processHeaderKey (0x19) : fbpay-android,pushsdk-android-subconfig,ardelivery-android,fdid-sync-android,urisigner-android,
ndroid,encrypted-backups-android,genai-imagine-android,genai-wwai-android,genai-intents-android,bloks-android,iap-android,sdb-android,
ndroid,xe-android,xe-growth-android,subs-android
[1745] Tigon.processHeaderKey (0xd) : GET,POST,OPTIONS,DELETE,HEAD,CONNECT,CONNECT-UDP,PUT,TRACE,PATCH,SUB,PUB,UNSUB
appNetSessionInterceptor init :  pc=0x6dfd80ec78 sp=0x6ea5e06830 lr=0x6dfd80ec6c pc=0x6ea5e06860

x0=0x6ea5e06c80 x1=0x6ea5e06848 x2=0x6ea5e06898 x3=0x10
x4=0x0  x5=0xb40000709fa02cca   x6=0x10 x7=0x7f7f7f7f7f7f7f7f
x8=0x6ea5e06860 x9=0x101        x10=0x45c1d71   x11=0x20000709fa02620
x12=0x20002     x13=0x38        x14=0x0 x15=0xebad6a89
x16=0x720b2b6000        x17=0x7200650ee0        x18=0x6ea3adc000        x19=0xb4000070ff91afd0
x20=0x6ea5e06c80        x21=0x100       x22=0x6ea5e08880        x23=0x1
x24=0x6ea5e073c0        x25=0x6ea5e073b4        x26=0xb40000704f92afd0  x27=0xb40000707f93f7a0
x28=0xb40000709fa02a10  x29=undefined   x30=undefined
Tigon.addHeader >  x-graphql-client-library graphservice
Tigon.addHeader >  X-FB-Friendly-Name FbStoriesLightBucketsQuery
Tigon.addHeader >  Host graph.facebook.com
Tigon.addHeader >  User-Agent [FBAN/FB4A;FBAV/554.0.0.57.70;FBBV/918990857;FBDM/{density=2.625,width=1080,height=2424};FBLC/en_US;FBRV
FBOP/1;FBCA/arm64-v8a:;]
Tigon.addHeader >  x-fb-net-hni 310260
Tigon.addHeader >  x-fb-sim-hni 310260
Tigon.addHeader >  Authorization OAuth EAAAAUaZA8jlABRNrHjZC6Gf9mx9ZAlQYwUZBXou4U09pbeGwJ0IKmve2QZBSJvnlX53CQbNGCQj3uK6K3XqDH6OZAZANmf
UZB1yWx3PCUNpF7yYRh9eHZAmgS7XTgvYwZDZD
Tigon.addHeader >  X-ZERO-STATE unknown
Tigon.addHeader >  app-scope-id-header b39cfef2-526c-49aa-b064-92840cefae94
Tigon.addHeader >  X-ZERO-F-DEVICE-ID 5b1c42d6-9891-4cae-8be5-e6c388422a1a
Tigon.addHeader >  X-Meta-ZCA empty_token
x-fb-sessions-id >
Before NID appended (assign nidValue from sp) >             0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F  0123456789ABCDEF
6de47dd0c0  6e 2f 69 67 46 4c 43 2f 78 2b 79 74 00 00 00 00  n/igFLC/x+yt....
6de47dd0d0  48 94 07 8f 6e 00 00 0c 0d 00 00 00 00 00 00 00  H...n...........
[1800] Tigon.processHeaderKey (0x2) : nid,n/igFLC/x+yt
[1800] Tigon.processHeaderKey (0x2) : tid,1
[1800] Tigon.processHeaderKey (0x2) : nc,0
[1800] Tigon.processHeaderKey (0x2) : fc,0
[1800] Tigon.processHeaderKey (0x2) : bc,0
```

Hooks are deployed at very early stage of libcoldstart.so, prior to initializing and 1st request

FRIDA

```
0x5e0c0c: std_string_construct >  RequestHeaderValidationInterceptor
0x5e0c0c: std_string_construct >  RequestAnnotationInterceptor
0x5e0c0c: std_string_construct >  AppNetSessionIdInfoInterceptor
0x5edbdc: createRandomDevice (LR=  7eb3ec  )>
0x5edbdc: generate NID (LR=  7eb3ec  )>
0x5e0c0c: std_string_construct (LR=  6edbf8  )>  /dev/urandom
0x5edbdc: transformed NID (LR=  7eb43c  )>
            0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F  0123456789ABCDEF
735a9df000  79 0a 9f dc b5 ce b9 8c f6 1f 93 42 70 a1 ae 2e  y..........Bp...
0x5e0c0c: std_string_construct >  RequestIdInterceptor
0x5e0c0c: std_string_construct >  ForceThirdPartyHttpsInterceptor
0x5e0c0c: std_string_construct >  AuthHeaderAndDomainInterceptor
0x5e0c0c: std_string_construct >  VPSTraceEventLoggingInterceptor
```

PROBLEM ORIGIN

Burp Suite

Request

Pretty Raw Hex

```
GET /zero/balance/pixel/ HTTP/2
Host: mobile.facebook.com
X-Fb-Request-Analytics-Tags:
{"network_tags":{"product":"350685531728","purpose":"none","retry_attempt":"0"},"application_tags":"unknown"}
X-Fb-Friendly-Name: ZeroBalancePingRequestMethod
X-Fb-Net-Hni: 310260
X-Fb-Appnetsession-Nid: aca530797516ddf222455d2599fb45d0,Wifi
X-Meta-Tasos-Tlbwe-Config: quic_transport_bwe:config_33
X-Fb-Network-Properties: Wifi;
X-Fb-Session-Gk: v1:gk:fb_android_tnts_dcu_experiments:@pass;v1:gk:fb_android_tasos_congestion_signal:@pass;
User-Agent: FB-ZeroBalance
X-Fb-Sim-Hni: 310260
X-Meta-Zca: empty_token
App-Scope-Id-Header: b1da0934-6076-4539-a74e-84d93643a963
X-Fb-Connection-Type: WIFI
X-Fb-Session-Id: nid=g0WY9Vi9lh4+;tid=83;nc=0;fc=0;bc=0
X-Fb-Device-Group: 4399
X-Fb-Integrity-Machine-Id: NW_NaduImCtPbxB7pVDA4_87
X-Zero-F-Device-Id: ff41efd3-9967-4d18-9757-b4ea8841e859
X-Tigon-Is-Retry: False
X-Fb-Qpl-Active-Flows-Json: {"schema_version":"v3","inprogress_qpls":[],"snapshot_attributes":{}}
Priority: u=3, i
X-Meta-Usdid:
f7f45378-4ccf-45c9-bd13-ce6aa620d81e.1775574015.MEUCIDe8HhE5qngJmydcM2ZWUmV4peVuiUfvlaWteYH6My6vAiEA-Z9IulNIJMla0
92HFVr-uI1P4iyVqvHhRwHaRpUdDX0
X-Fb-Rmd: state=URL_ELIGIBLE
X-Meta-Enable-Tasos-Ss-Bwe: 1
X-Fb-Tasos-Experimental: 1
X-Fb-Appnetsession-Sid: 4852c6a939a2796e55d3b4af57f7060b
Accept-Encoding: gzip, deflate, br
X-Fb-Http-Engine: Tigon/Liger
X-Fb-Client-Ip: True
X-Fb-Server-Cluster: True
X-Fb-Conn-Uuid-Client: /7gCdF5tmkbBY9YSqOzkZg==
Connection: close
```

Response

Pretty Raw Hex Render

```
Report-To:
{"max_age":2592000,"endpoints":[{"url":"https:\/\/www.facebook.com\/browser_reporting\/coop\/?minimize=0"}],"grou
p":"coop_report","include_subdomains":true},
{"max_age":86400,"endpoints":[{"url":"https:\/\/www.facebook.com\/browser_reporting\/coep\/?minimize=0"}],"group"
:"coep_report"},
{"max_age":259200,"endpoints":[{"url":"https:\/\/mobile.facebook.com\/ajax\/mtouch_error_reports\/?device_level=u
nknown&brsid=7626017319985174651&cpp=C3&cv=1036810159&st=1775570521737"}]},
{"max_age":21600,"endpoints":[{"url":"https:\/\/www.facebook.com\/ajax\/browser_error_reports\/"}],"group":"permi
ssions_policy"}
X-Frame-Options: DENY
Content-Security-Policy: default-src blob: 'self' https://*.fbsbx.com *.facebook.com *.fbcdn.net *.whatsapp.com
*.whatsapp.net;script-src *.facebook.com *.fbcdn.net 127.0.0.1:* 'nonce-iKwdiujb' blob: 'self'
'unsafe-eval';style-src *.fbcdn.net data: *.facebook.com 'unsafe-inline';connect-src *.facebook.com facebook.com
*.fbcdn.net wss://*.facebook.com:* wss://*.whatsapp.com:* wss://*.fbcdn.net attachment.fbsbx.com ws://localhost:*
 blob: *.cdninstagram.com 'self' http://localhost:3103 wss://gateway.facebook.com wss://edge-chat.facebook.com
wss://snaptu-d.facebook.com wss://kaios-d.facebook.com/ v.whatsapp.net *.fbsbx.com *.fb.com *.instagram.com
*.messenger.com;font-src data: *.facebook.com *.fbcdn.net *.fbsbx.com;img-src *.fbcdn.net *.facebook.com data:
https://*.fbsbx.com facebook.com *.cdninstagram.com fbsbx.com fbcdn.net blob: android-webview-video-poster:
*.whatsapp.net *.fb.com *.oculuscdn.com https://trustly.one/ https://*.trustly.one/ https://paywithmybank.com/
https://*.paywithmybank.com/;media-src *.cdninstagram.com blob: *.fbcdn.net *.fbsbx.com www.facebook.com
*.facebook.com data:;child-src data: blob: 'self' https://*.fbsbx.com *.facebook.com *.fbcdn.net;frame-src
*.facebook.com *.fbsbx.com fbsbx.com data: www.instagram.com *.fbcdn.net accounts.meta.com *.accounts.meta.com
https://trustly.one/ https://*.trustly.one/ https://paywithmybank.com/ https://*.paywithmybank.com/;manifest-src
data: blob: 'self' https://*.fbsbx.com *.facebook.com *.fbcdn.net;object-src data: blob: 'self'
https://*.fbsbx.com *.facebook.com *.fbcdn.net;worker-src blob: *.facebook.com data:;report-uri
https://www.facebook.com/csp/reporting/?m=c&minimize=0;
Document-Policy: force-load-at-top
Document-Policy: include-js-call-stacks-in-crash-reports
Permissions-Policy: accelerometer=(self), attribution-reporting=(), autoplay=(), bluetooth=(),
browsing-topics=(self), camera=(self), ch-device-memory=(), ch-downlink=(), ch-dpr=(), ch-ect=(), ch-rtt=(),
ch-save-data=(), ch-ua-arch=(), ch-ua-bitness=(), ch-viewport-height=(), ch-viewport-width=(), ch-width=(),
clipboard-read=(), clipboard-write=(self), compute-pressure=(), display-capture=(), encrypted-media=(),
fullscreen=(self), gamepad=(), geolocation=(self), gyroscope=(self), hid=(), idle-detection=(),
interest-cohort=(self), keyboard-map=(), local-fonts=(), magnetometer=(), microphone=(self), midi=(),
otp-credentials=(self), payment=(), picture-in-picture=(), private-state-token-issuance=(),
publickey-credentials-get=(self), screen-wake-lock=(), serial=(), shared-storage=(),
shared-storage-select-url=(), private-state-token-redemption=(), usb=(), unload=(self), window-management=(),
xr-spatial-tracking=();report-to="permissions_policy"
Cross-Origin-Resource-Policy: same-site
Cross-Origin-Embedder-Policy-Report-Only: require-corp;report-to="coep_report"
Cross-Origin-Opener-Policy: same-origin-allow-popups
Pragma: no-cache
Cache-Control: private, no-cache, no-store, must-revalidate
Expires: Sat, 01 Jan 2000 00:00:00 GMT
X-Content-Type-Options: nosniff
X-Xss-Protection: 0
Origin-Agent-Cluster: ?1
Accept-Ch-Lifetime: 4838400
Accept-Ch:
viewport-width,dpr,Sec-CH-Prefers-Color-Scheme,Sec-CH-UA-Full-Version-List,Sec-CH-UA-Platform-Version,Sec-CH-UA-M
odel
Content-Type: image/gif
Strict-Transport-Security: max-age=15552000; preload
X-Fb-Debug: T2UilQmLvzPXiuFBq7TFvP8QGbWERctVTCvktvQCRmC/tGMY71A7b9kBAbu2a3wmSzH3SY9+mo7eEaG/LhI7Gw==
Date: Tue, 07 Apr 2026 14:02:01 GMT
X-Fb-Client-Ip-Forwarded: 90.55.46.24
X-Fb-Server-Cluster-Forwarded: cdg4c03
X-Fb-Connection-Quality: EXCELLENT; q=0.9, rtt=25, rtx=0, c=10, mss=1380, tbw=3579, tp=-1, tpl=-1, uplat=84,
ullat=0
Alt-Svc: h3=":443"; ma=86400
```

PROBLEM ORIGIN

Application

Binary & code

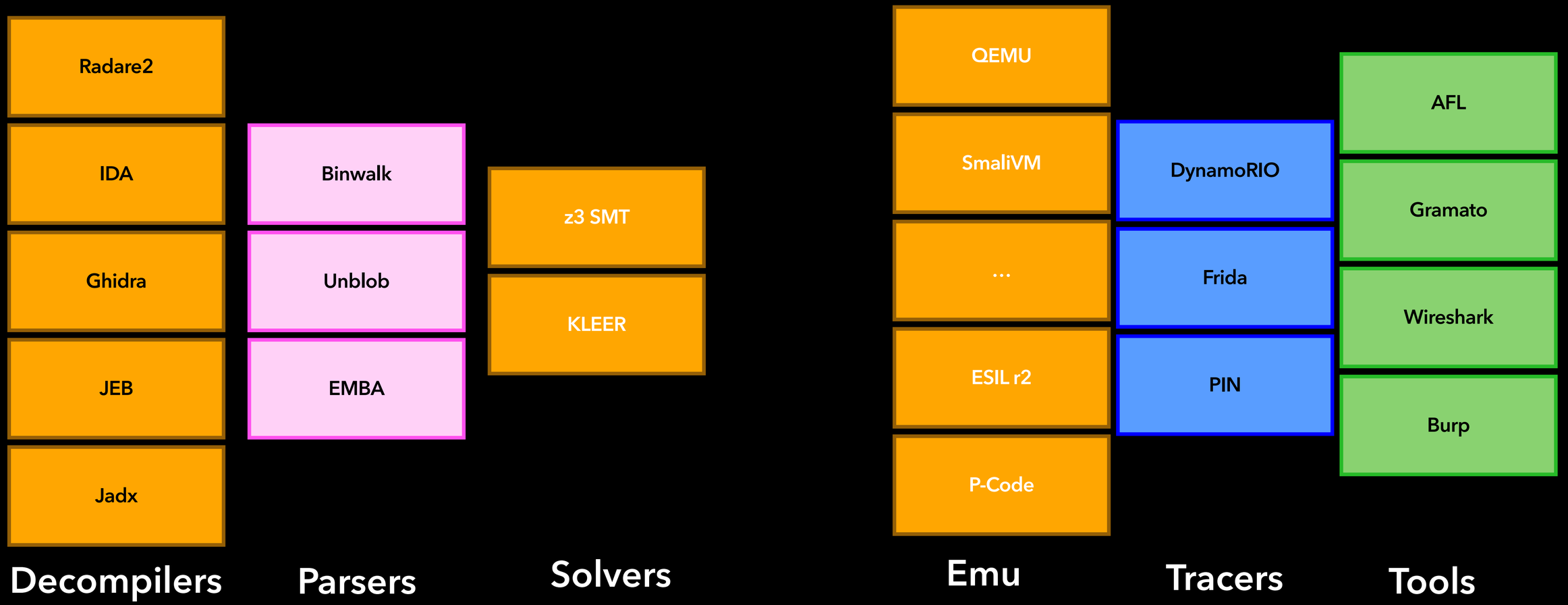
Runtime traces



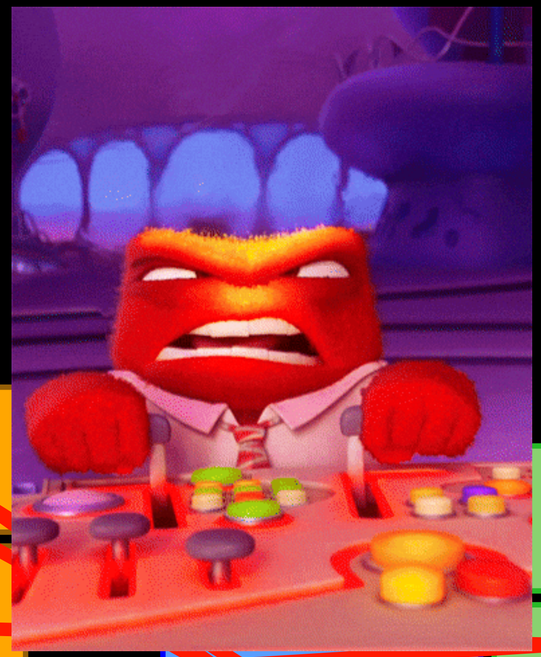
Ghidra | Binwalk | z3 SMT | QEMU | Frida | MITMProxy

Decompilers | Parsers | Solvers | Emu | Tracers | Tools

PROBLEM ORIGIN



PROBLEM ORIGIN



Radare2

IDA

Ghidra

JEB

Ja

Binwalk

Unblob

EMBA

KLEEr

QEMU

SmaliVM

...

ESIL r2

P-Code

Frida

PIN

AFL

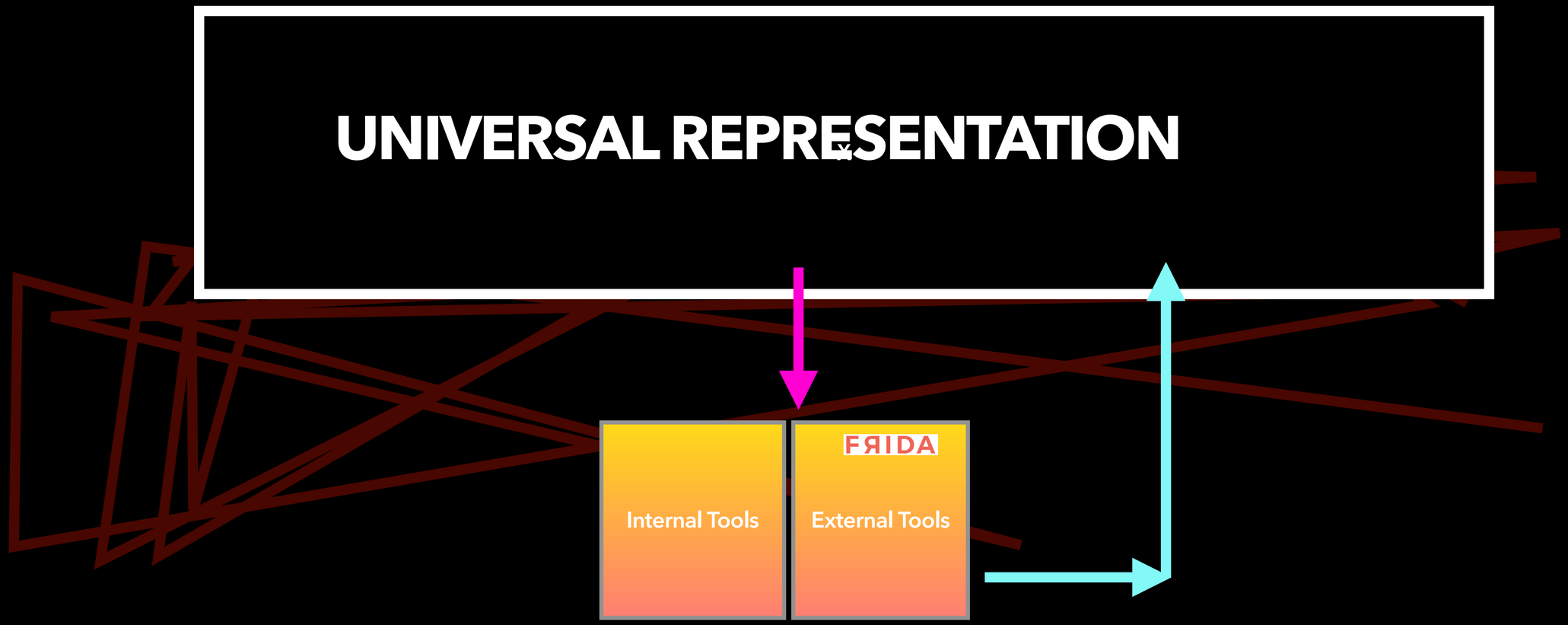
MITMProxy

Wireshark

Burp

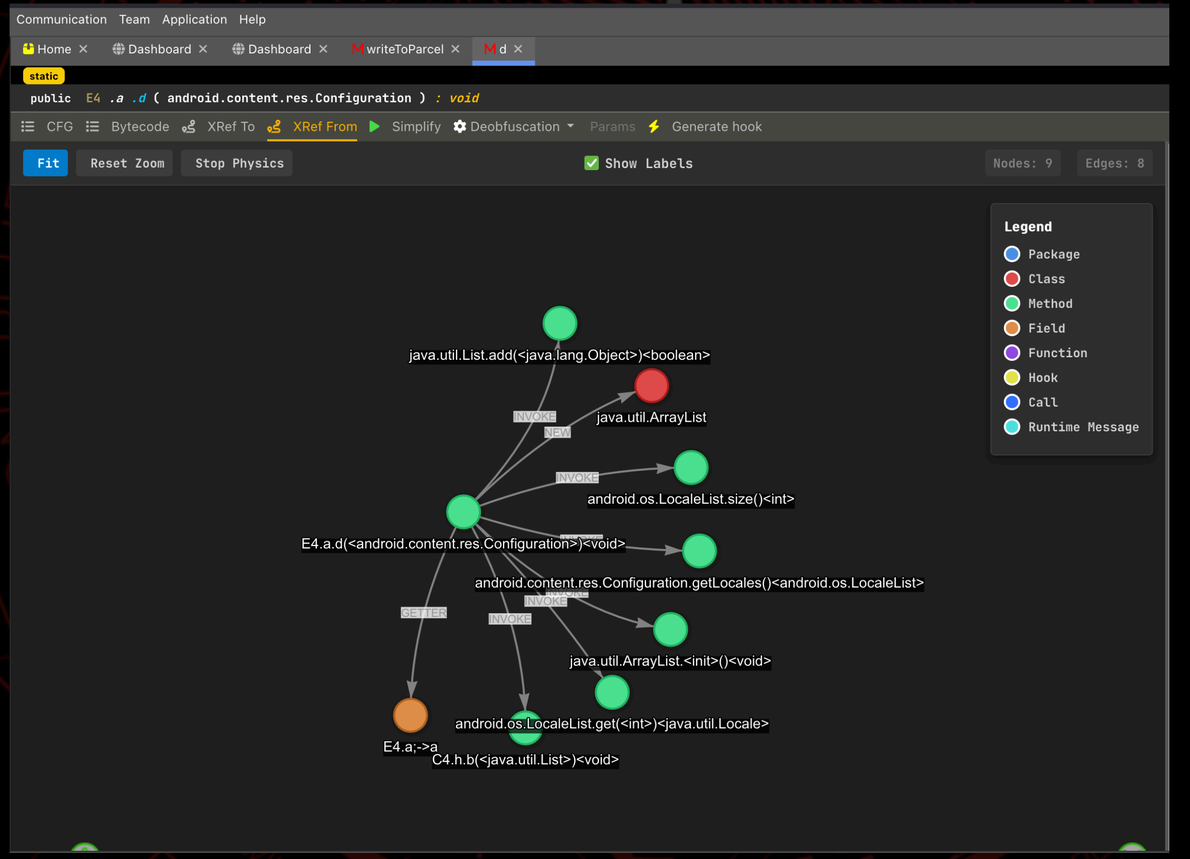
Decompilers Parsers Solvers Emu Tracers Tools

PROBLEM ORIGIN

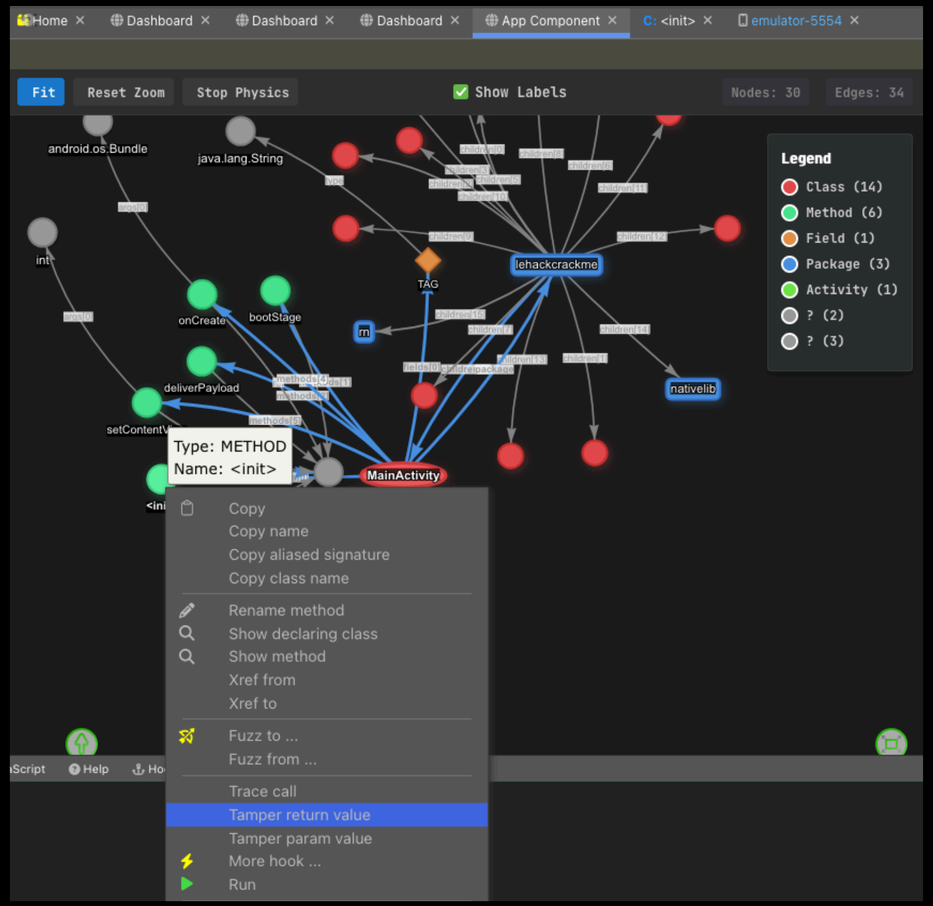


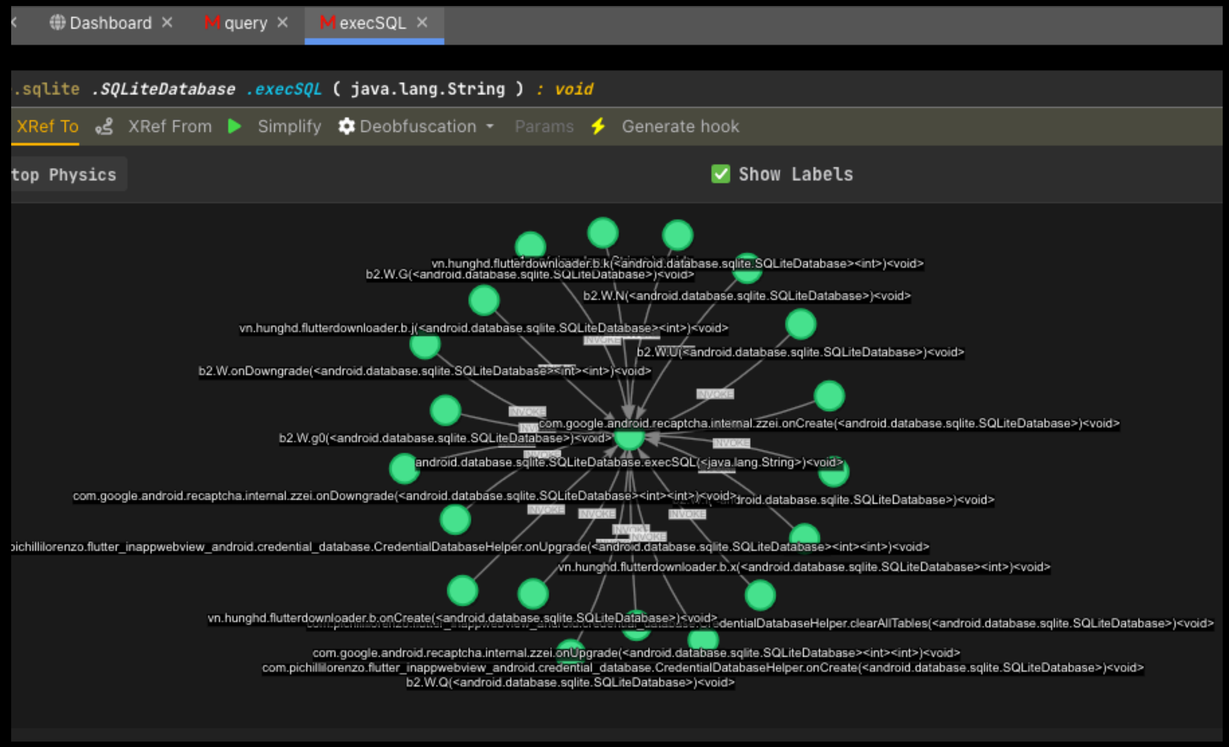
PART 2

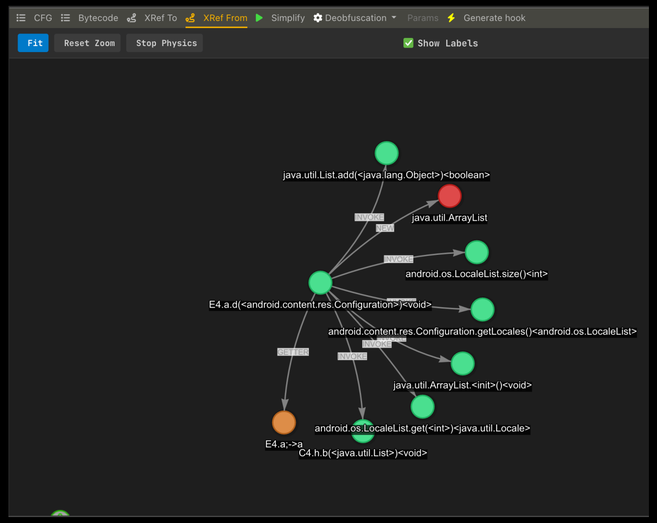
Principles & ElixIR



PRINCIPLE


Home
Dashboard
Dashboard
Dashboard
App Component
C: <init>
emulator-5554
Fit
Reset Zoom
Stop Physics
Show Labels
Nodes: 30
Edges: 34
Legend
Class (14)
Method (6)
Field (1)
Package (3)
Activity (1)
? (2)
? (3)
android.os.Bundle
java.lang.String
lehackcrackme
TAG
onCreate
bootStage
deliverPayload
setContentV
nativelib
MainActivity
Type: METHOD
Name: <init>
Copy
Copy name
Copy aliased signature
Copy class name
Rename method
Show declaring class
Show method
Xref from
Xref to
Fuzz to ...
Fuzz from ...
Trace call
Tamper return value
Tamper param value
More hook ...
Run
Help

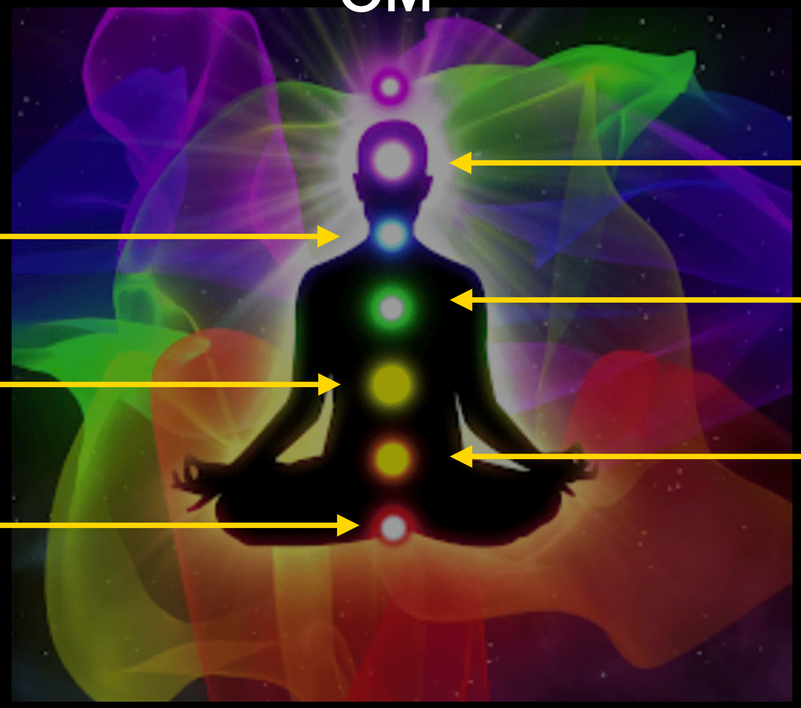

Dashboard
query
execSQL
.sqlite .SQLiteDatabase .execSQL (java.lang.String) : void
XRef To
XRef From
Simplify
Deobfuscation
Params
Generate hook
top Physics
Show Labels
vn.hunghd.flutterdownloader.b.k(<android.database.sqlite.SQLiteDatabase><int>)<void>
b2.W.G(<android.database.sqlite.SQLiteDatabase>)<void>
b2.W.N(<android.database.sqlite.SQLiteDatabase>)<void>
vn.hunghd.flutterdownloader.b.j(<android.database.sqlite.SQLiteDatabase><int>)<void>
b2.W.U(<android.database.sqlite.SQLiteDatabase>)<void>
b2.W.onDowngrade(<android.database.sqlite.SQLiteDatabase><int><int>)<void>
com.google.android.recaptcha.internal.zzei.onCreate(<android.database.sqlite.SQLiteDatabase>)<void>
b2.W.g0(<android.database.sqlite.SQLiteDatabase>)<void>
android.database.sqlite.SQLiteDatabase.execSQL(<java.lang.String>)<void>
com.google.android.recaptcha.internal.zzei.onDowngrade(<android.database.sqlite.SQLiteDatabase><int><int>)<void>
android.database.sqlite.SQLiteDatabase)<void>
vn.hunghd.flutterdownloader.b.x(<android.database.sqlite.SQLiteDatabase><int>)<void>
vn.hunghd.flutterdownloader.b.onCreate(<android.database.sqlite.SQLiteDatabase>)<void>
com.google.android.recaptcha.internal.zzei.onUpgrade(<android.database.sqlite.SQLiteDatabase><int><int>)<void>
com.pichillilorenzo.flutter_inappwebview_android.credential_database.CredentialDatabaseHelper.onCreate(<android.database.sqlite.SQLiteDatabase>)<void>
b2.W.Q(<android.database.sqlite.SQLiteDatabase>)<void>


CFG
Bytecode
XRef To
XRef From
Simplify
Deobfuscation
Params
Generate hook
Fit
Reset Zoom
Stop Physics
Show Labels
java.util.List.add(<java.lang.Object>)<boolean>
java.util.ArrayList
android.os.LocaleList.size()<int>
E4.a.d(<android.content.res.Configuration>)<void>
android.content.res.Configuration.getLocales()<android.os.LocaleList>
java.util.ArrayList.<init>()<void>
android.os.LocaleList.get(<int>)<java.util.Locale>
E4.a;->a
C4.h.b(<java.util.List>)<void>

PRINCIPLE

(Workflows)

OM



Network & Processes

Filesystem activity

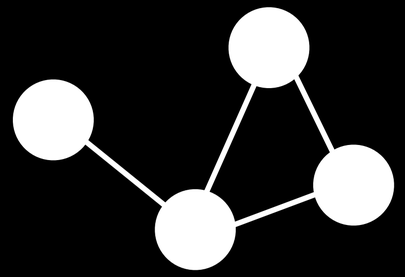
Binary Code

UI

Memory

Interpreted Code

PRINCIPLE



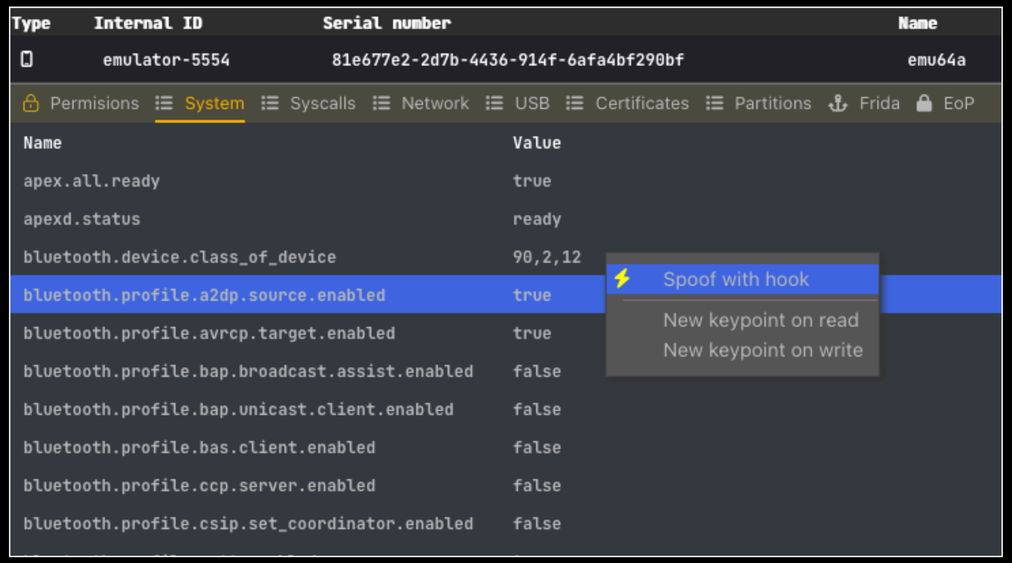
System API

Vendors API

Libs, TEE, drivers

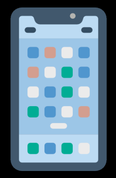
SE Policy

Permissions

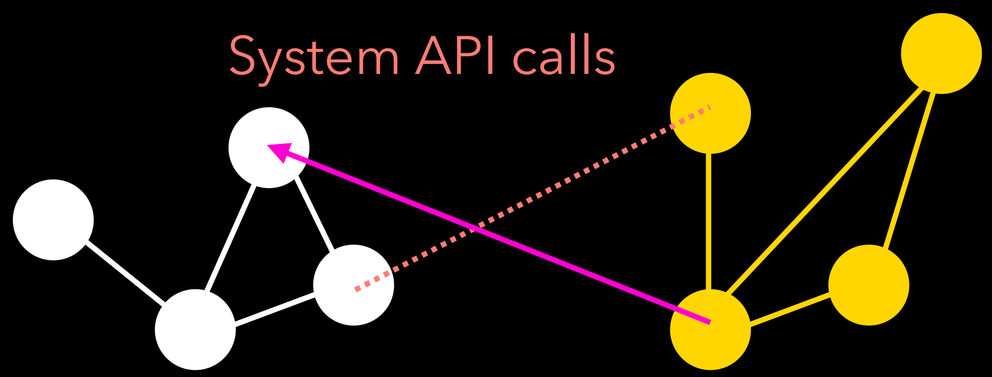


BUILT-IN SPOOFING OF DEVICE PROPERTIES

Device & Userland Profiling



PRINCIPLE



UI Components *

App components

Files / Data / Certificates / Keystores

Bytecode (dalvik, hermes,…)

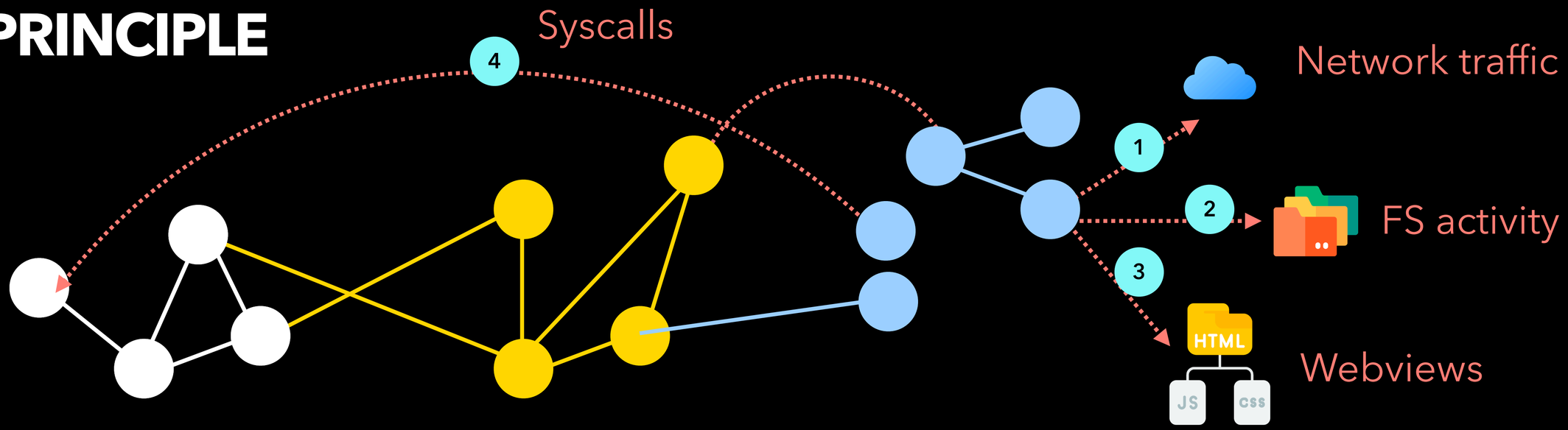
Interpreted code (AST)

Native functions

Basic Analysis



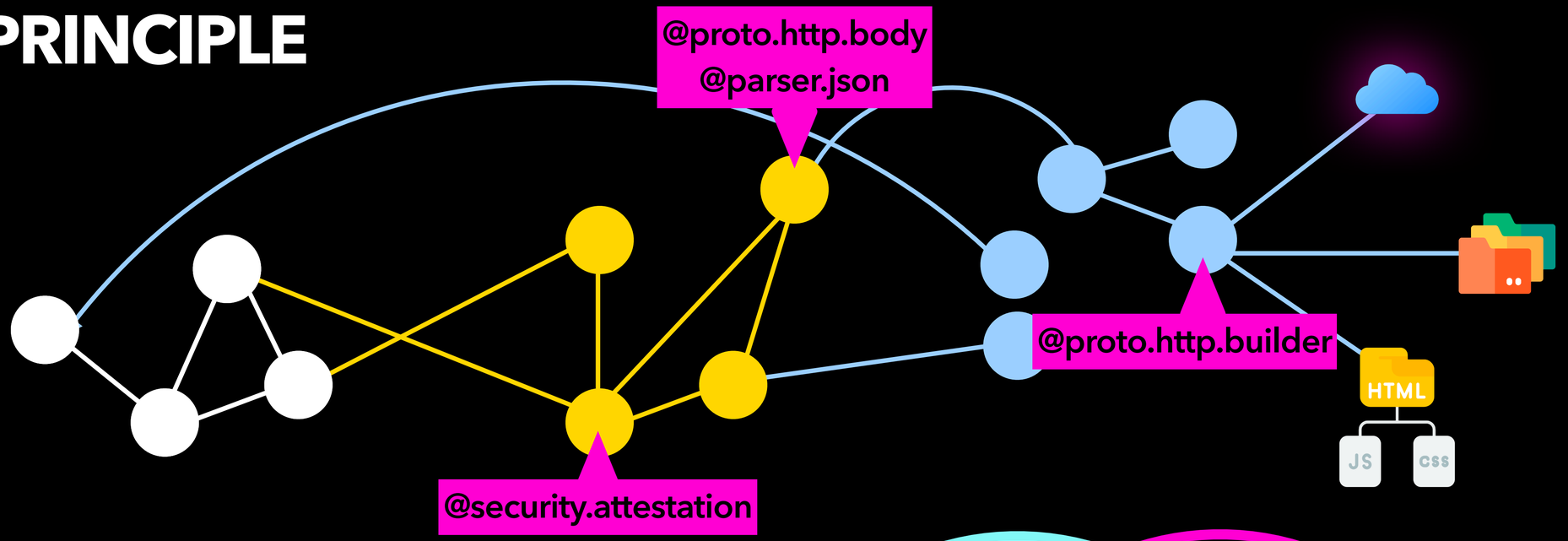
PRINCIPLE

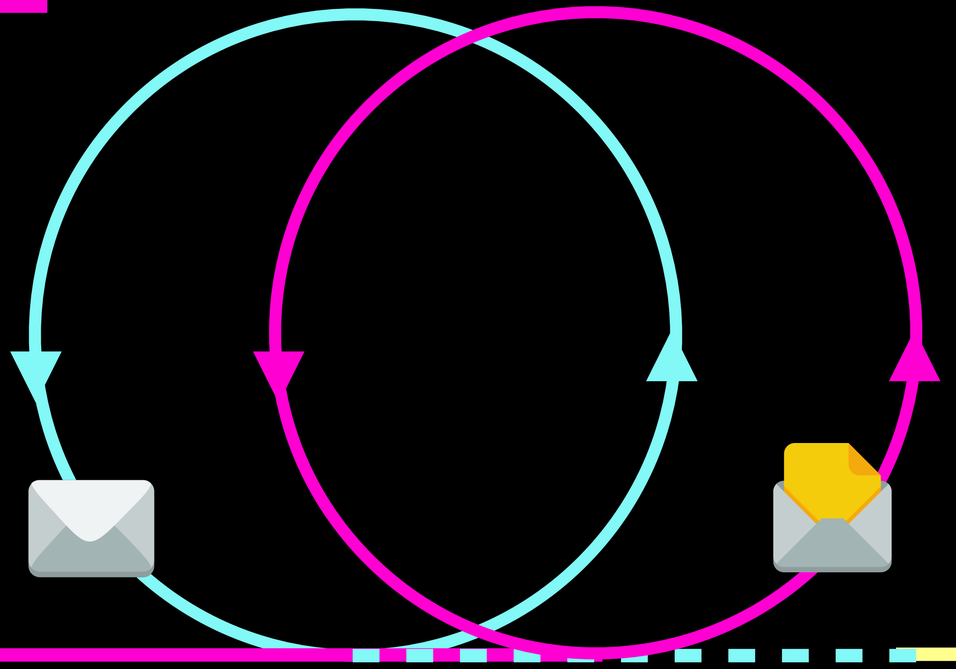




Hook Messages +
backtrace bound to IR

PRINCIPLE

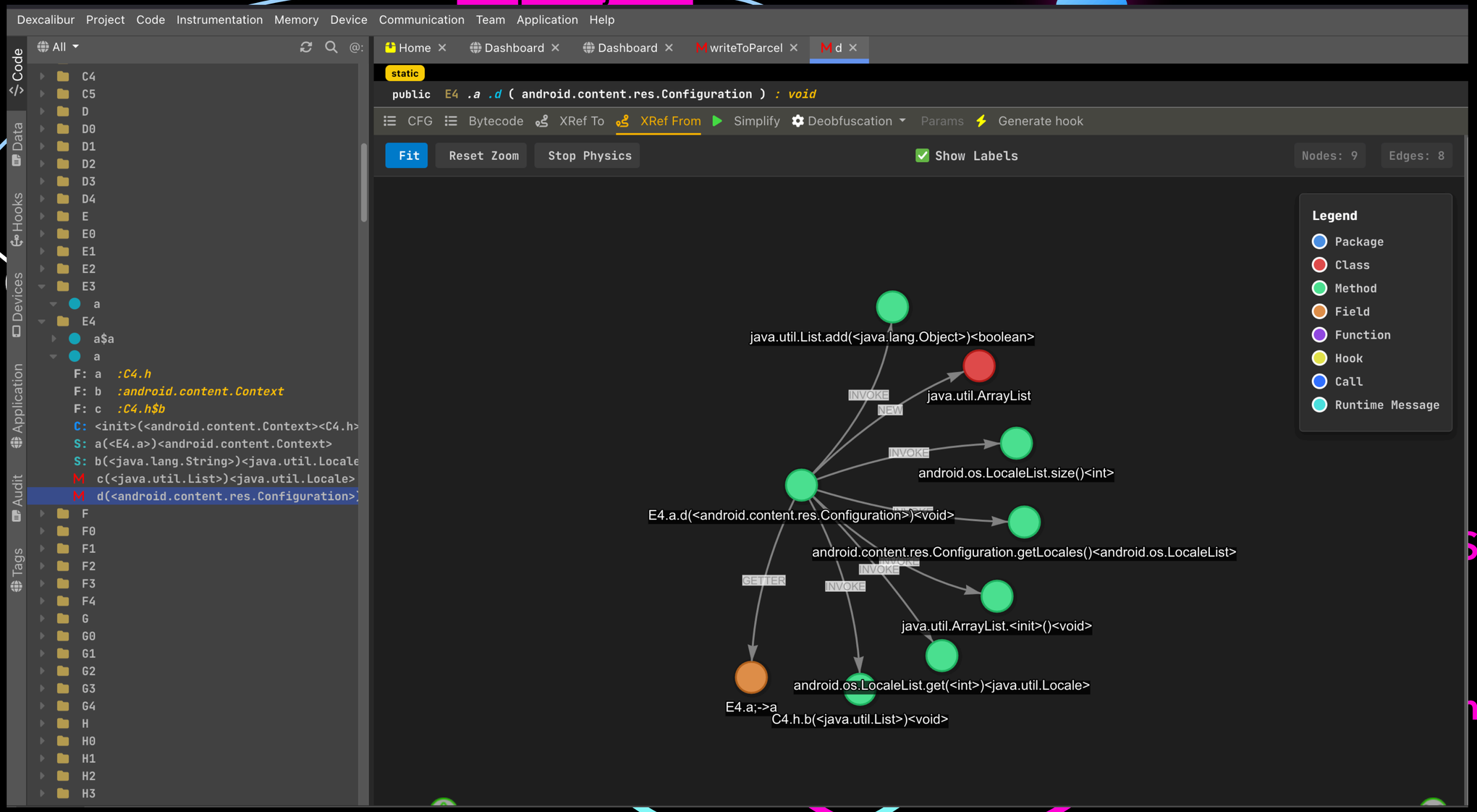




Continous Static Analysis, Tagging & Scheduling

PRINCIPLE

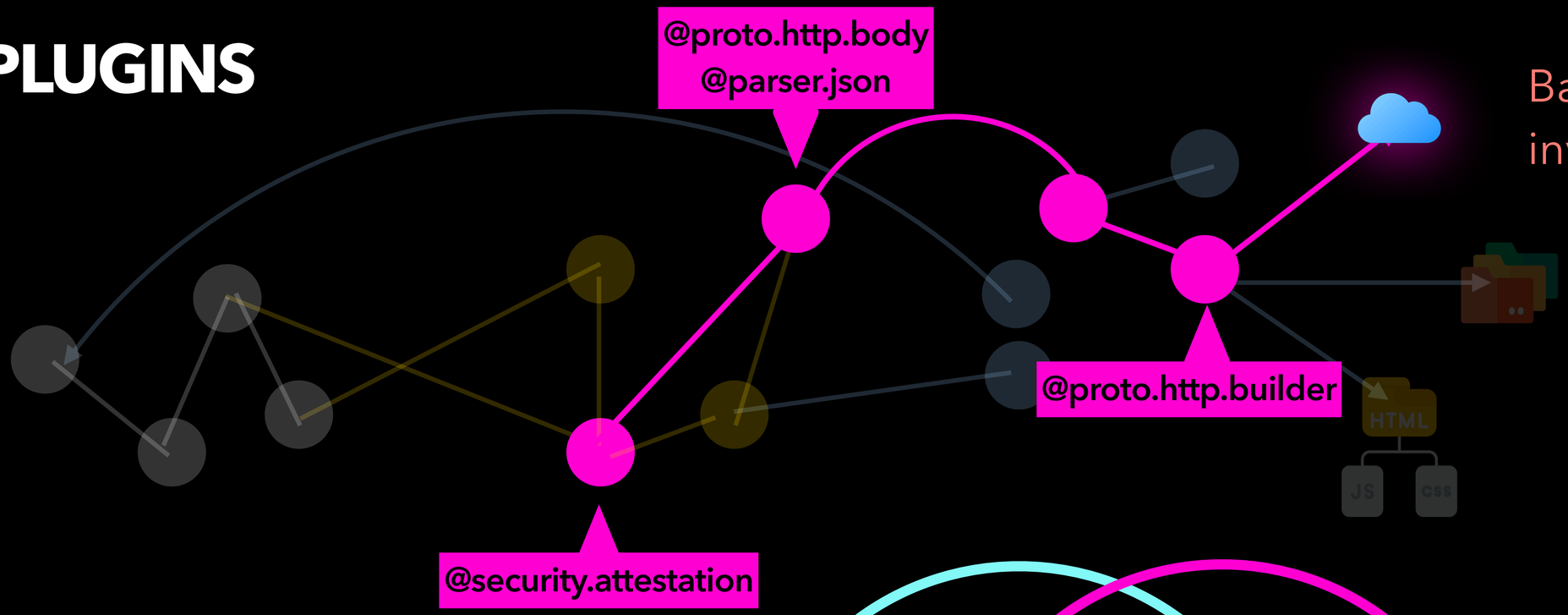
@proto.http.body
@parser.json



Static

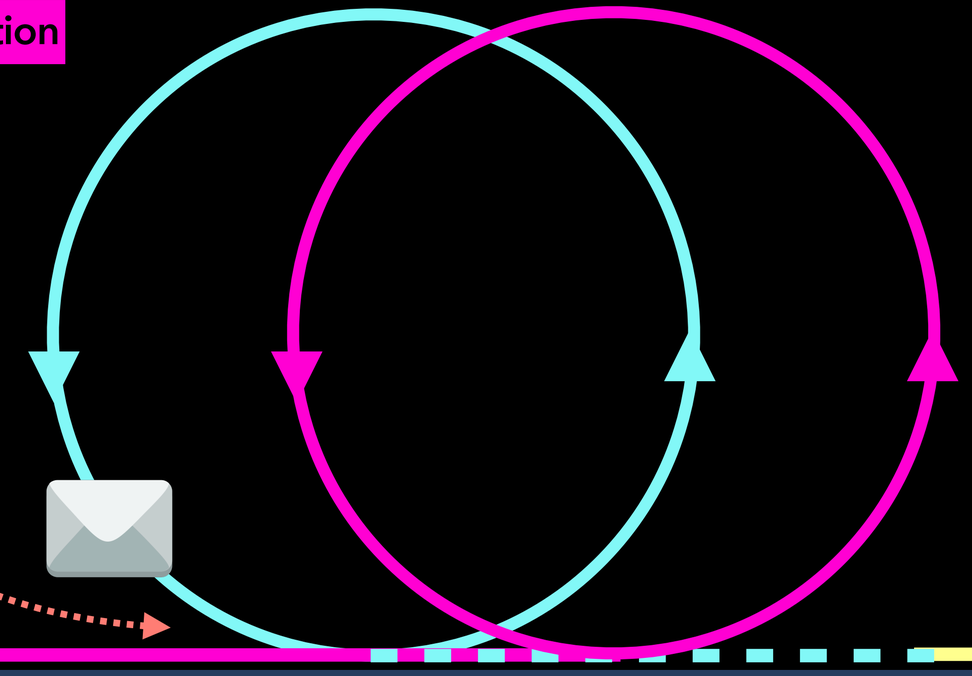
ng

PLUGINS



Backward taint-analysis involved by traces

New hooks & re-run

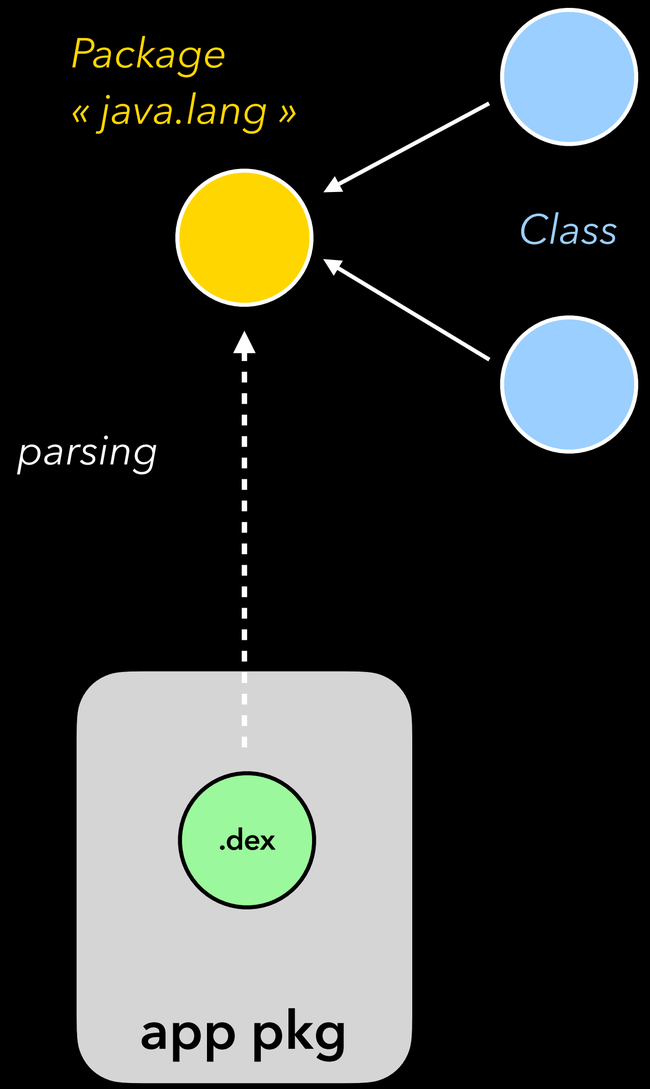


Scan/share/explore

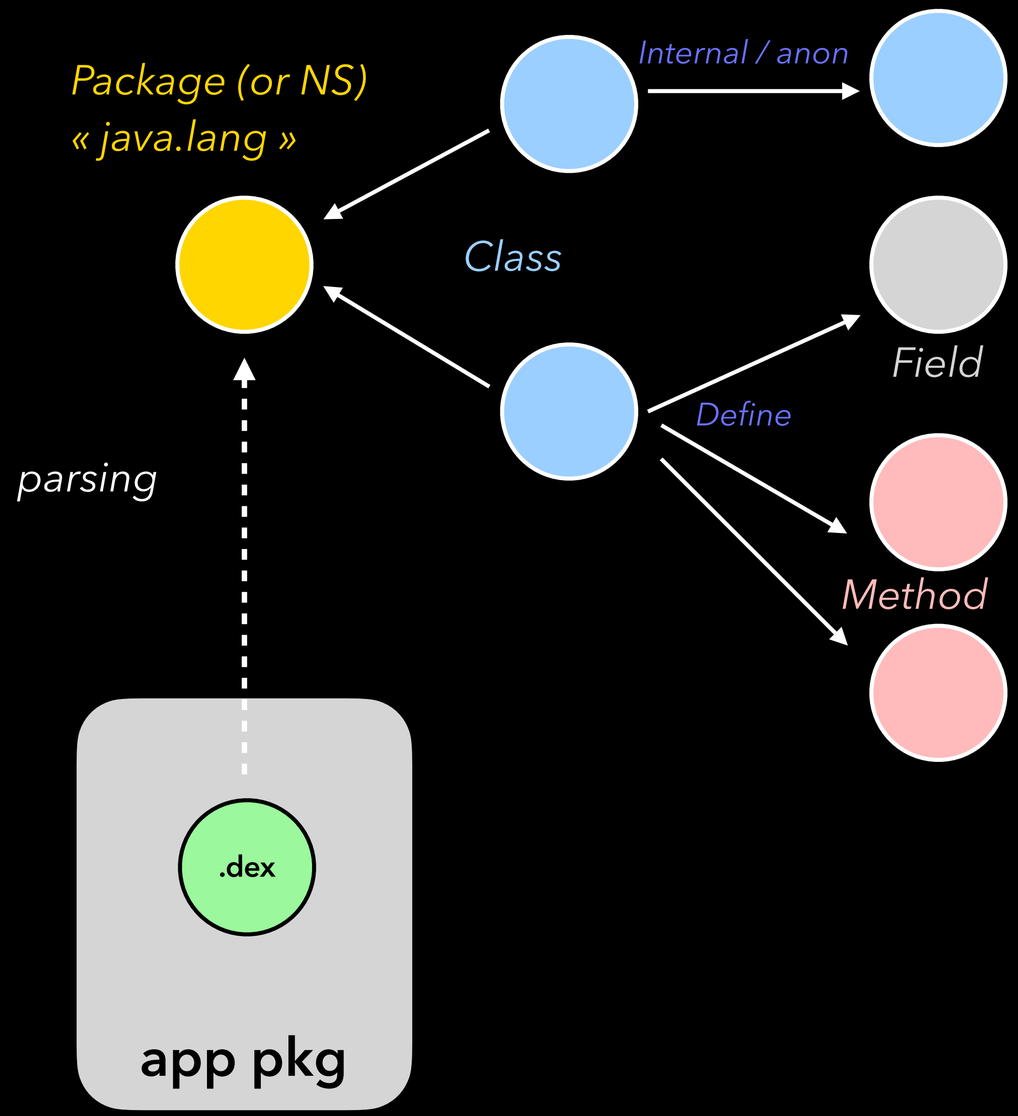


```
from(method('@security.attestation')).to(call(callee@protocol.http.builder))
```

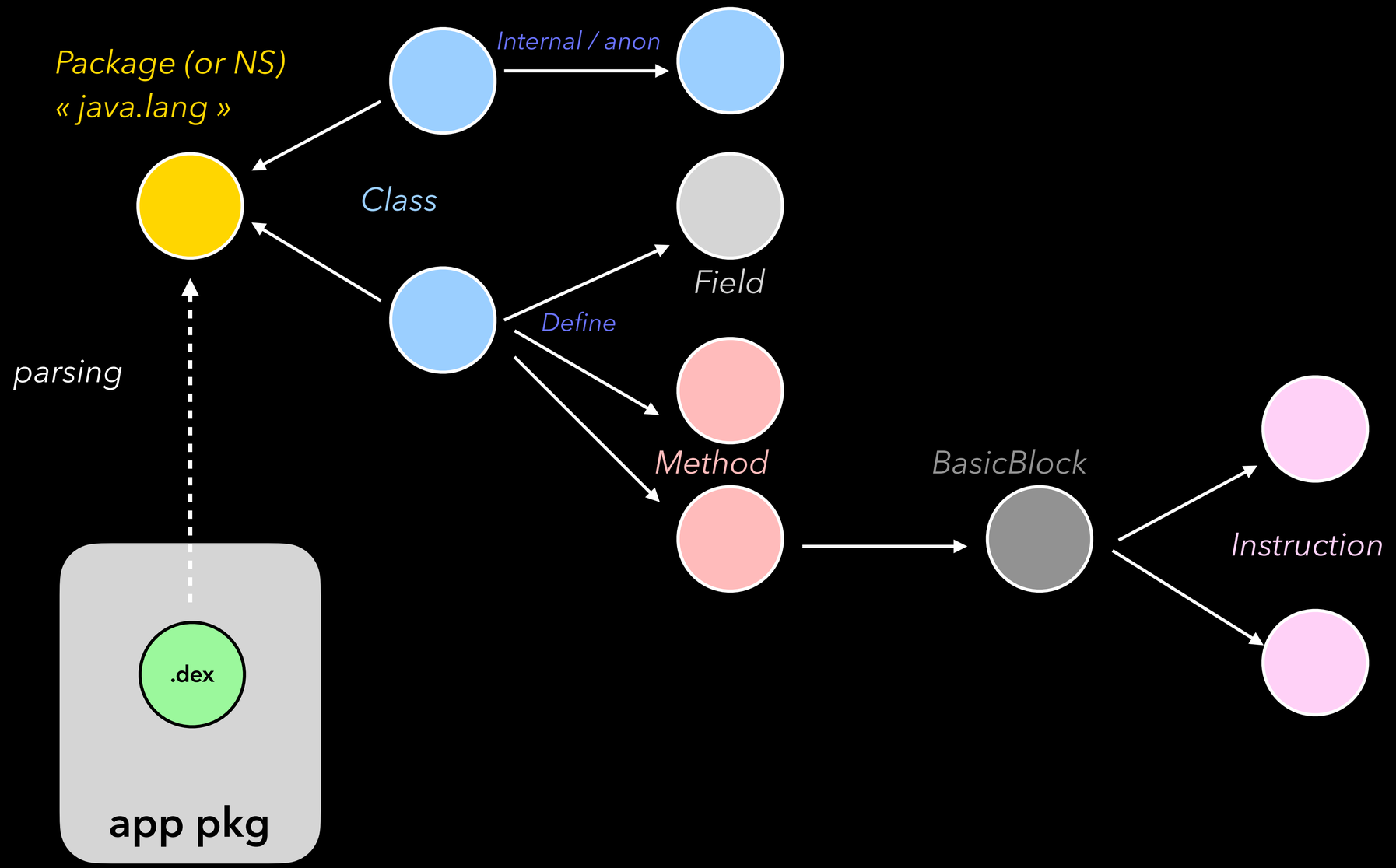
ELIXIR : OBJECTS



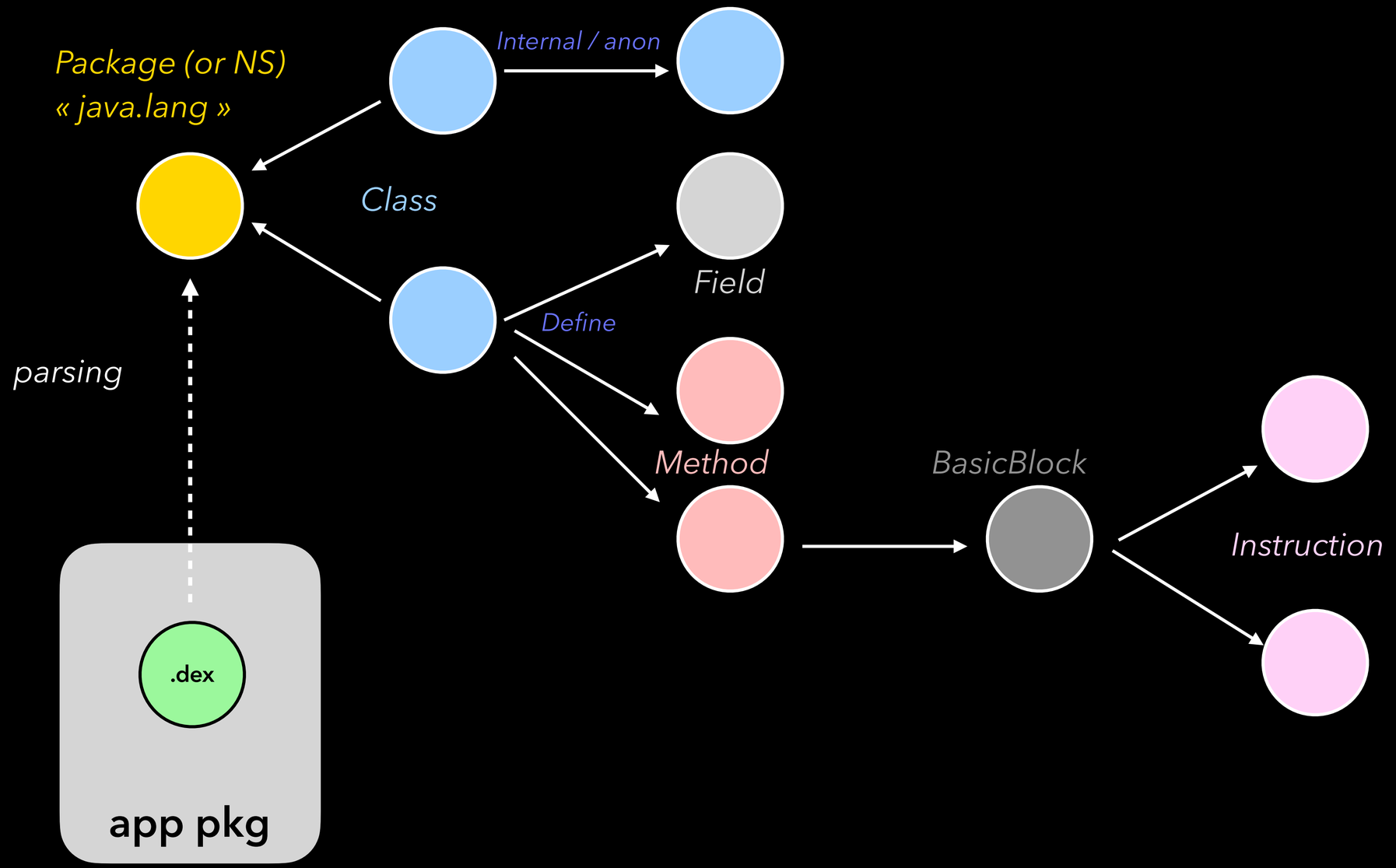
ELIXIR : OBJECTS



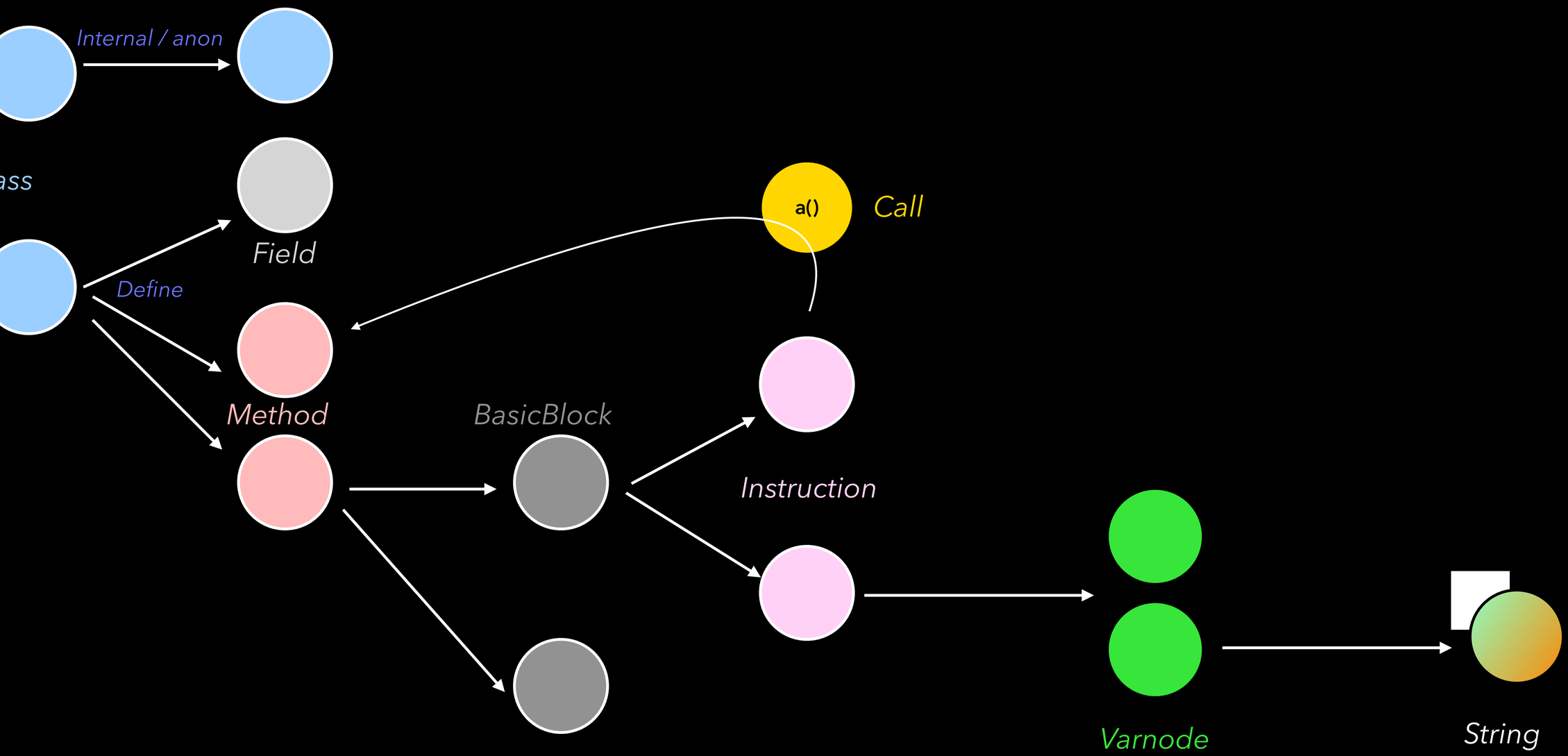
ELIXIR : OBJECTS



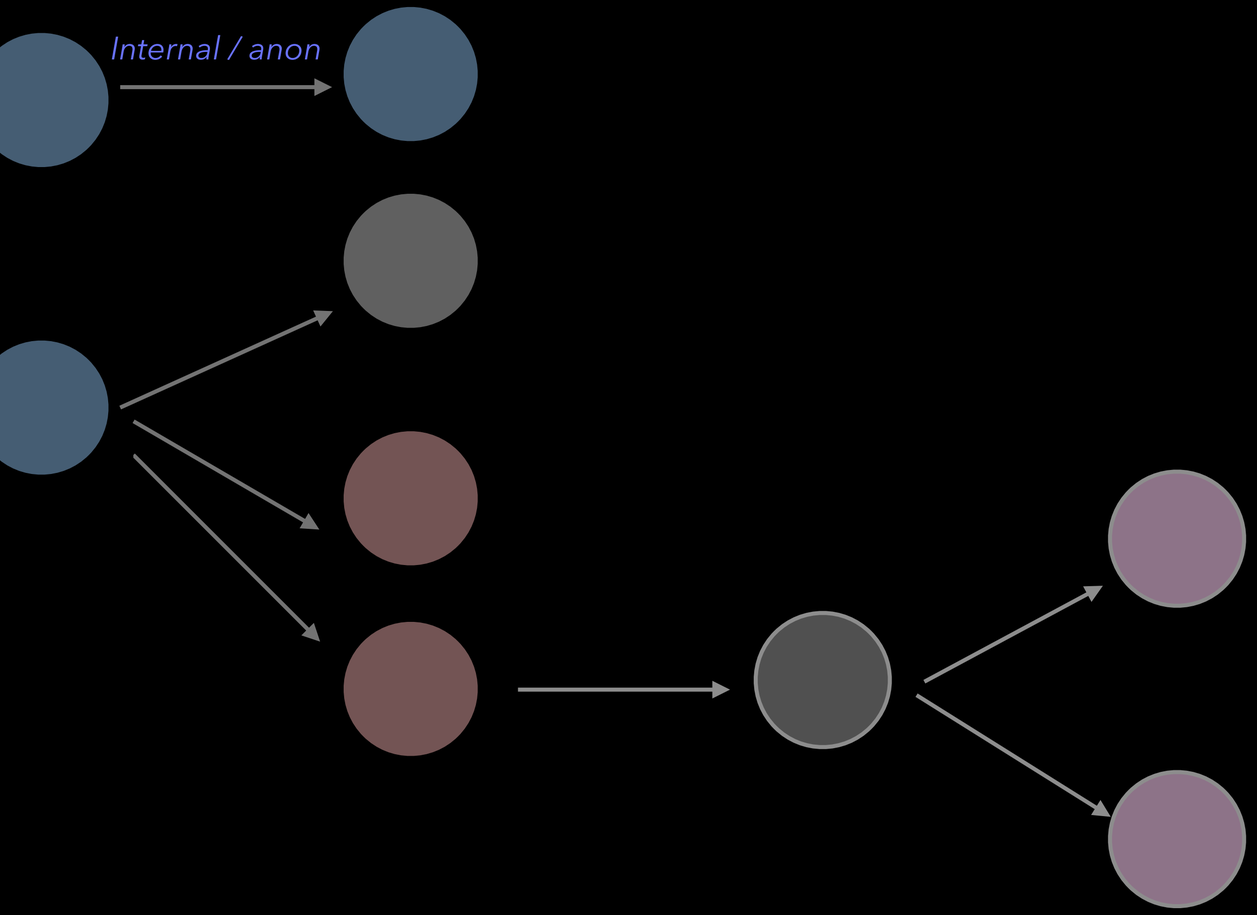
ELIXIR : OBJECTS

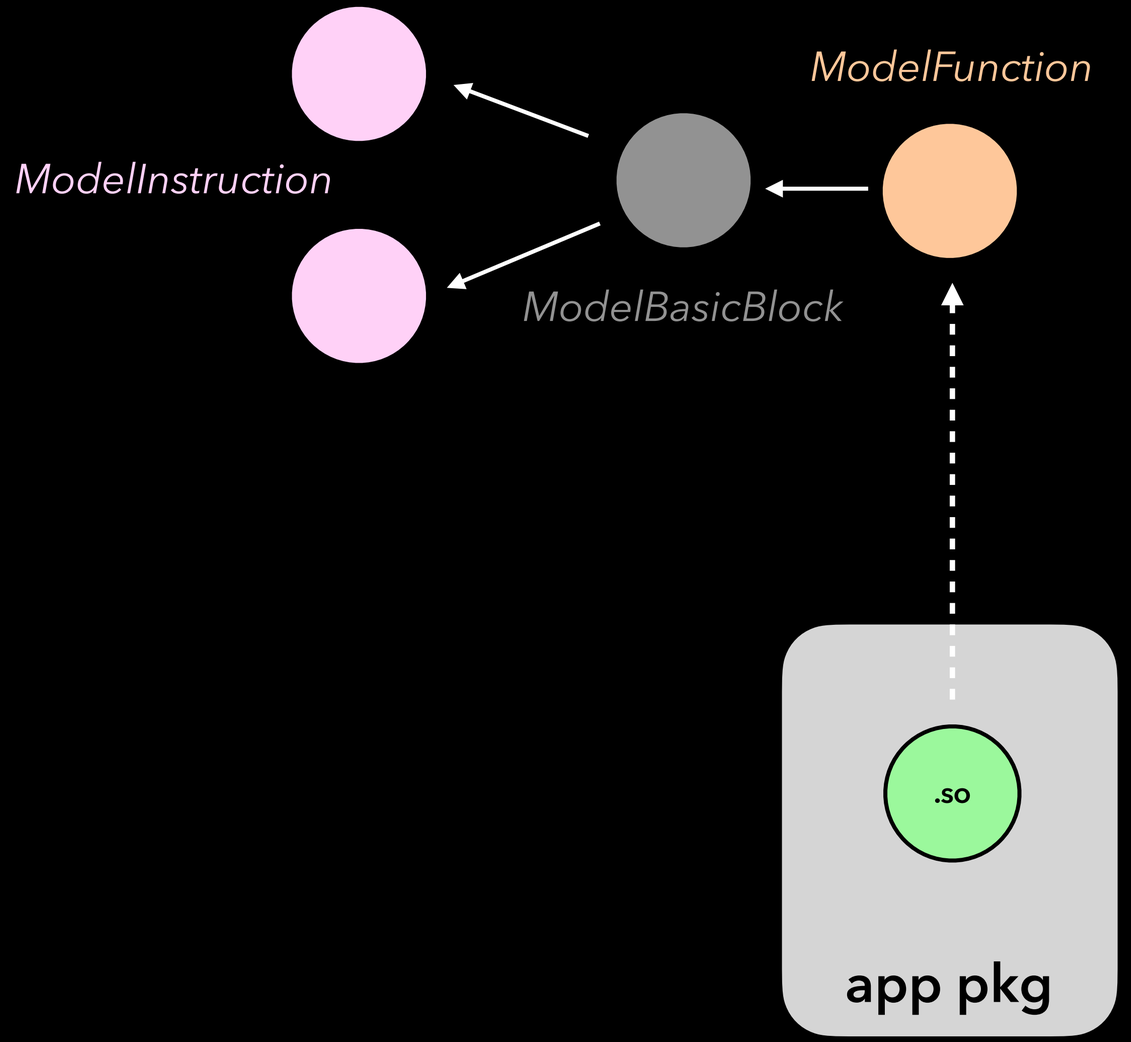


ELIXIR : OBJECTS

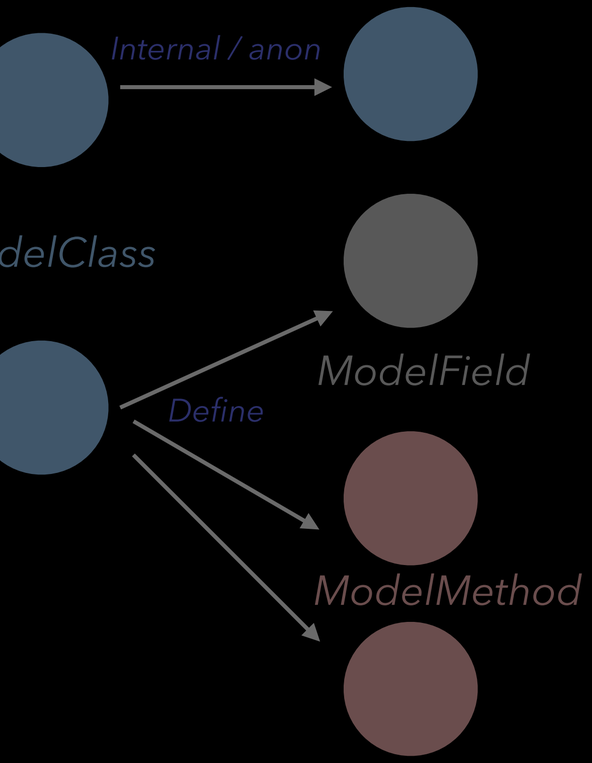


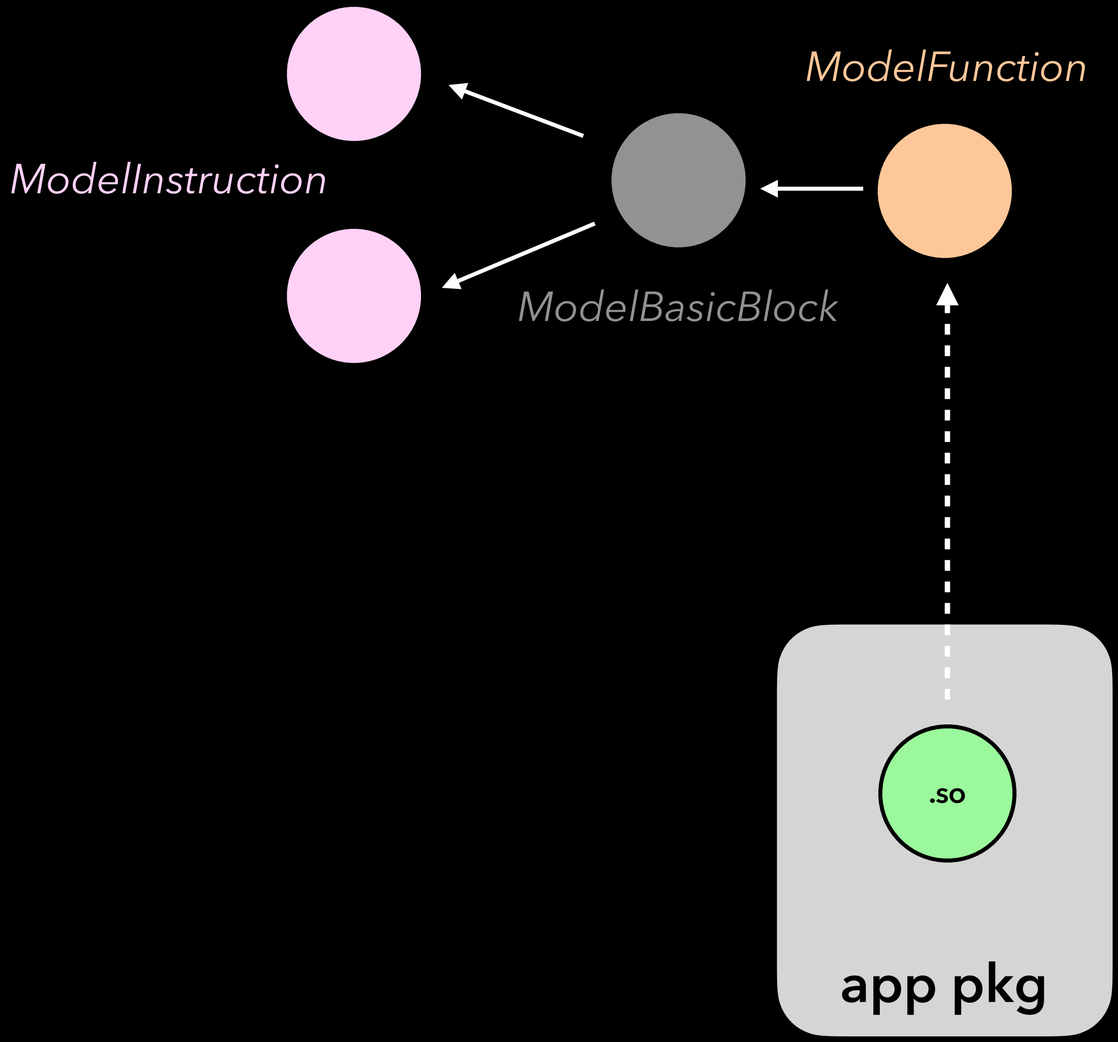
ELIXIR : CROSS-REFERENCES



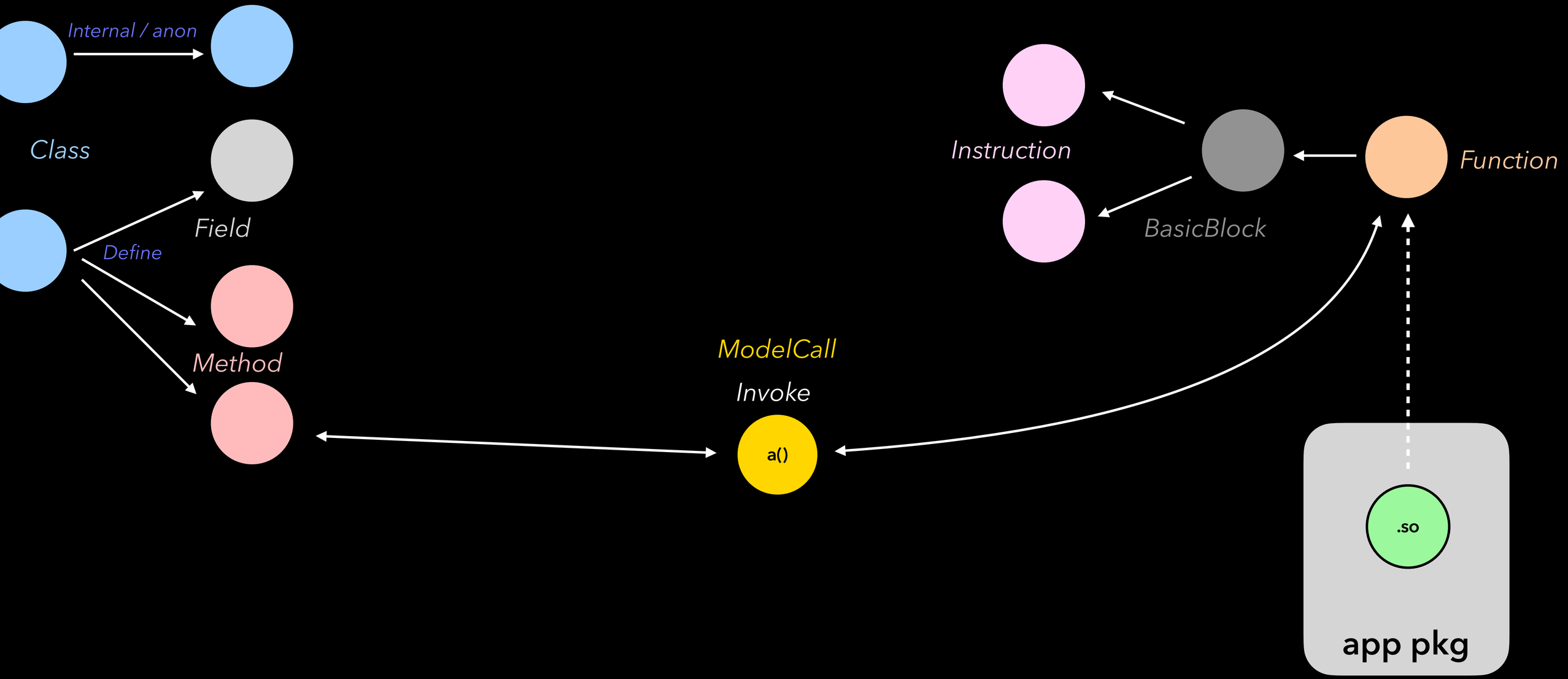


ELIXIR : CROSS-REFERENCES

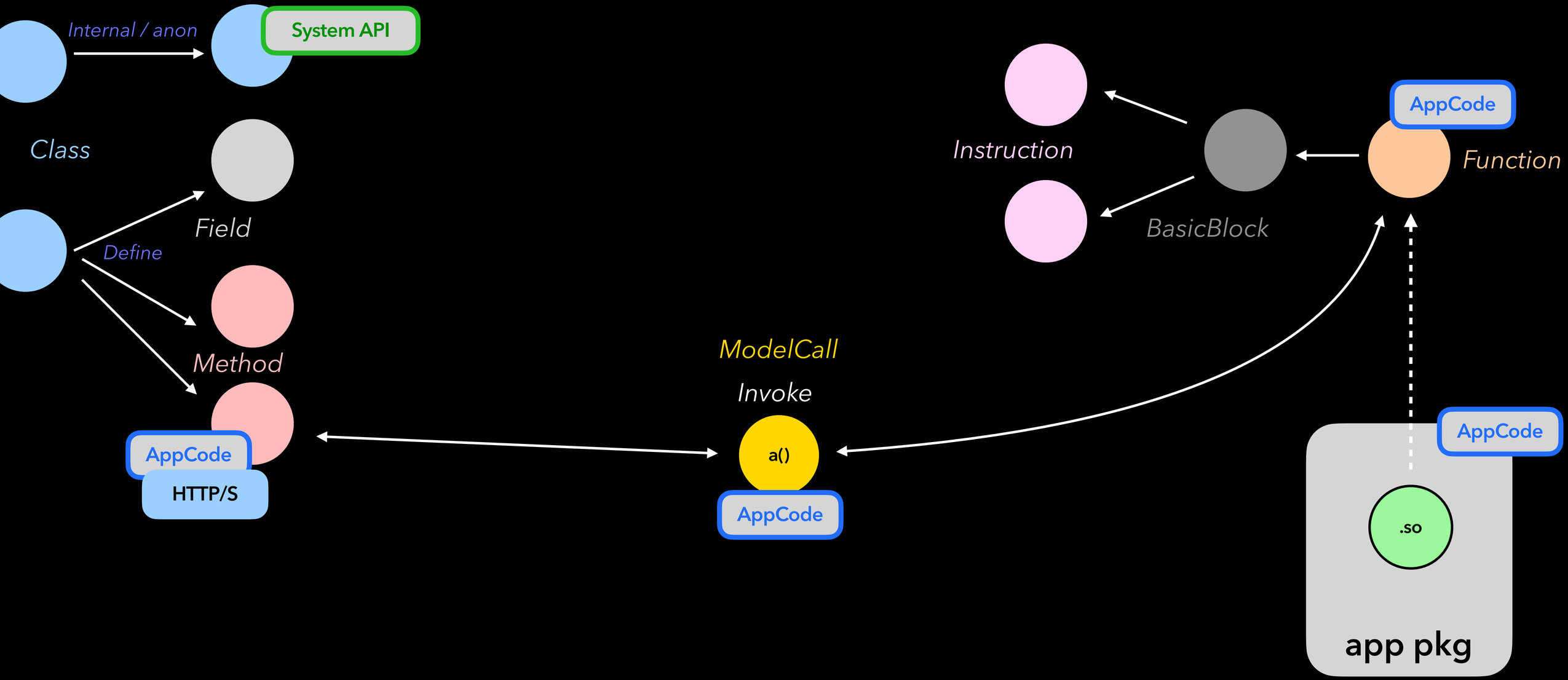




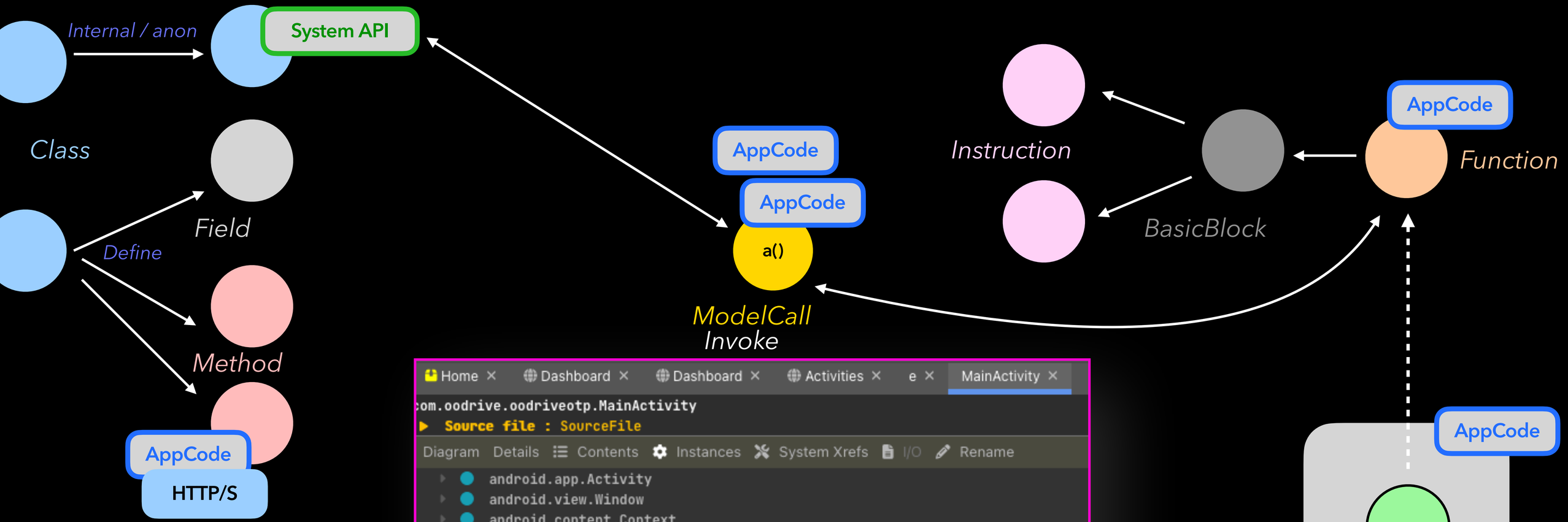
ELIXIR : CROSS-REFERENCES

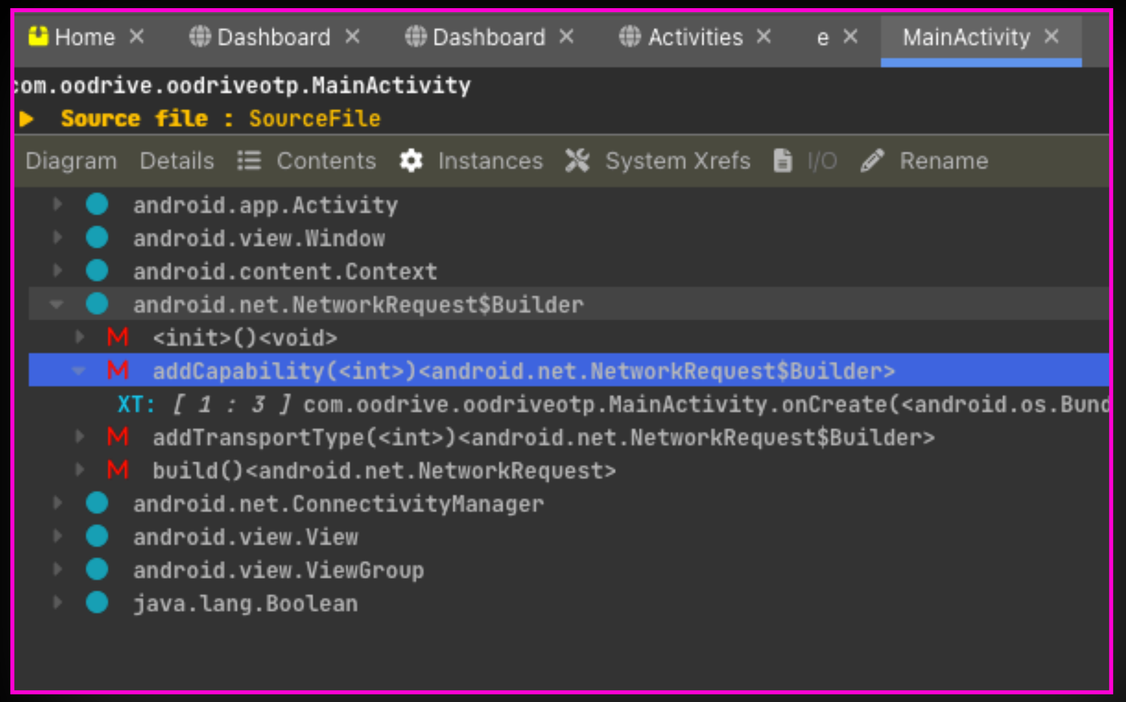


ELIXIR : CROSS-REFERENCES

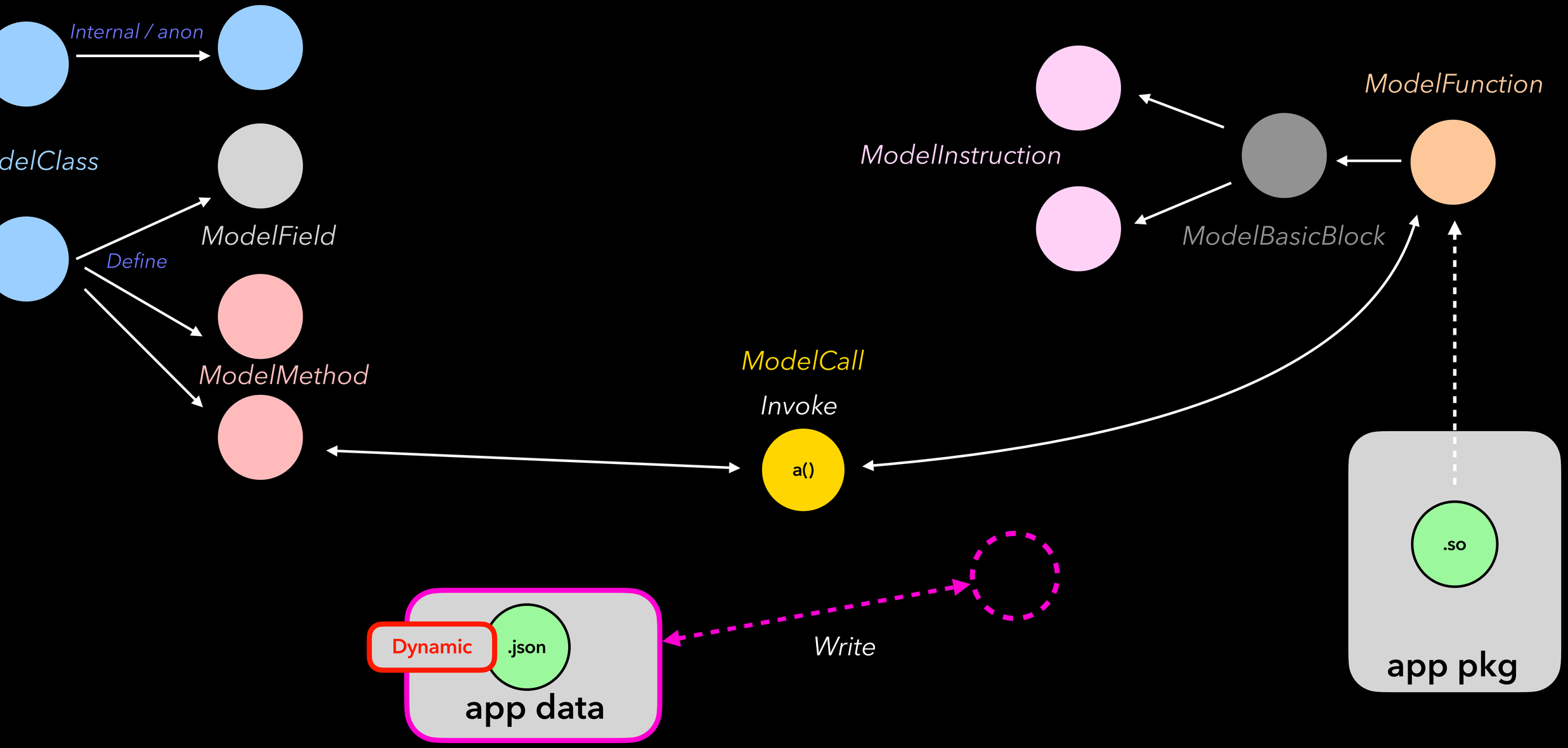


ELIXIR : CROSS-REFERENCES

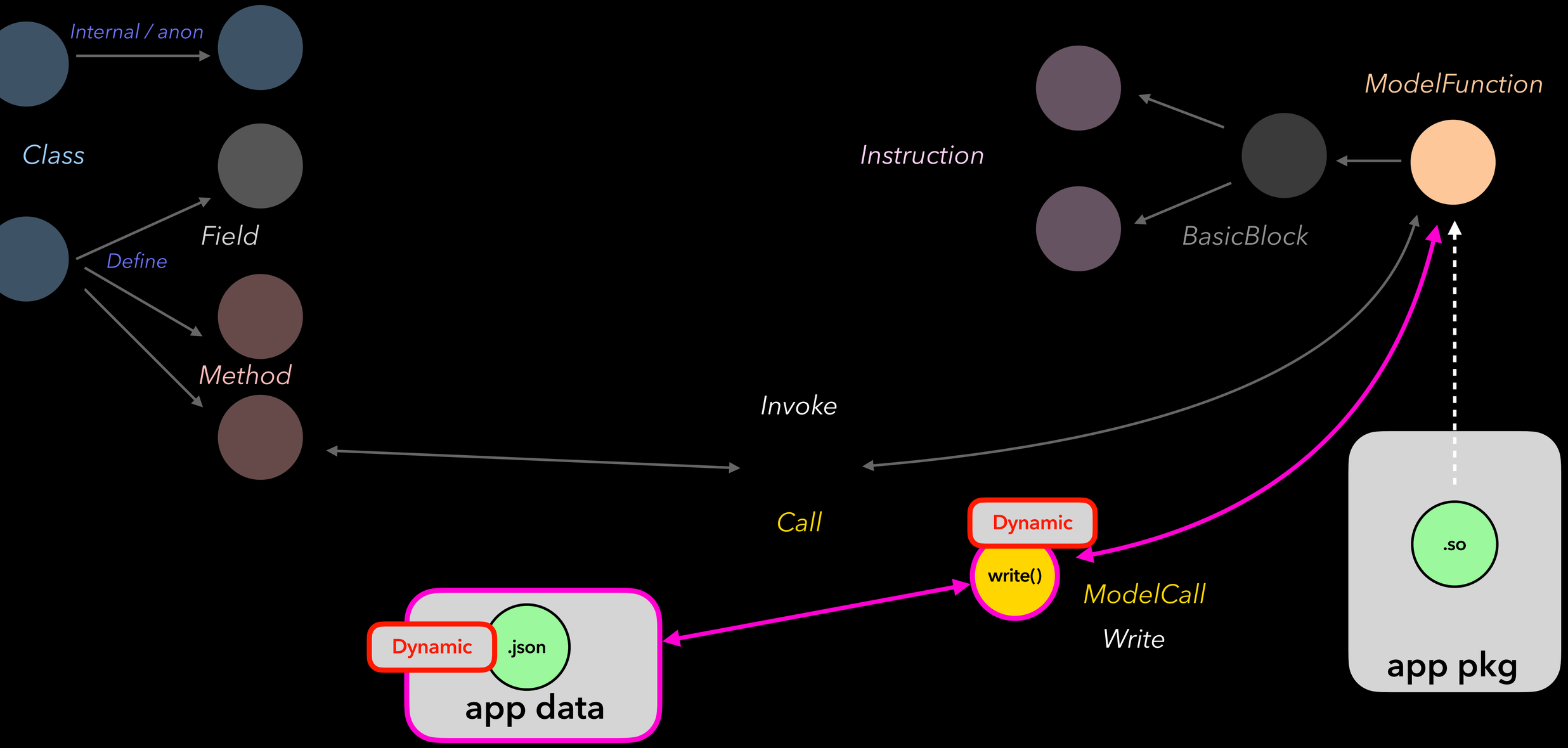




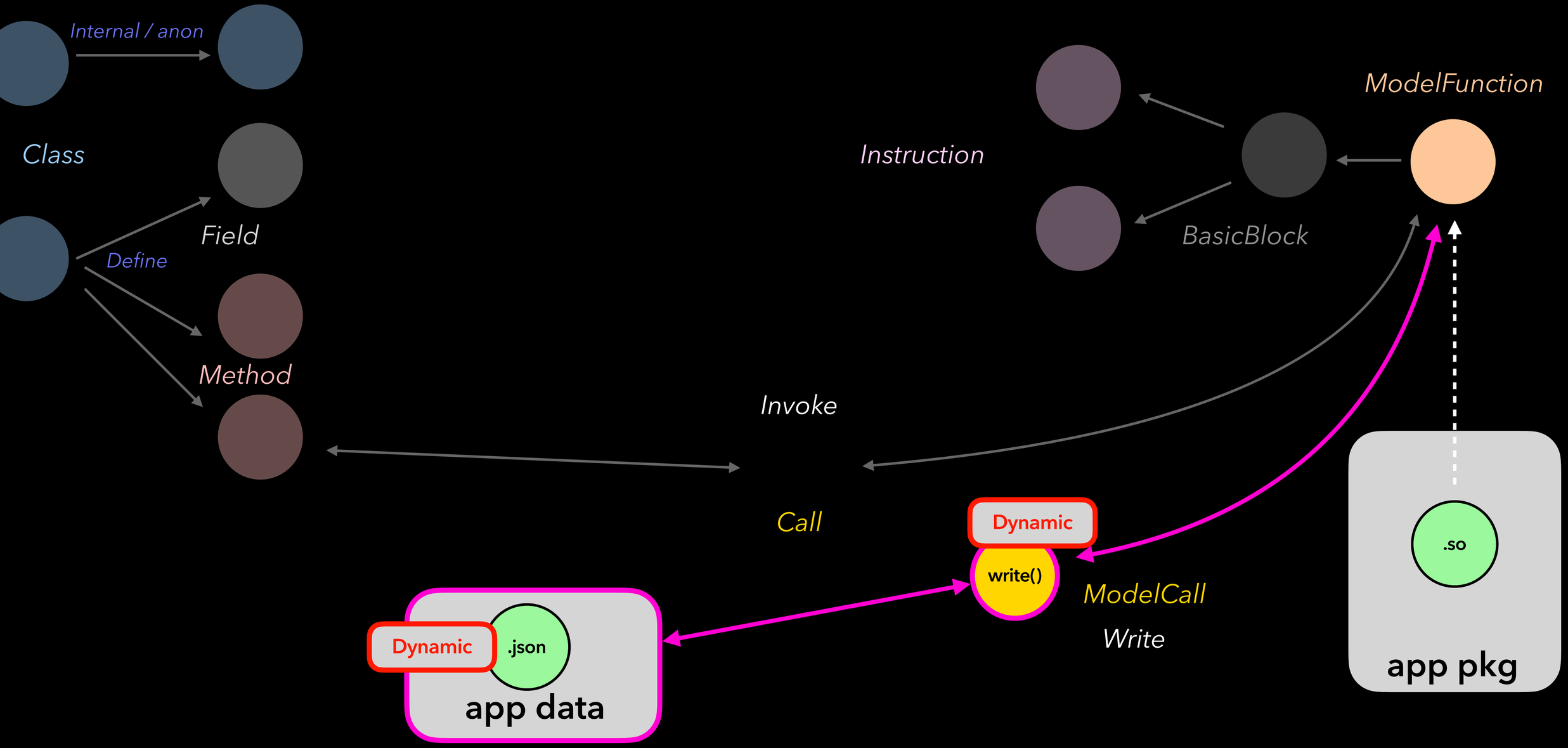
ELIXIR : CROSS-REFERENCES | RUNTIME



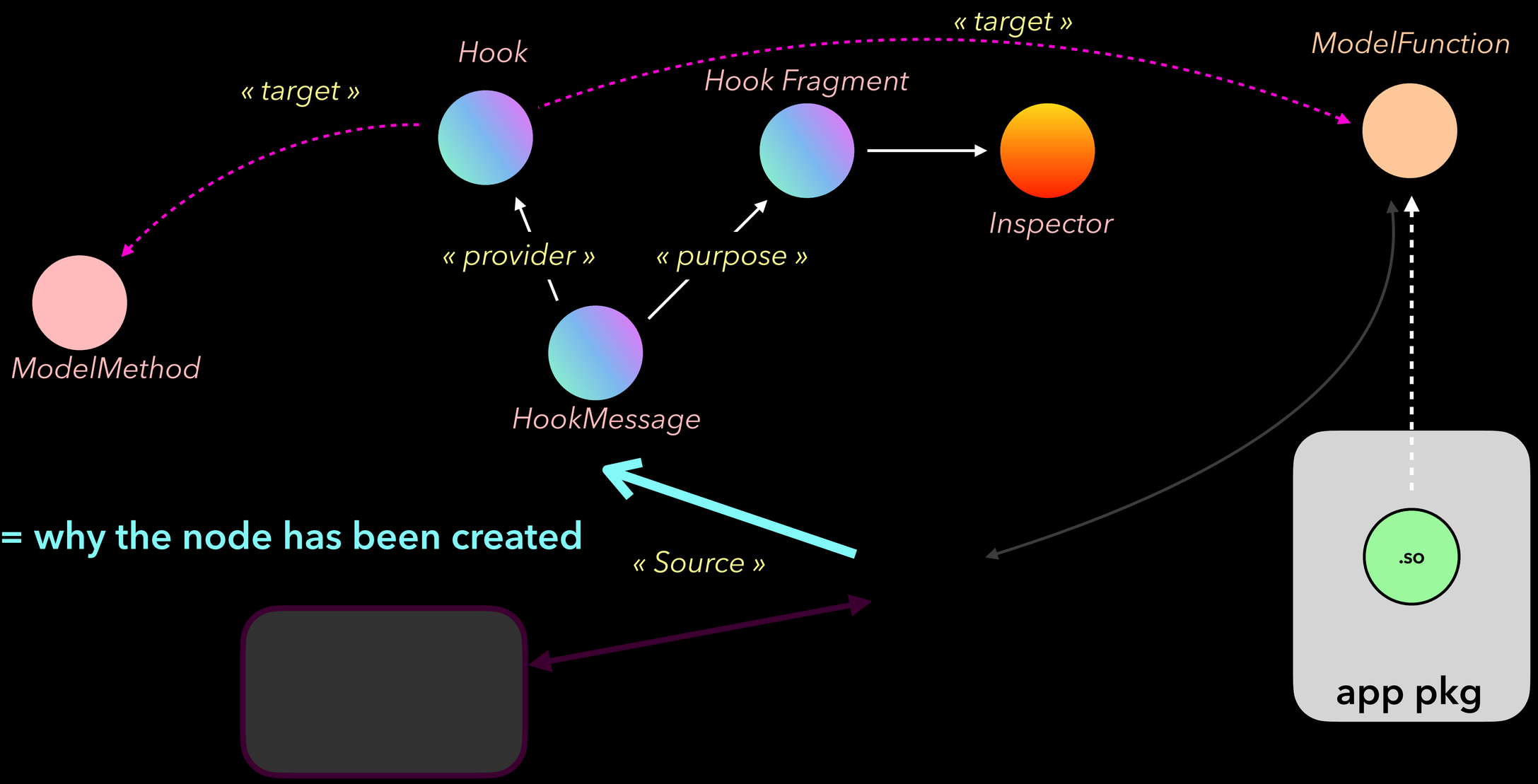
ELIXIR : CROSS-REFERENCES | RUNTIME



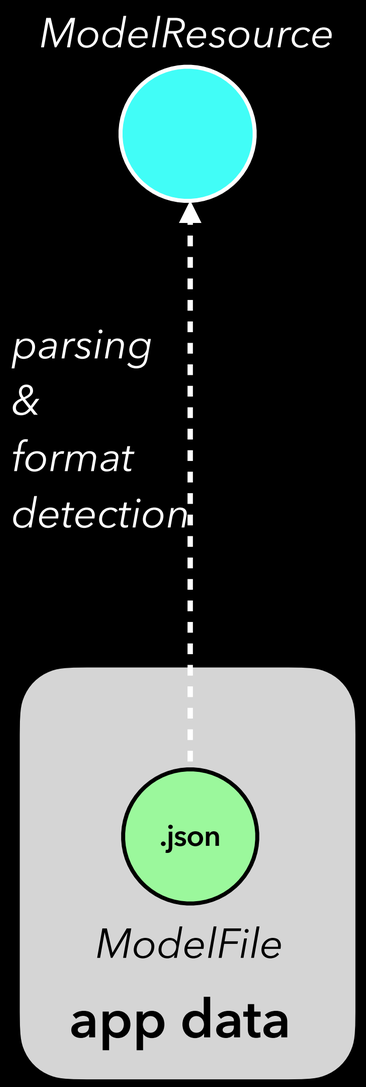
ELIXIR : CROSS-REFERENCES | RUNTIME



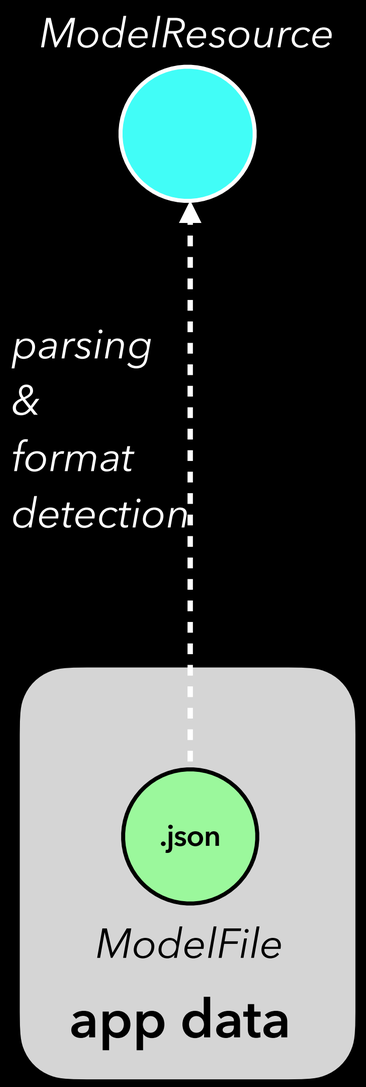
ELIXIR : CROSS-REFERENCES



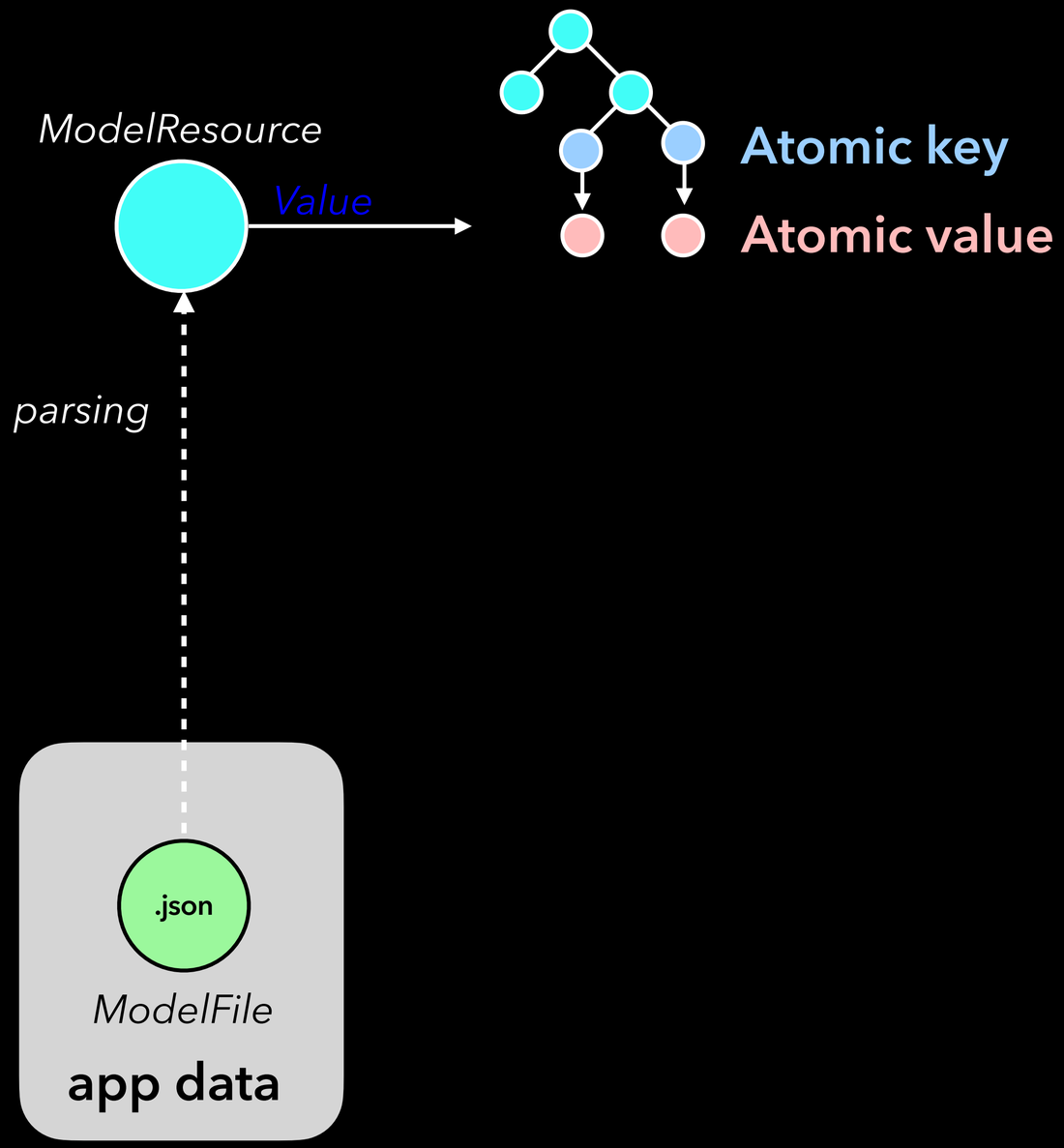
ELIXIR : STRINGS



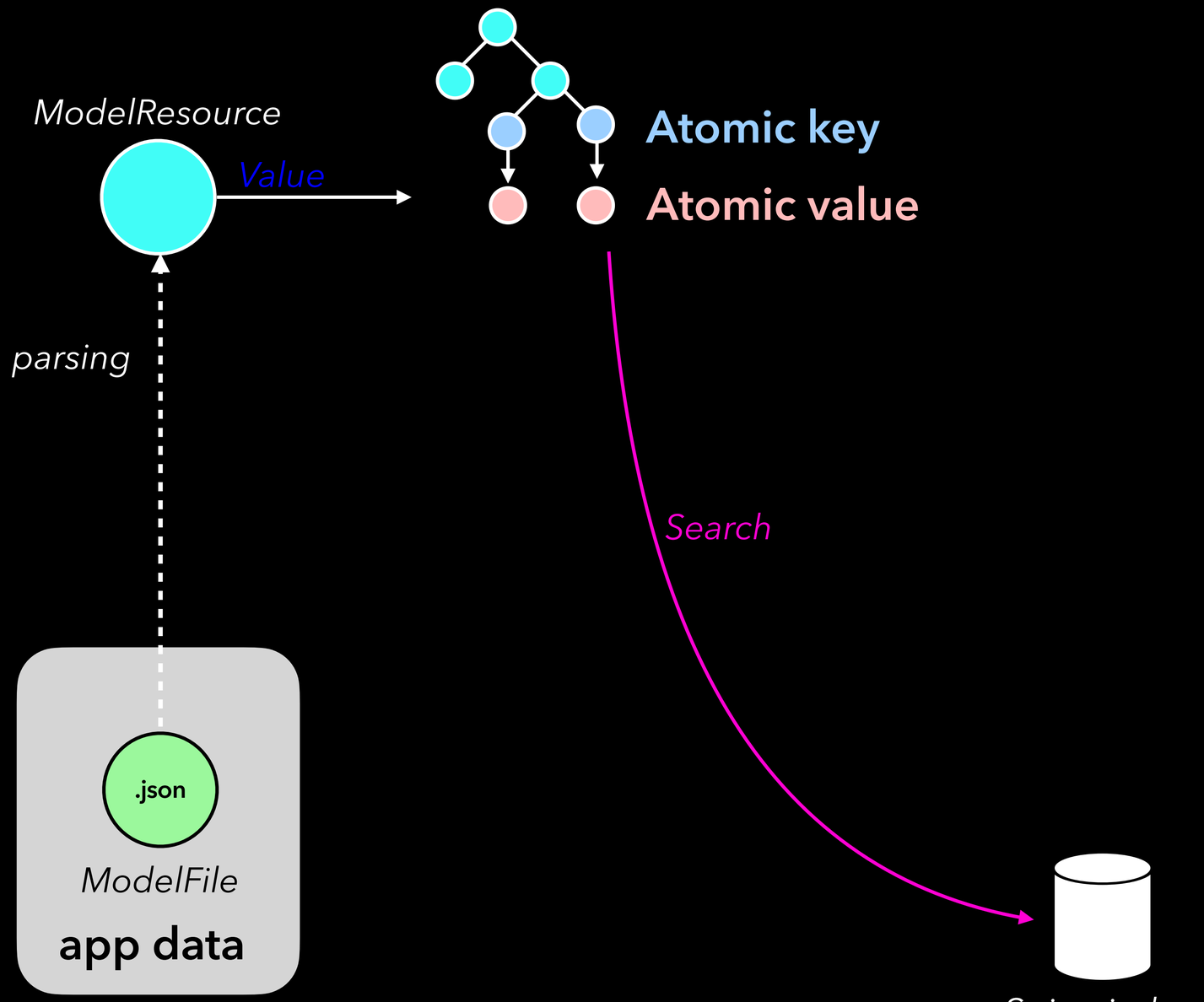
ELIXIR : STRINGS



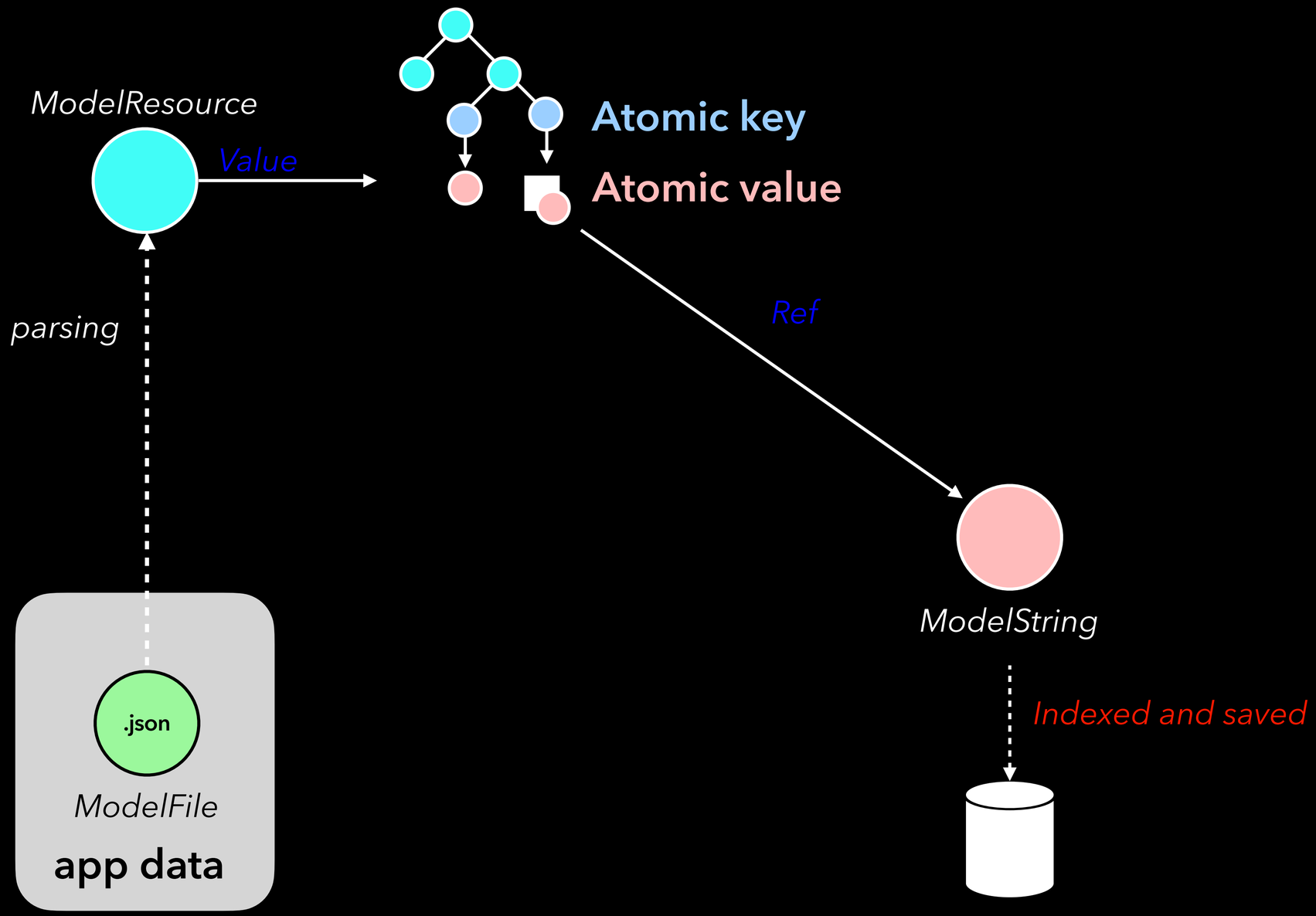
ELIXIR : STRINGS



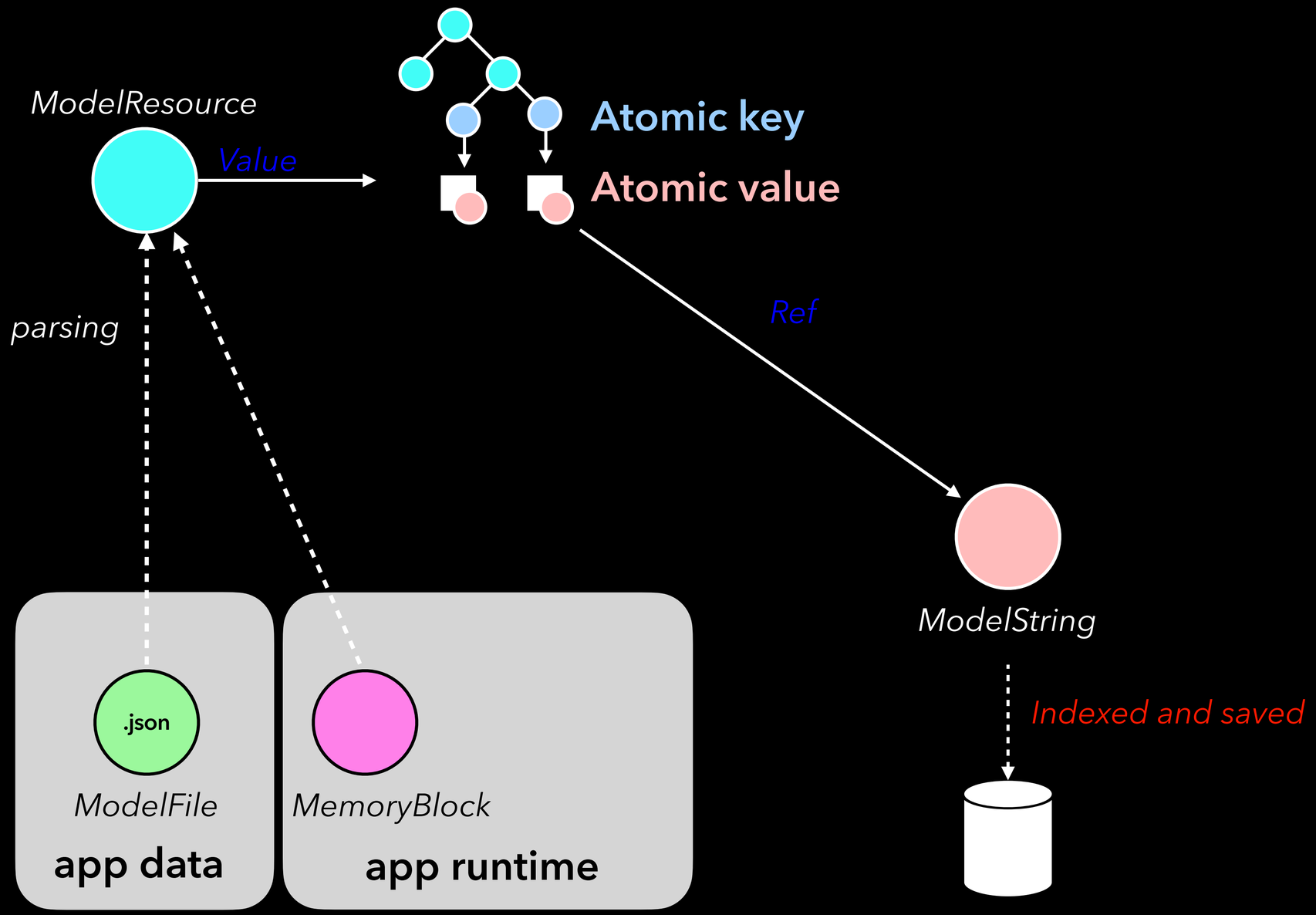
ELIXIR : STRINGS



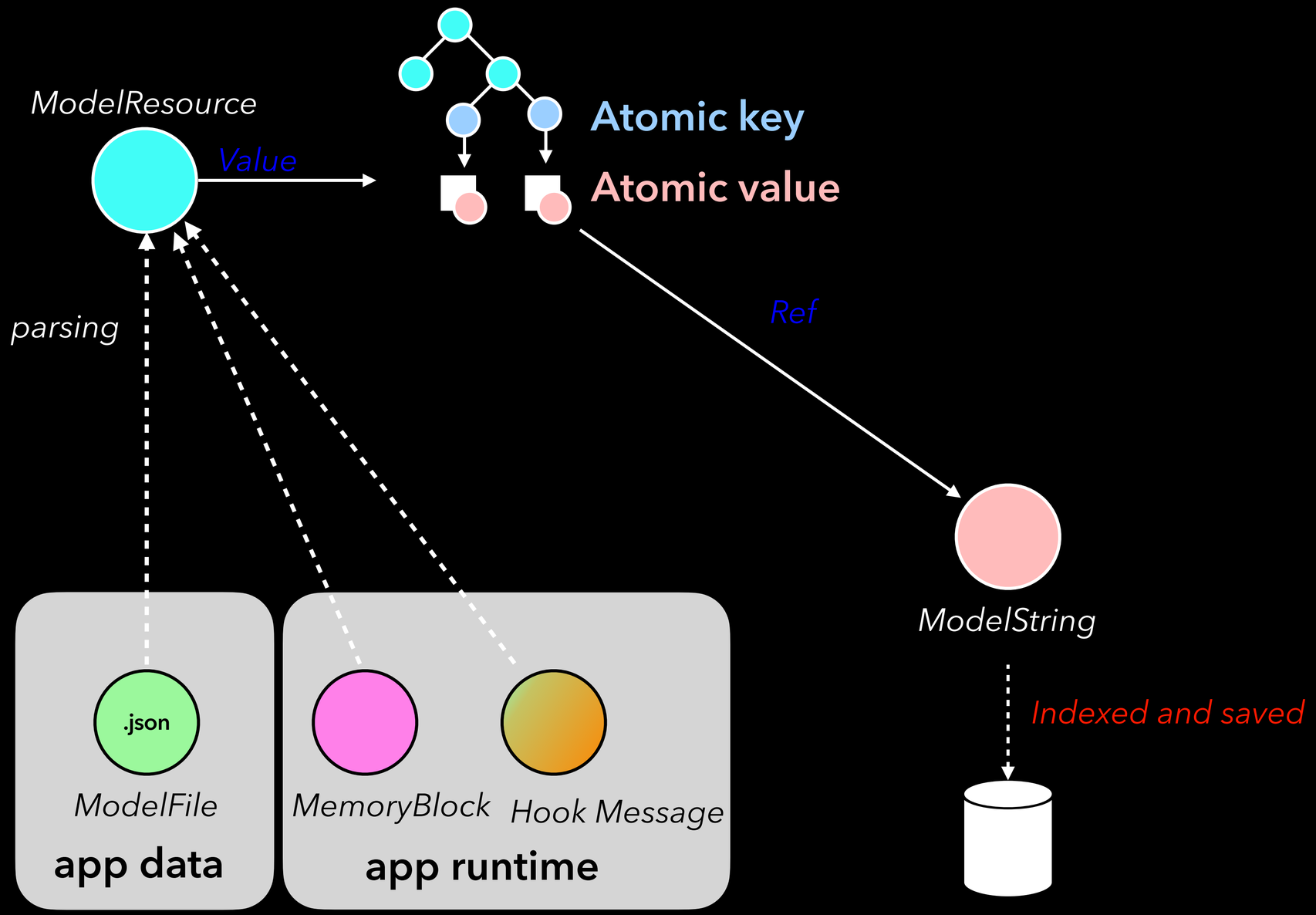
ELIXIR : STRINGS



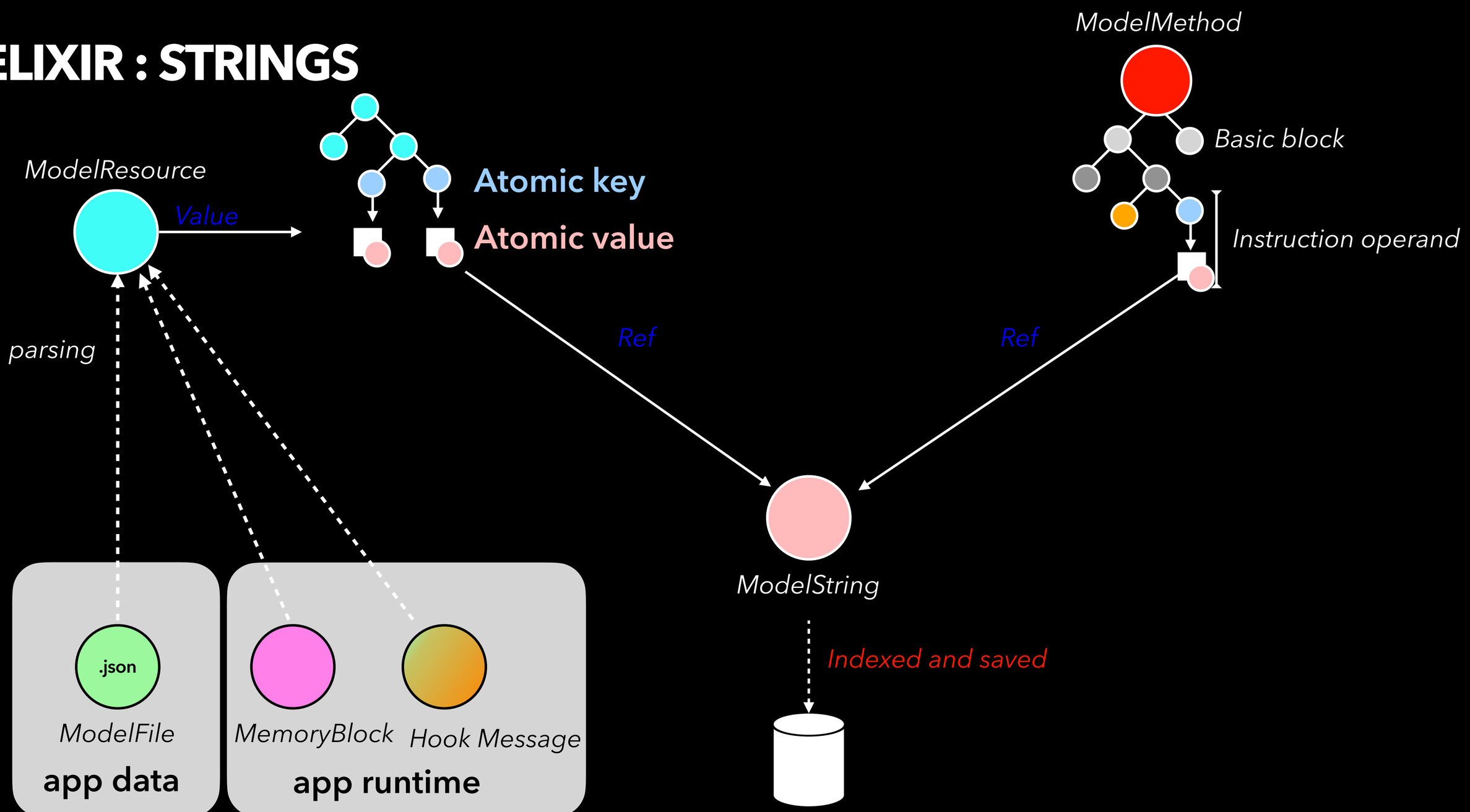
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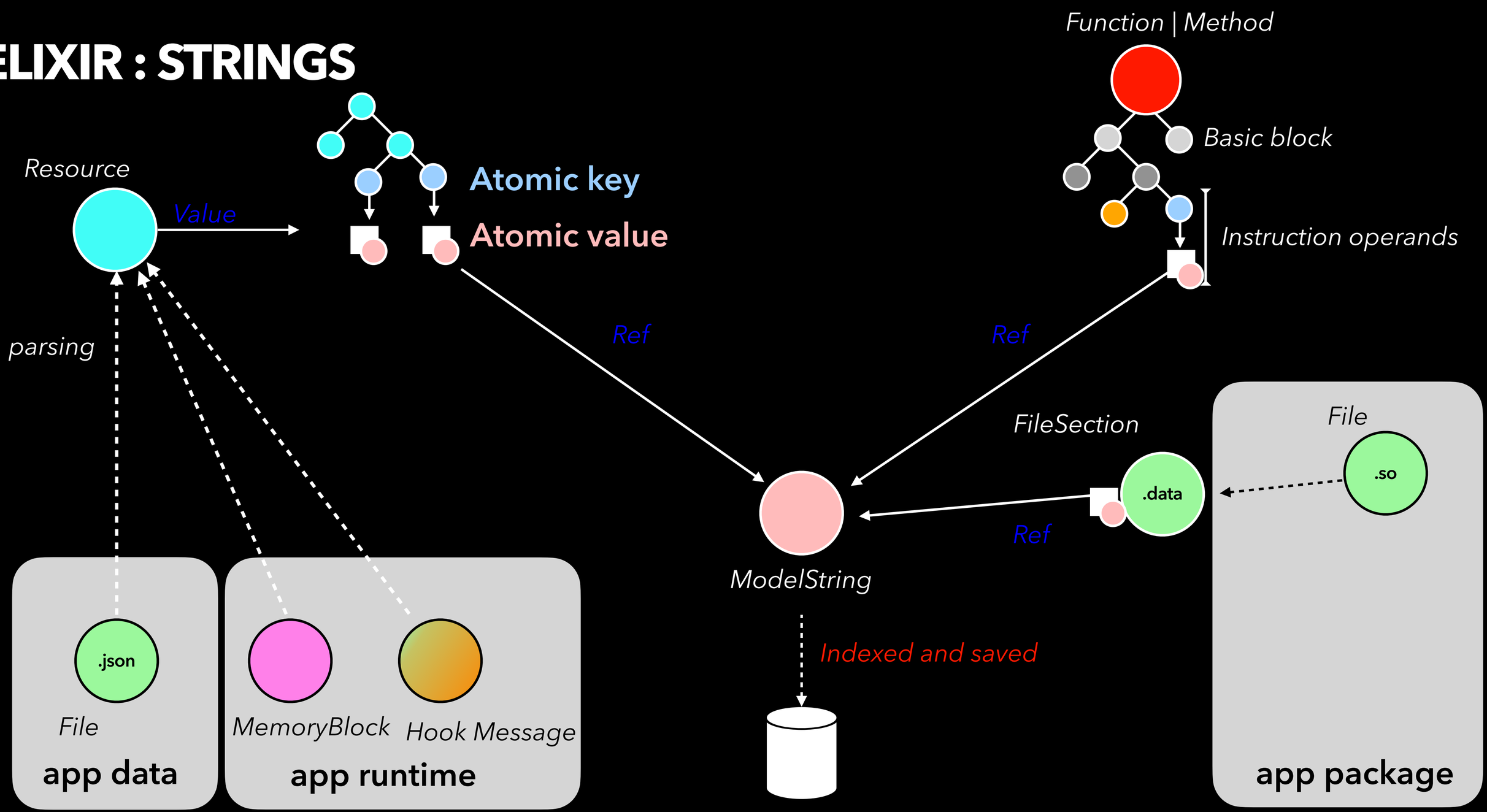
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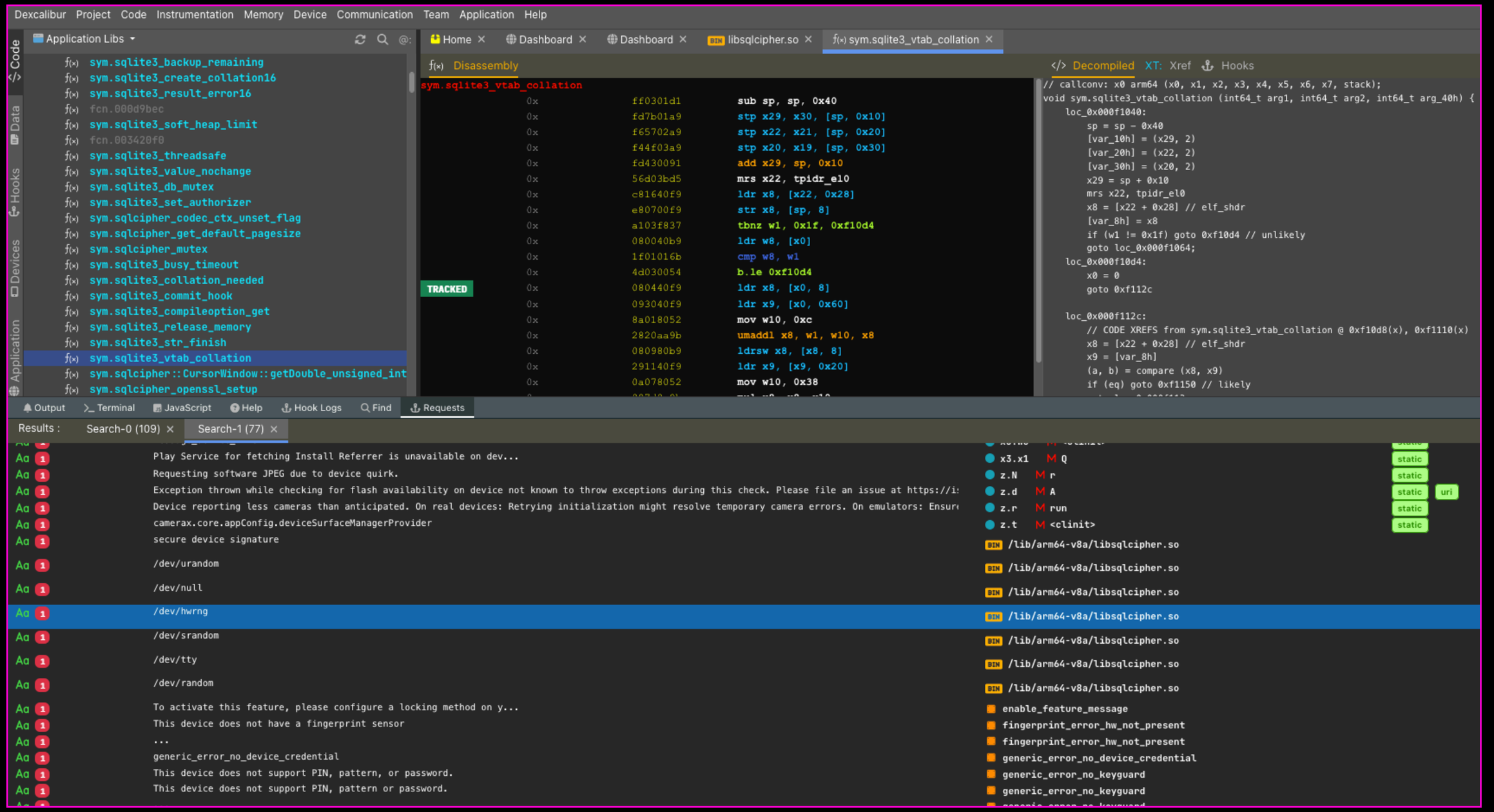


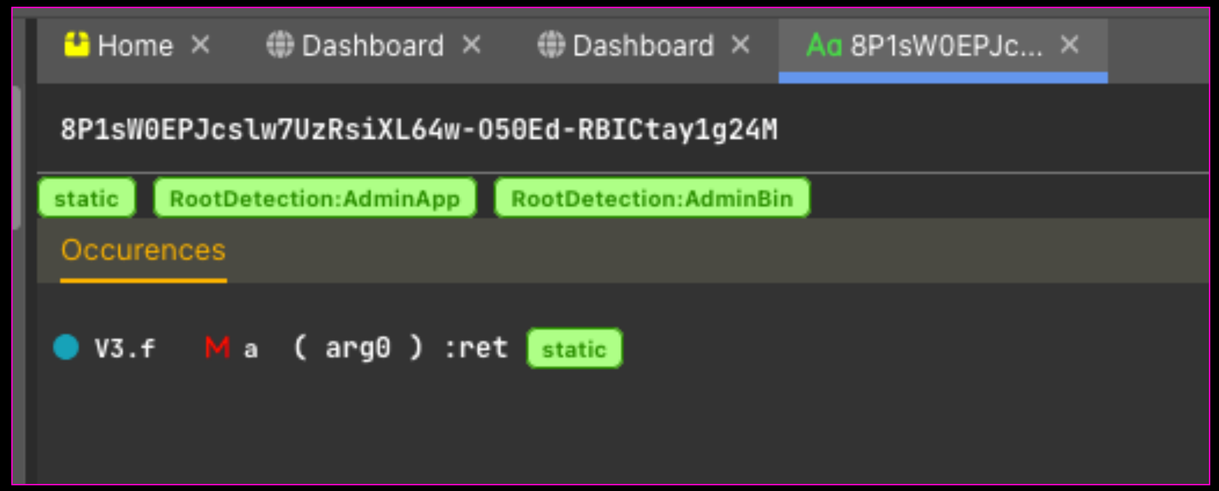
ELIXIR : STRINGS



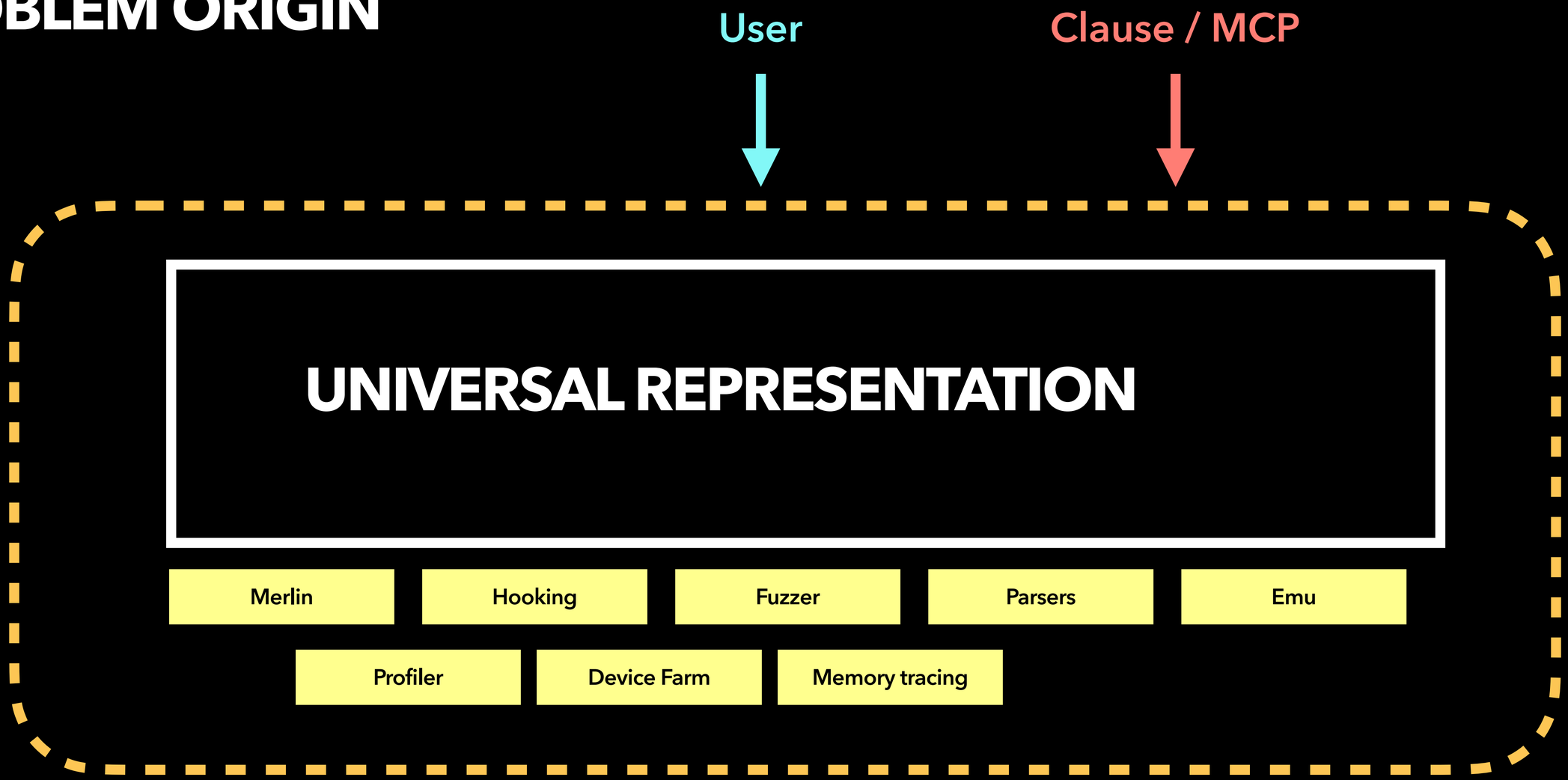
ELIXIR : STRINGS




Dexcalibur Project Code Instrumentation Memory Device Communication Team Application Help
Application Libs
sym.sqlite3_backup_remaining
sym.sqlite3_create_collation16
sym.sqlite3_result_error16
fcn.000d9bec
sym.sqlite3_soft_heap_limit
fcn.003420f0
sym.sqlite3_threadsafe
sym.sqlite3_value_nochange
sym.sqlite3_db_mutex
sym.sqlite3_set_authorizer
sym.sqlcipher_codec_ctx_unset_flag
sym.sqlcipher_get_default_pagesize
sym.sqlcipher_mutex
sym.sqlite3_busy_timeout
sym.sqlite3_collation_needed
sym.sqlite3_commit_hook
sym.sqlite3_compileoption_get
sym.sqlite3_release_memory
sym.sqlite3_str_finish
sym.sqlite3_vtab_collation
sym.sqlcipher::CursorWindow::getDouble_unsigned_int
sym.sqlcipher_openssl_setup
Home
Dashboard
Dashboard
libsqlcipher.so
sym.sqlite3_vtab_collation
Disassembly
Decompiled
XT: Xref
Hooks
sym.sqlite3_vtab_collation
0x ff0301d1 sub sp, sp, 0x40
0x fd7b01a9 stp x29, x30, [sp, 0x10]
0x f65702a9 stp x22, x21, [sp, 0x20]
0x f44f03a9 stp x20, x19, [sp, 0x30]
0x fd430091 add x29, sp, 0x10
0x 56d03bd5 mrs x22, tpidr_el0
0x c81640f9 ldr x8, [x22, 0x28]
0x e80700f9 str x8, [sp, 8]
0x a103f837 tbnz w1, 0x1f, 0xf10d4
0x 080040b9 ldr w8, [x0]
0x 1f01016b cmp w8, w1
0x 4d030054 b.le 0xf10d4
TRACKED
0x 080440f9 ldr x8, [x0, 8]
0x 093040f9 ldr x9, [x0, 0x60]
0x 8a018052 mov w10, 0xc
0x 2820aa9b umaddl x8, w1, w10, x8
0x 080980b9 ldrsw x8, [x8, 8]
0x 291140f9 ldr x9, [x9, 0x20]
0x 0a078052 mov w10, 0x38
// callconv: x0 arm64 (x0, x1, x2, x3, x4, x5, x6, x7, stack);
void sym.sqlite3_vtab_collation (int64_t arg1, int64_t arg2, int64_t arg_40h) {
loc_0x000f1040:
sp = sp - 0x40
[var_10h] = (x29, 2)
[var_20h] = (x22, 2)
[var_30h] = (x20, 2)
x29 = sp + 0x10
mrs x22, tpidr_el0
x8 = [x22 + 0x28] // elf_shdr
[var_8h] = x8
if (w1 != 0x1f) goto 0xf10d4 // unlikely
goto loc_0x000f1064;
loc_0x000f10d4:
x0 = 0
goto 0xf112c
loc_0x000f112c:
// CODE XREFS from sym.sqlite3_vtab_collation @ 0xf10d8(x), 0xf1110(x)
x8 = [x22 + 0x28] // elf_shdr
x9 = [var_8h]
(a, b) = compare (x8, x9)
if (eq) goto 0xf1150 // likely
Output
Terminal
JavaScript
Help
Hook Logs
Find
Requests
Results :
Search-0 (109)
Search-1 (77)
Play Service for fetching Install Referrer is unavailable on dev...
Requesting software JPEG due to device quirk.
Exception thrown while checking for flash availability on device not known to throw exceptions during this check. Please file an issue at https://i
Device reporting less cameras than anticipated. On real devices: Retrying initialization might resolve temporary camera errors. On emulators: Ensur
camerax.core.appConfig.deviceSurfaceManagerProvider
secure device signature
/dev/urandom
/dev/null
/dev/hwrng
/dev/srandom
/dev/tty
/dev/random
To activate this feature, please configure a locking method on y...
This device does not have a fingerprint sensor
...
generic_error_no_device_credential
This device does not support PIN, pattern, or password.
This device does not support PIN, pattern or password.
x3.x1 Q
z.N r
z.d A
z.r run
z.t <clinit>
static
uri
/lib/arm64-v8a/libsqlcipher.so
enable_feature_message
fingerprint_error_hw_not_present
fingerprint_error_hw_not_present
generic_error_no_device_credential
generic_error_no_keyguard
generic_error_no_keyguard


Home
Dashboard
Dashboard
Aa 8P1sW0EPJc...
8P1sW0EPJcslw7UzRsiXL64w-050Ed-RBICtay1g24M
static
RootDetection:AdminApp
RootDetection:AdminBin
Occurences
V3.f
M a (arg0) :ret
static

PROBLEM ORIGIN

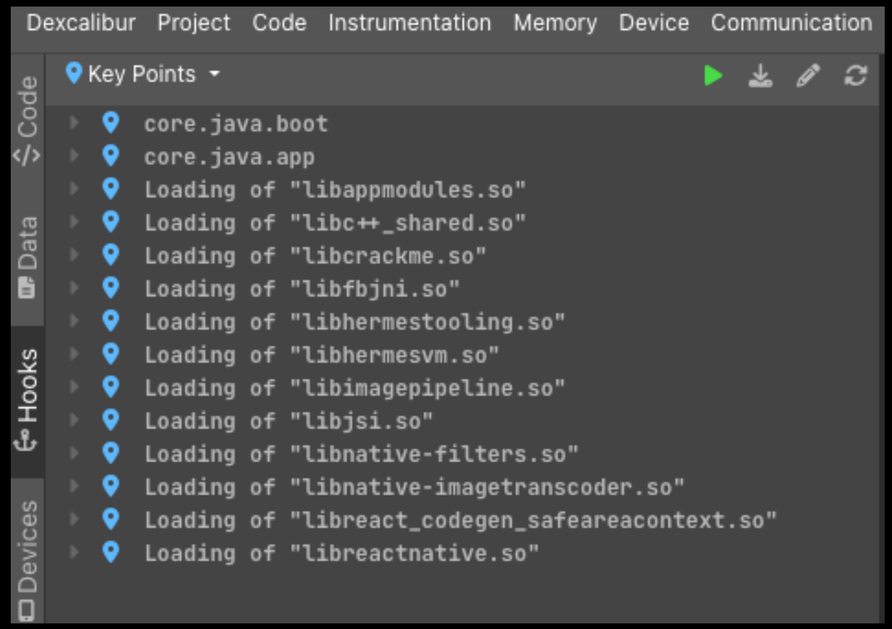


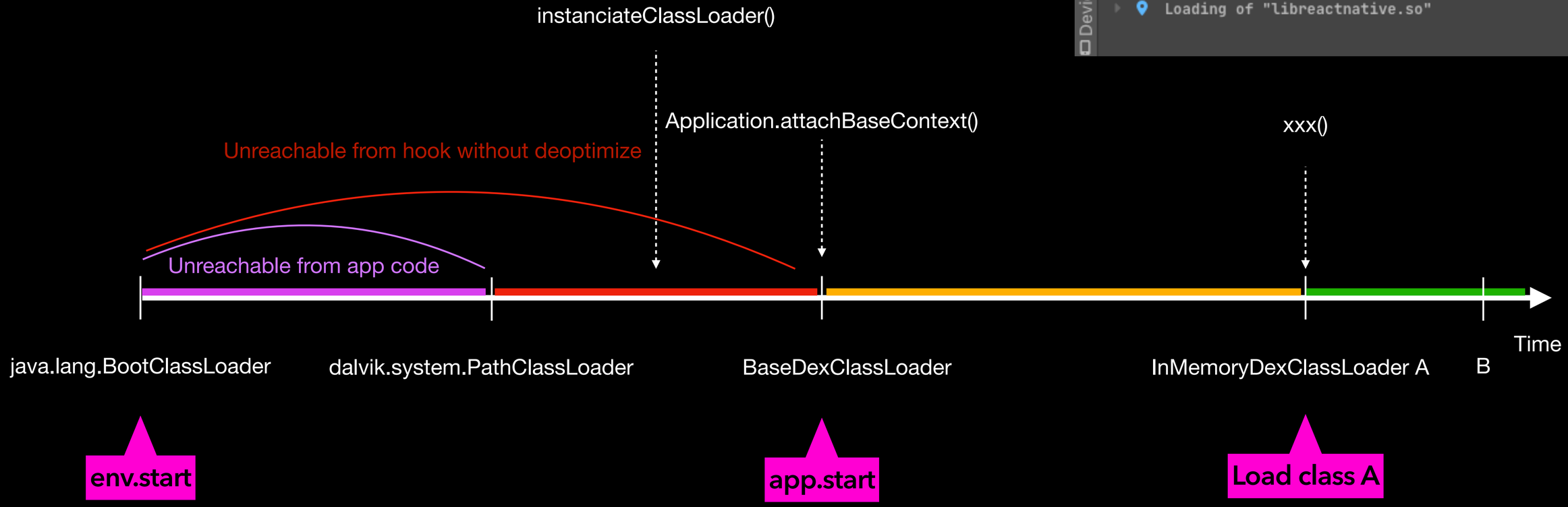
PART 3

Features & Demo

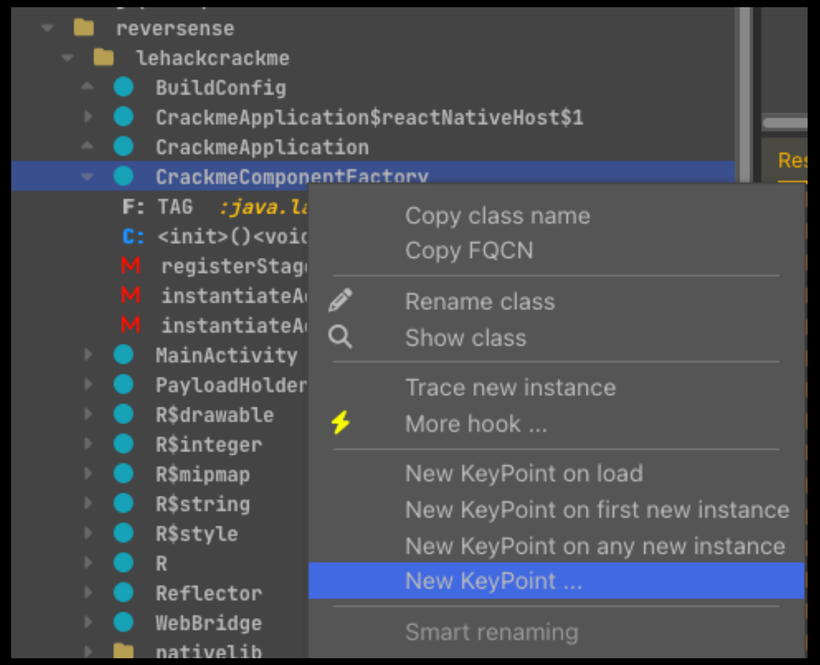


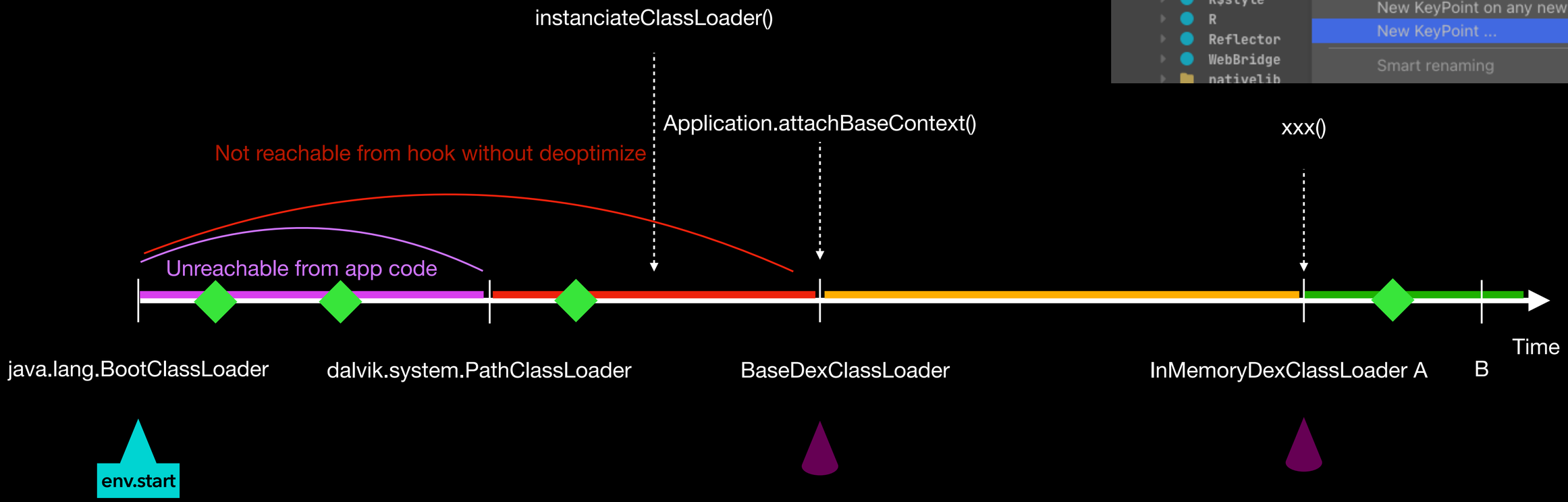
KEY POINT


Dexcalibur Project Code Instrumentation Memory Device Communication
Key Points
core.java.boot
core.java.app
Loading of "libappmodules.so"
Loading of "libc++_shared.so"
Loading of "libcrackme.so"
Loading of "libfbjni.so"
Loading of "libhermestooling.so"
Loading of "libhermesvm.so"
Loading of "libimagepipeline.so"
Loading of "libjsi.so"
Loading of "libnative-filters.so"
Loading of "libnative-imagetranscoder.so"
Loading of "libreact_codegen_safeareacontext.so"
Loading of "libreactnative.so"
Code
Data
Hooks
Devices

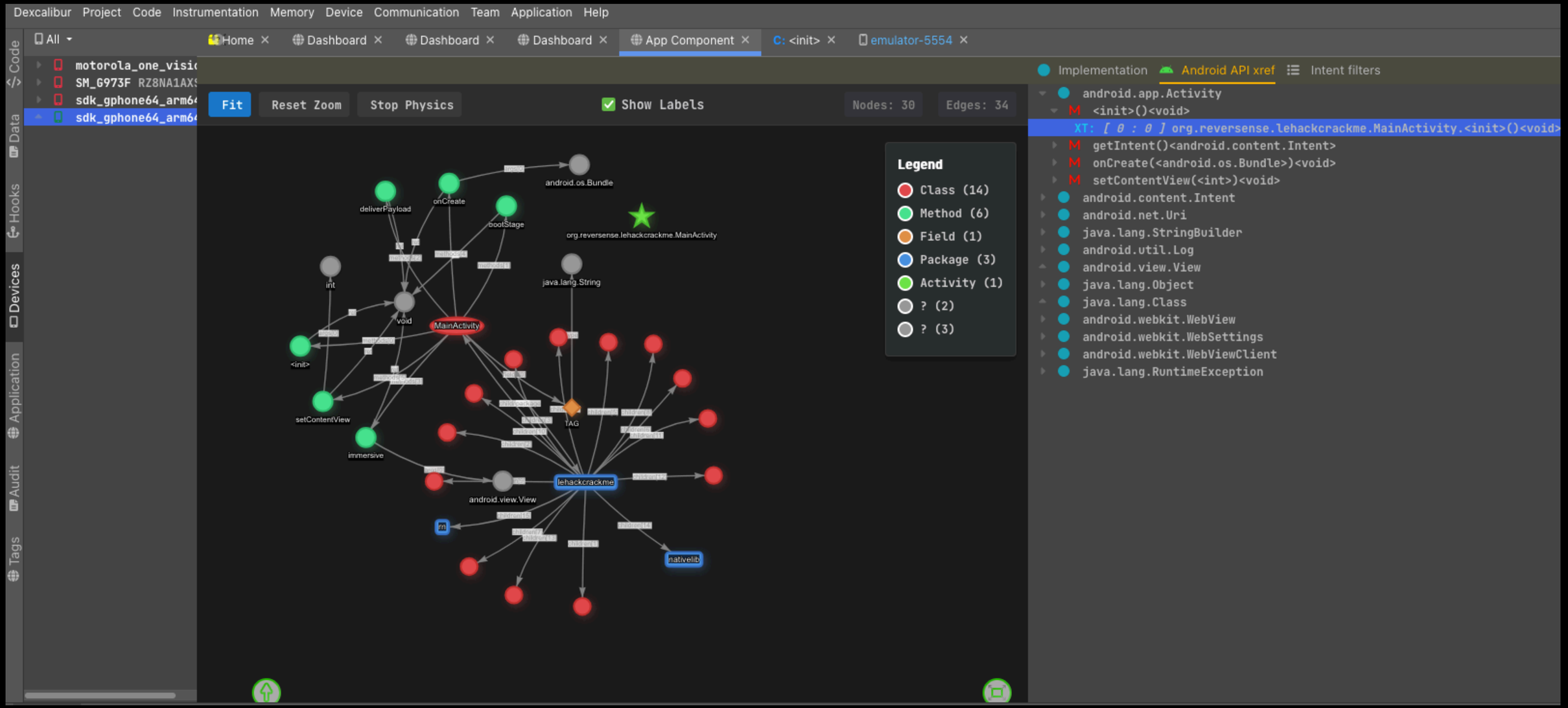

instanciateClassLoader()
Application.attachBaseContext()
xxx()
Unreachable from hook without deoptimize
Unreachable from app code
Time
java.lang.BootClassLoader
dalvik.system.PathClassLoader
BaseDexClassLoader
InMemoryDexClassLoader A
B
env.start
app.start
Load class A

KEY POINT

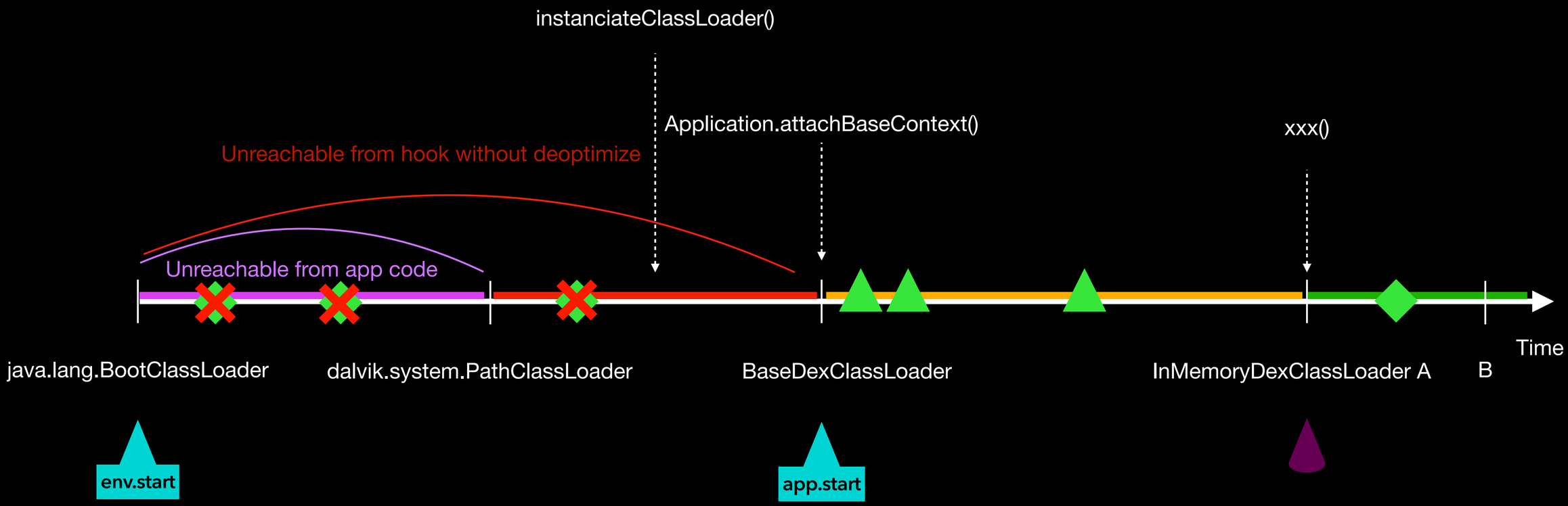

reversense
lehackcrackme
BuildConfig
CrackmeApplication$reactNativeHost$1
CrackmeApplication
CrackmeComponentFactory
F: TAG
C: <init>()<voi
M registerStag
M instantiateA
M instantiateA
MainActivity
PayloadHolder
R$drawable
R$integer
R$mipmap
R$string
R$style
R
Reflector
WebBridge
nativelib
Copy class name
Copy FQCN
Rename class
Show class
Trace new instance
More hook ...
New KeyPoint on load
New KeyPoint on first new instance
New KeyPoint on any new instance
New KeyPoint ...
Smart renaming


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env.start

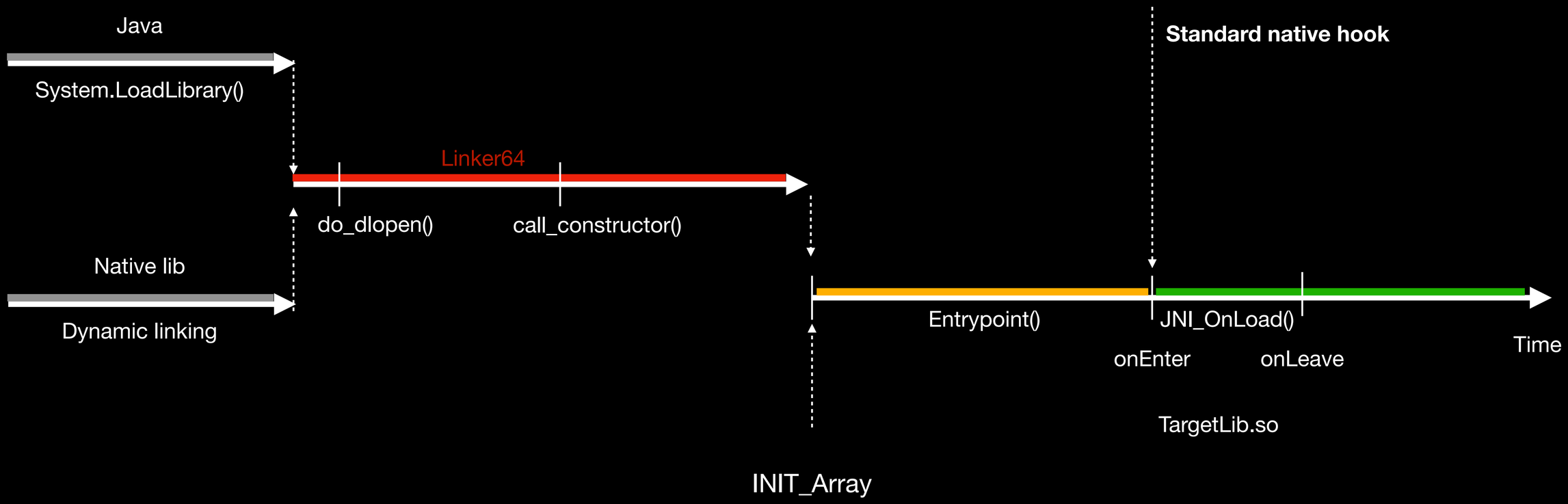
KEY POINT


Dexcalibur Project Code Instrumentation Memory Device Communication Team Application Help
Home
Dashboard
Dashboard
Dashboard
App Component
C: <init>
emulator-5554
Fit
Reset Zoom
Stop Physics
Show Labels
Nodes: 30
Edges: 34
Legend
Class (14)
Method (6)
Field (1)
Package (3)
Activity (1)
? (2)
? (3)
android.os.Bundle
deliverPayload
onCreate
bootStage
org.reversense.lehackcrackme.MainActivity
int
java.lang.String
void
MainActivity
<init>
setContentView
TAG
immersive
lehackcrackme
android.view.View
nativelib
Implementation
Android API xref
Intent filters
android.app.Activity
<init>()<void>
XT: [0 : 0] org.reversense.lehackcrackme.MainActivity.<init>()<void>
getIntent()<android.content.Intent>
onCreate(<android.os.Bundle>)<void>
setContentView(<int>)<void>
android.content.Intent
android.net.Uri
java.lang.StringBuilder
android.util.Log
android.view.View
java.lang.Object
java.lang.Class
android.webkit.WebView
android.webkit.WebSettings
android.webkit.WebViewClient
java.lang.RuntimeException

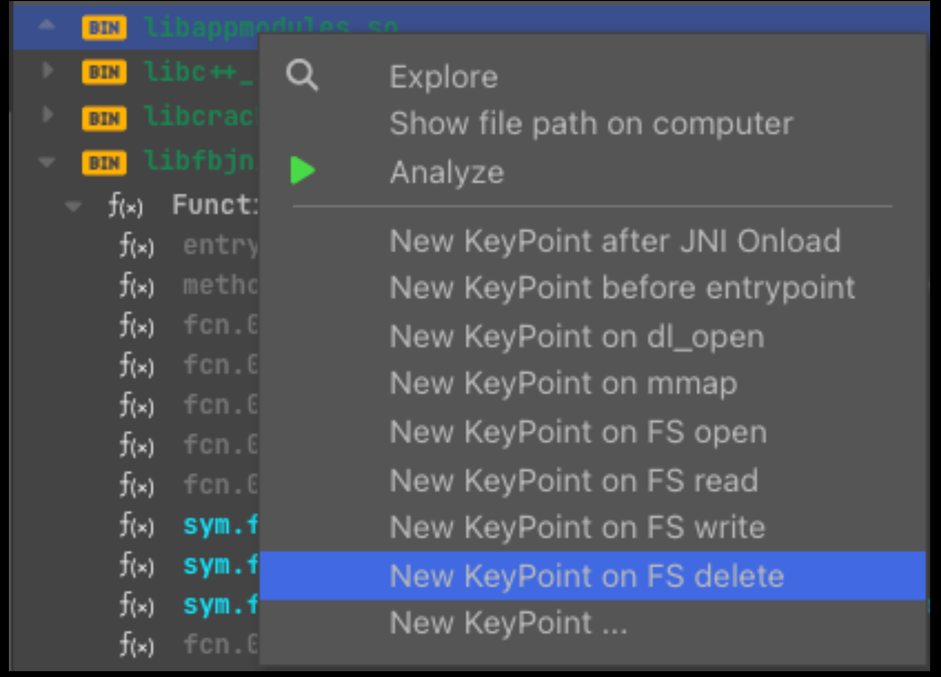
KEY POINT

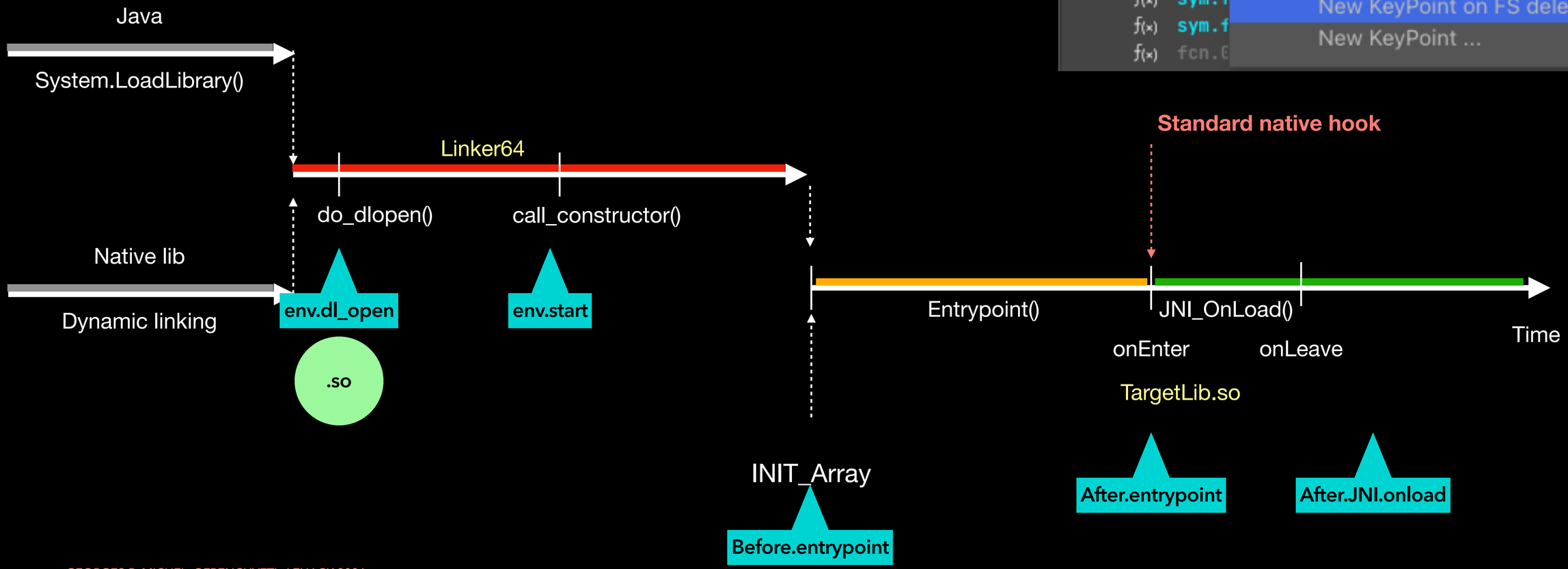

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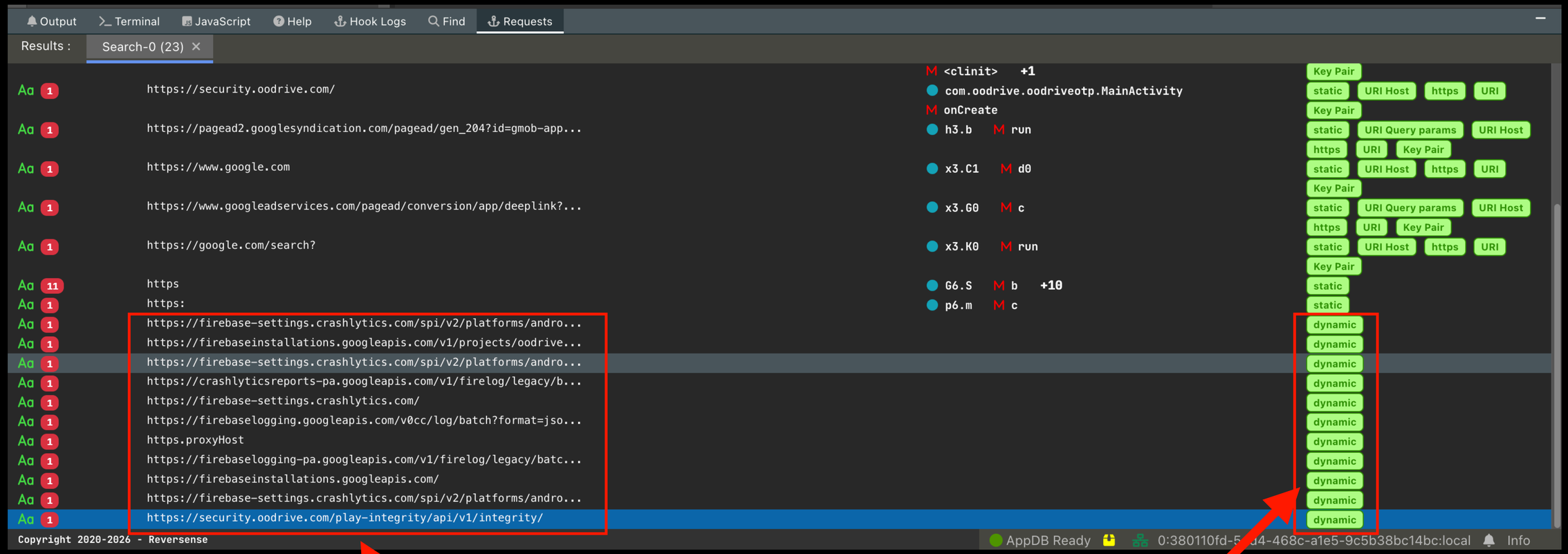

Java
System.LoadLibrary()
Linker64
do_dlopen()
call_constructor()
Native lib
Dynamic linking
Standard native hook
Entrypoint()
JNI_OnLoad()
onEnter
onLeave
Time
TargetLib.so
INIT_Array

KEY POINT


Explore
Show file path on computer
Analyze
New KeyPoint after JNI Onload
New KeyPoint before entrypoint
New KeyPoint on dl_open
New KeyPoint on mmap
New KeyPoint on FS open
New KeyPoint on FS read
New KeyPoint on FS write
New KeyPoint on FS delete
New KeyPoint ...


Java
System.LoadLibrary()
Linker64
do_dlopen()
call_constructor()
Native lib
Dynamic linking
env.dl_open
env.start
.so
Standard native hook
Entrypoint()
JNI_OnLoad()
onEnter
onLeave
Time
TargetLib.so
INIT_Array
Before.entrypoint
After.entrypoint
After.JNI.onload

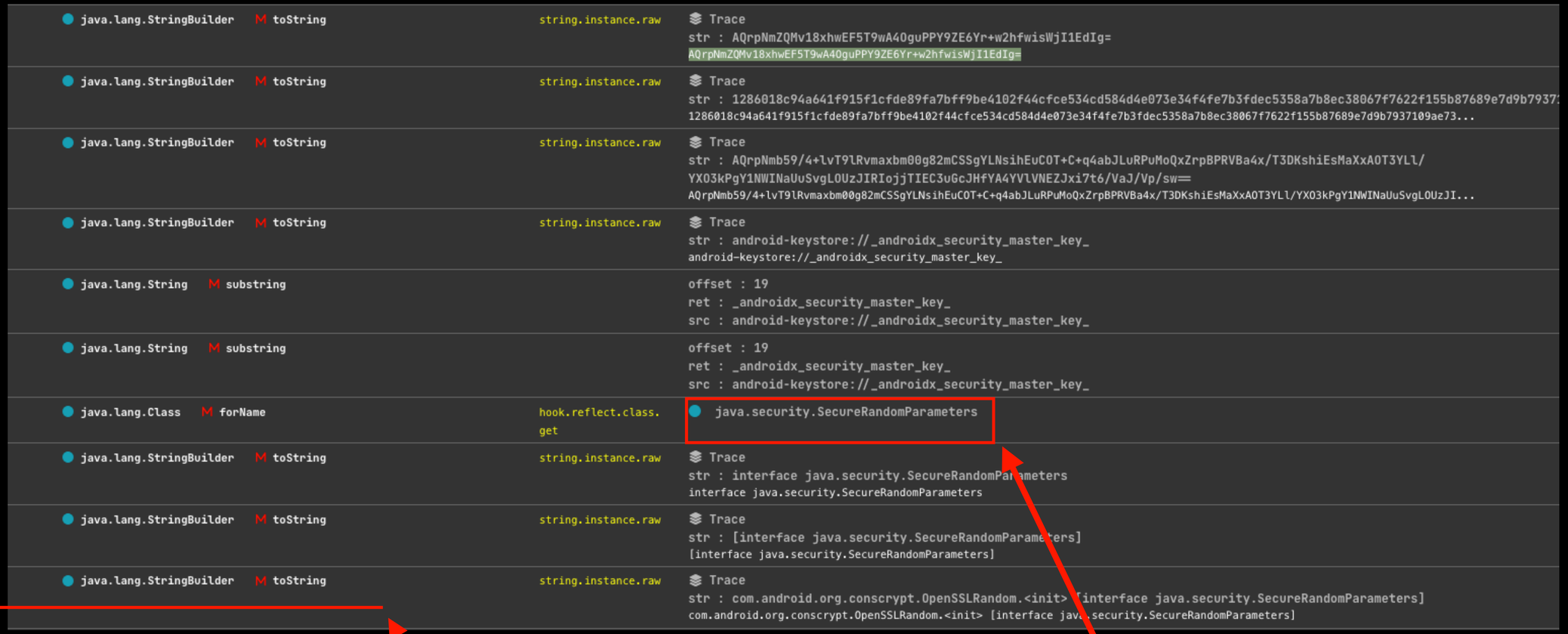
DYNAMIC DISCOVERY



String value

« dynamic » tag means « gathered at runtime »

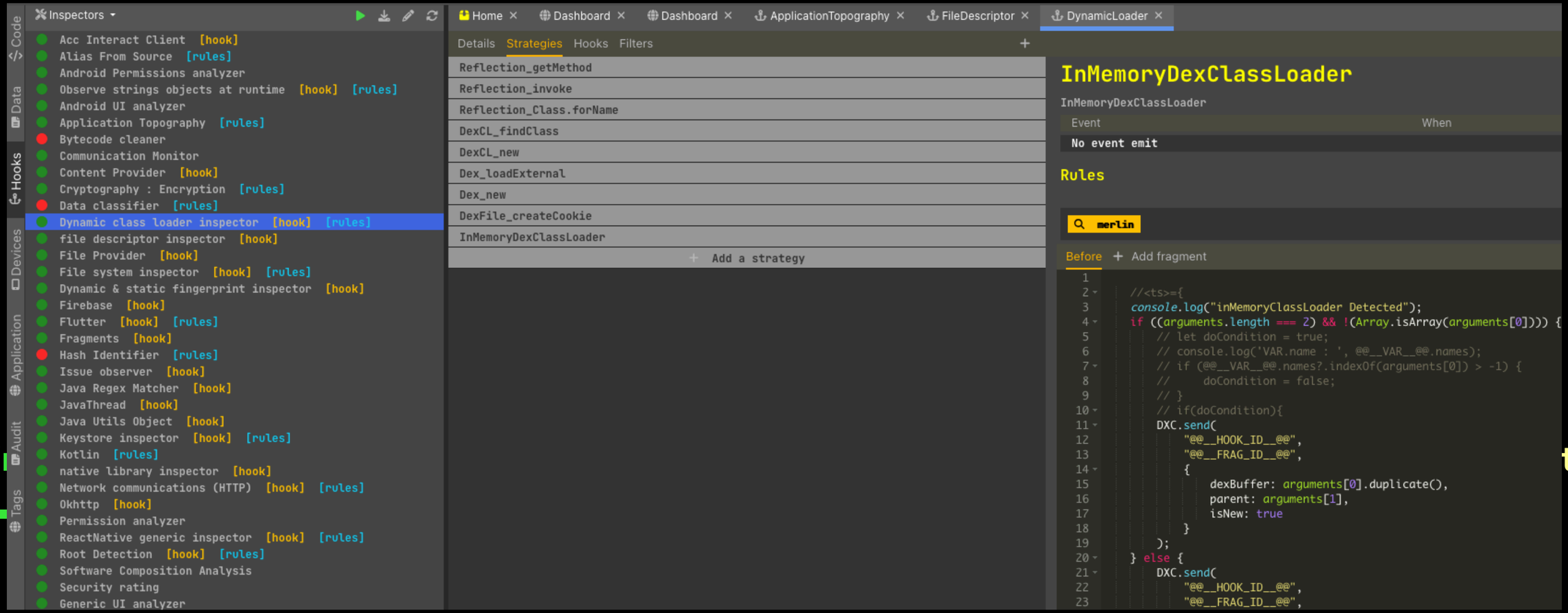
DYNAMIC DISCOVERY



Node where the hook message has been emitted

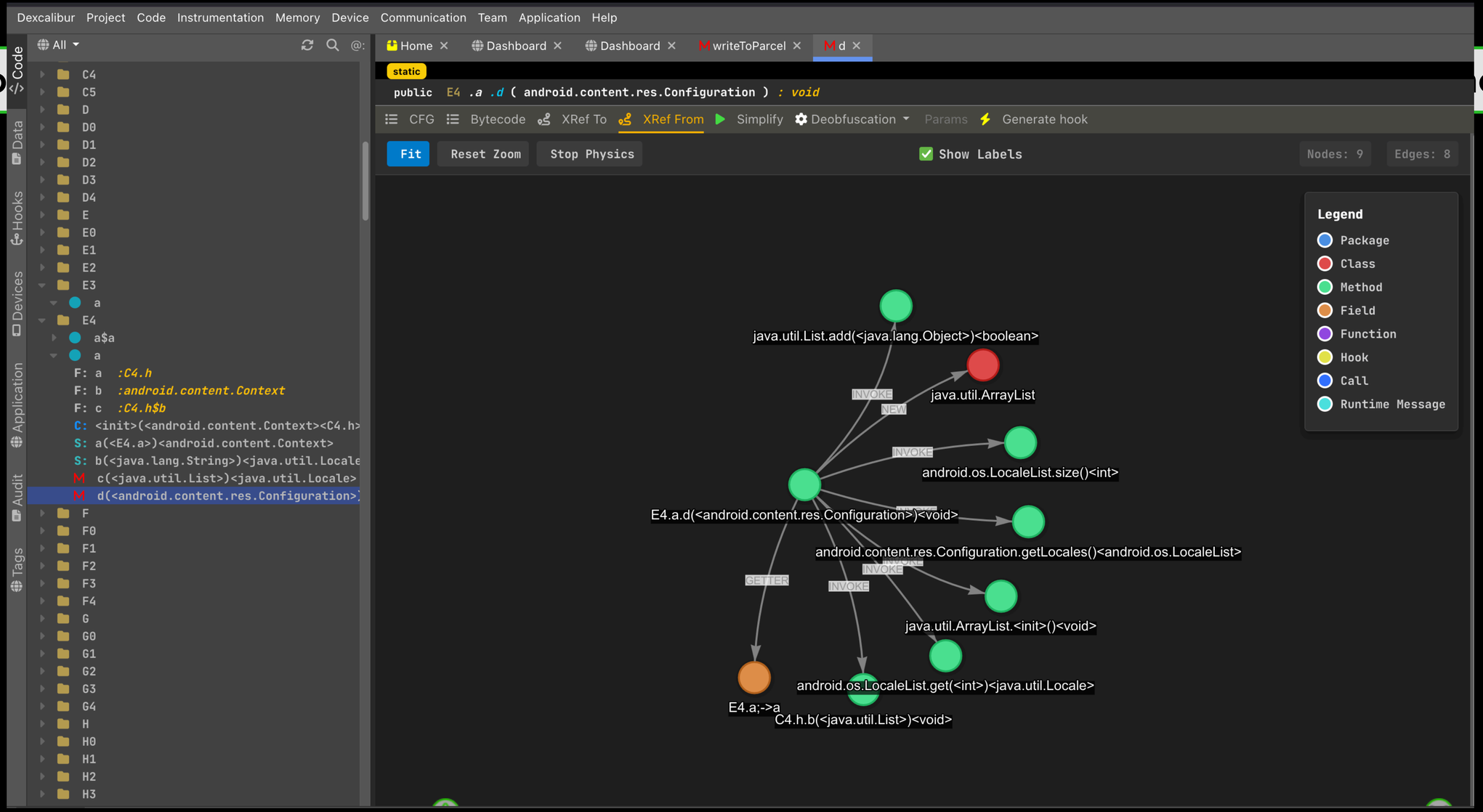
« dynamic » tag means « gathered at runtime »

INSPECTORS



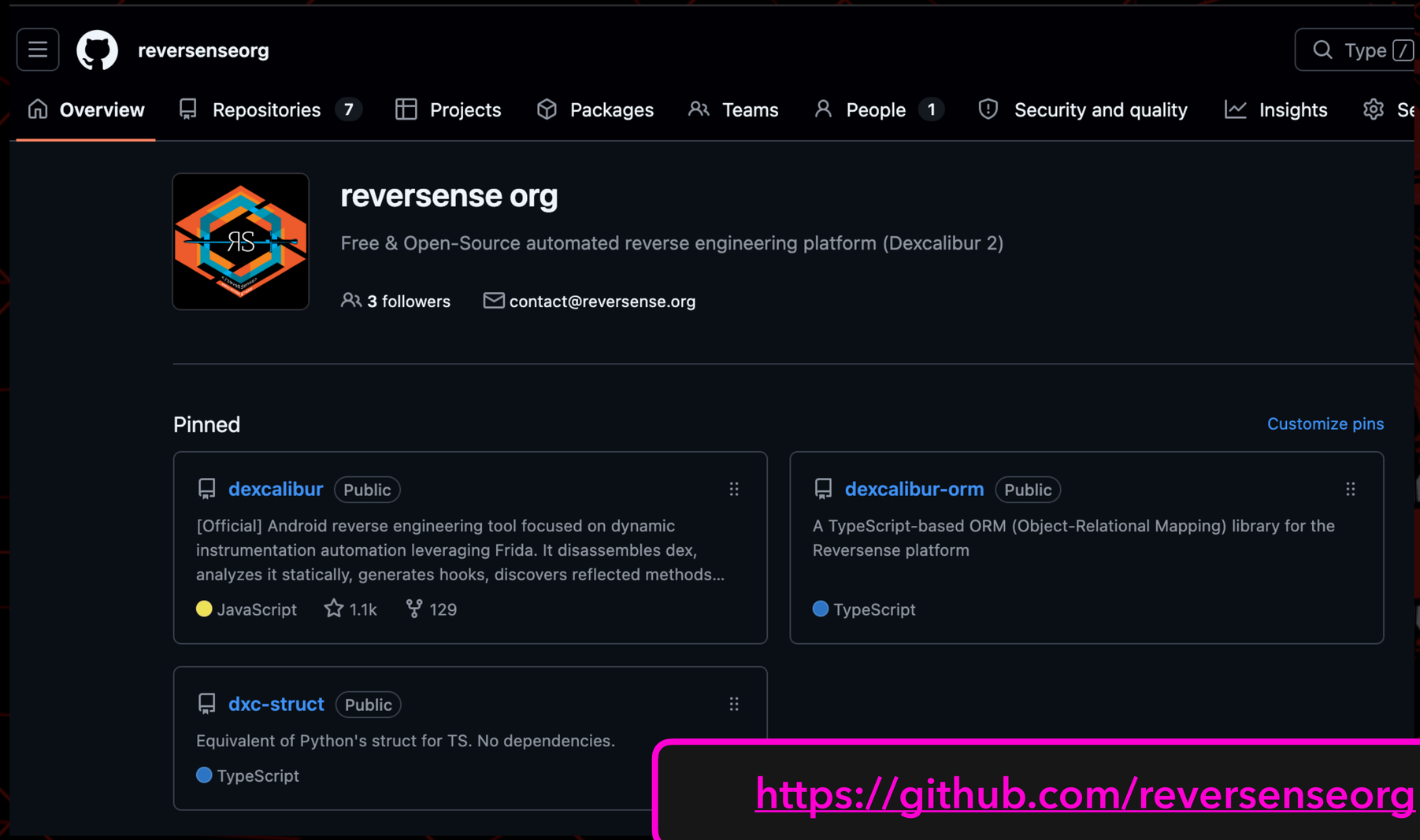
Engi t)

HOOK MANAGEMENT



DEMO REVERSENSE

REPO

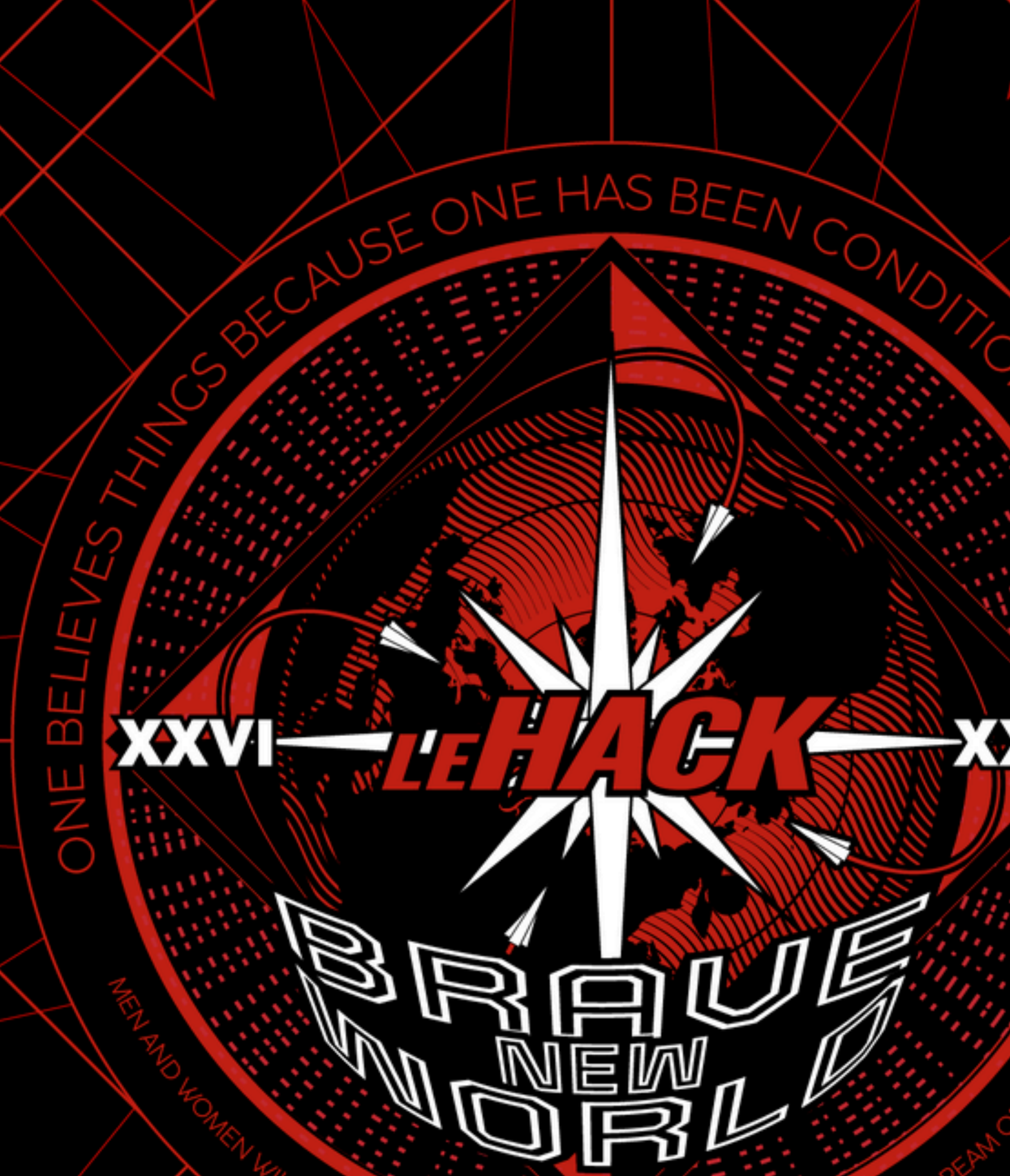


https://github.com/reversenseorg

THANKS :)

frenchyeti@reversense.org

www.reversense.com



Workshop Info

0 - Get started (~30 min)

1 - Bypass anti-hook countermeasure with Reversense (~60 min)

=== Coffee Break ? ===

2 - Uncover commands in Samsung Debug Bridge (SDB) (~60 min)

3 - Enumerate the attack surface with AI (~30 min)

